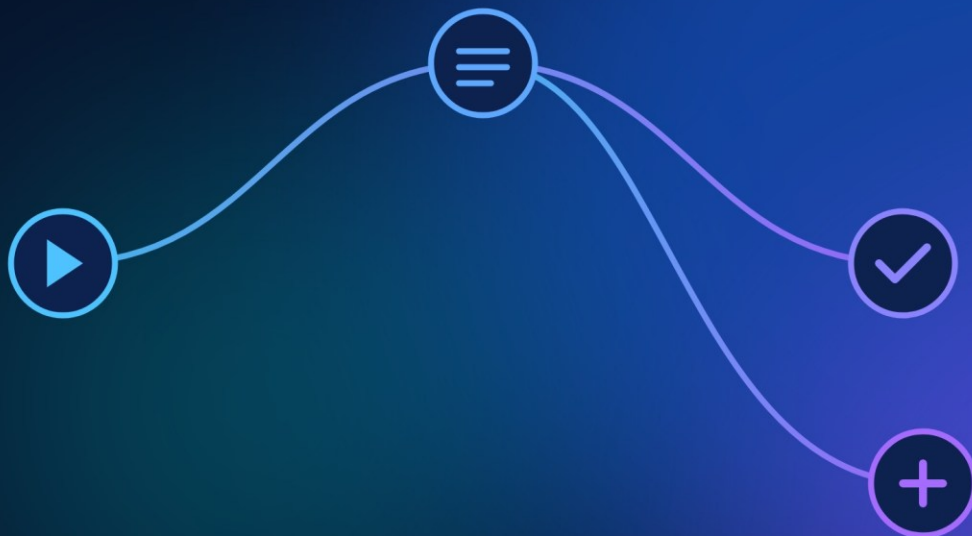




Microsoft Power Automate for End Users

Two-day participant manual

Build practical cloud flows with Microsoft 365 connectors, review their results, plan automation with Copilot, and package completed work for controlled transfer.

How to use this manual

This manual is written for participants who are new to Power Automate. Follow the chapters in order during the two-day course. Each chapter explains the business purpose, shows the configuration, provides checkpoints, and ends with independent practice. Screenshots show the important states to confirm; your environment may use slightly different labels or navigation.

Use only fictional training data and trainer approved destinations. Keep flows off until their connections, recipients, and expected effects have been reviewed. Testing a flow can create files, send messages, start approvals, or change shared data.

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Chapter 1: Understand Power Automate

Estimated time: 45 minutes

Your result: A simple automation plan and familiarity with the Power Automate home and creation pages.

What you will learn

By the end of this chapter, you will be able to:

- Identify a repetitive business task that could benefit from automation.
- Explain the trigger, actions, data, condition, and result of a flow.
- Choose between an automated, instant, and scheduled cloud flow.
- Open Power Automate and find Home, Create, and My flows.
- Write a short plan before building an automation.

Before you begin

You need a Windows computer with Microsoft Edge, an internet connection, and your training work or school account. Have your sign-in method ready. Your trainer will identify the environment to use.

For this chapter, you do not need an Excel workbook, a Form, or any test files. You will prepare those resources in Chapter 2. Have a place to record short answers, such as the notes area in your manual.

Your business scenario

You coordinate training for a small organization. Course materials arrive by email, session details are stored in Excel, and participants need reminders. You repeatedly download attachments, copy information into messages, and check whether requests have been approved.

These tasks have recognizable starting points and repeatable steps. During this course, you will use Power Automate to carry out some of those steps for you.

1.1 What is automation?

Automation means setting up a repeatable process so that software performs steps when a defined event occurs or when you ask it to run.

For example, you might normally open an email, download its attachment, and save the file to a training folder. An automation can follow the same rule whenever a suitable email arrives.

Power Automate calls an automation a **flow**. In this course, you will build **cloud flows** that work with online services such as Outlook and OneDrive for Business.

Automation can reduce repeated work, apply rules consistently, and make routine tasks easier to track. It still needs a clear rule and testing. If you choose the wrong folder or recipient, the flow can repeat that mistake.

Tip: Start with a small task whose result you can check easily. A successful first flow might save one test attachment in the correct folder.

1.2 Recognize the parts of a flow

Consider this requirement:

When an email with the subject marker [PA TRAINING] arrives, examine each attachment. If the attachment is a PDF, save it to the training attachments folder.

Part	Plain-language meaning	Example
Trigger	What starts the flow	A matching email arrives.
Action	A step the flow carries out	Create a file in OneDrive for Business.
Data	Information a step receives or produces	The attachment's name and contents.
Condition	A question that selects a path	Is this attachment a PDF?
Loop	Steps repeated for each item	Check each attachment in the email.
Result	The business outcome you verify	Matching files appear in the intended folder.

The basic pattern is **something starts the flow > the flow follows its steps > you check the result**. A flow does not need a condition or a loop unless the process calls for one.

An **action** is an instruction in the flow. A **result** is what you wanted the process to achieve. Checking both will help you troubleshoot later.

Guided activity: Trace an attachment

1. Imagine a training email arrives with two attachments: **CourseOutline.pdf** and **TrainerPhoto.jpg**.
2. Identify what starts the flow. Write **A matching email arrives**.
3. Identify what the flow must examine. Write **Each attachment**.
4. Apply the PDF rule to **CourseOutline.pdf**. Record **Save the file**.
5. Apply the same rule to **TrainerPhoto.jpg**. Record **Skip the file**.
6. State the expected result in the folder. Record the filename that should appear.

Checkpoint: The folder should receive CourseOutline.pdf. The image should be skipped. One incoming email can contain several items that a loop processes separately.

Think about it: If the email contains no PDF attachments, the intended result is that no file is created. A flow can run successfully without creating a file when its rules exclude every attachment.

1.3 Choose how the flow starts

Cloud-flow type	When it starts	Training example
Automated	When a specified event occurs	Save attachments when a training email arrives.
Instant	When you deliberately start it	Create an Excel summary when you are ready to share it.
Scheduled	At a time and frequency you configure	Send training reminders every Monday morning.

Choose the type by asking: **What should make this process begin?** The fact that a flow sends an email does not tell you its type. All three types could send an email; their starting points differ.

Quick check

Choose a flow type for each request before reading the answers.

1. Whenever a new training request is submitted, ask someone to approve it.
2. At 9:00 AM every Friday, prepare a weekly reminder.
3. Let me generate a summary whenever I need one.

Answers: 1. Automated. 2. Scheduled. 3. Instant.

1.4 Understand connectors and connections

A **connector** gives Power Automate a set of triggers and actions for a service. For example, Office 365 Outlook provides email operations, and OneDrive for Business provides file operations.

A **connection** links those operations to an account. When a later lesson asks you to sign in to a connector, check that you are using the training account. The account must have permission to use the mailbox, folder, or other resource involved.

This course uses standard connectors for its core exercises. Microsoft documents that Power Automate Free can create and run cloud flows with standard connectors, but does not include flow sharing. You will create your own flows. [Microsoft: Power Automate license types](https://learn.microsoft.com/en-us/power-platform/admin/power-automate-licensing/types).

Your training account also needs access to the Microsoft 365 services used by each exercise. Having Power Automate Free does not by itself tell you whether you have a mailbox, Teams access, or OneDrive for Business.

Tip: Office 365 Outlook and Outlook.com are different connectors. Follow the connector name in the lesson rather than choosing by icon alone.

1.5 Explore Power Automate

In this activity, you will find the pages you will use throughout the course. You will build your first flow in Chapter 3.

Most numbered software steps have a screenshot beneath them. Where two consecutive steps act on the same screen, one figure serves both and its caption says so. Sign-in screens are not illustrated. Focused views keep small labels readable.

Open the training workspace

1. Open **Microsoft Edge**. In the address bar, enter **https://make.powerautomate.com** and press **Enter**. If you already have Power Automate open, select its tab.
2. If prompted, sign in with your training work or school account and complete the sign-in verification. Your organization's sign-in and verification screens are not illustrated here, because they differ between organizations and may show account details.

Steps 1 and 2 share one expected result, shown below.

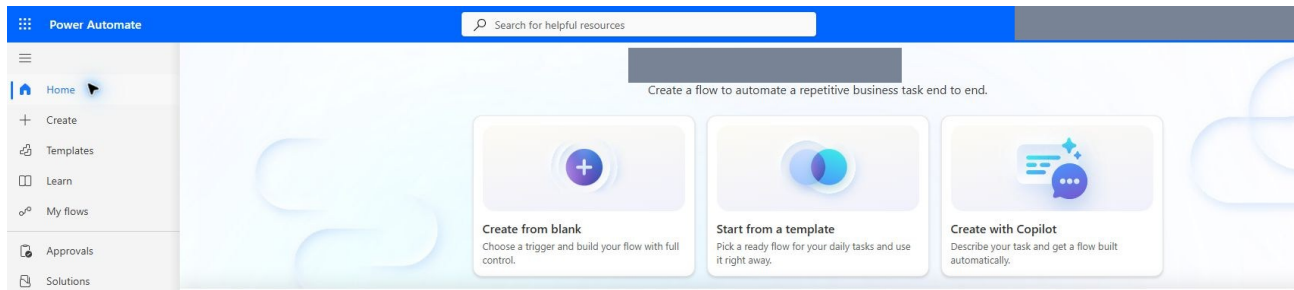


Figure 1.1. The signed-in Power Automate Home page. This is the expected result of steps 1 and 2.

3. Check that the page identifies the product as **Power Automate**.

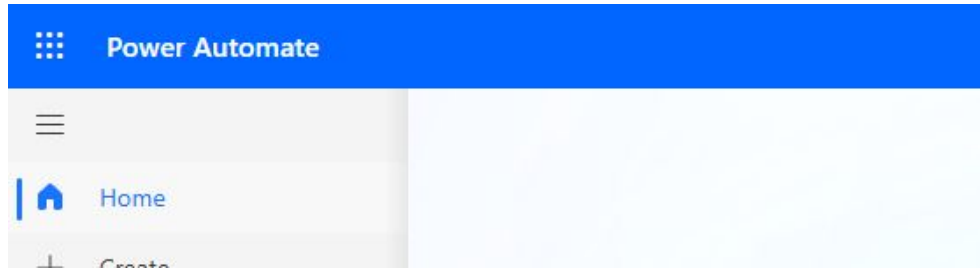


Figure 1.2. Look for Power Automate at the upper left of the page.

4. Locate the environment name near the top of the page, under **Environments**. An environment is a workspace that holds Power Platform resources, including flows.
5. Compare the displayed name with the environment supplied by your trainer. If they differ, ask your trainer which environment to select before building anything.

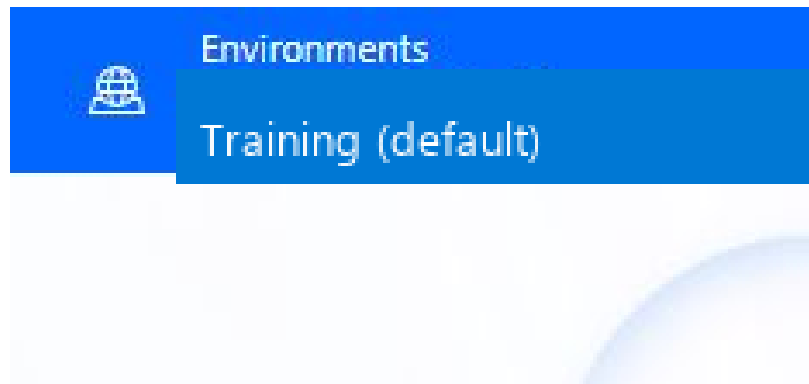


Figure 1.3. The Environments selector shows the active workspace. Steps 4 and 5 both read this one control. The name in your class will differ from this example.

Record your environment: _____

Checkpoint: You can identify Power Automate and the intended training environment. Use the environment given to your class rather than automatically copying the example.

Find the three cloud-flow choices

1. Locate the navigation on the left. If it is collapsed, use the navigation button to expand it.

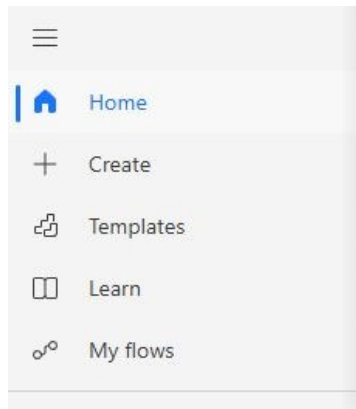


Figure 1.4. The expanded navigation shows Home, Create, and My flows; the menu button is above Home.

2. Select **Create**.

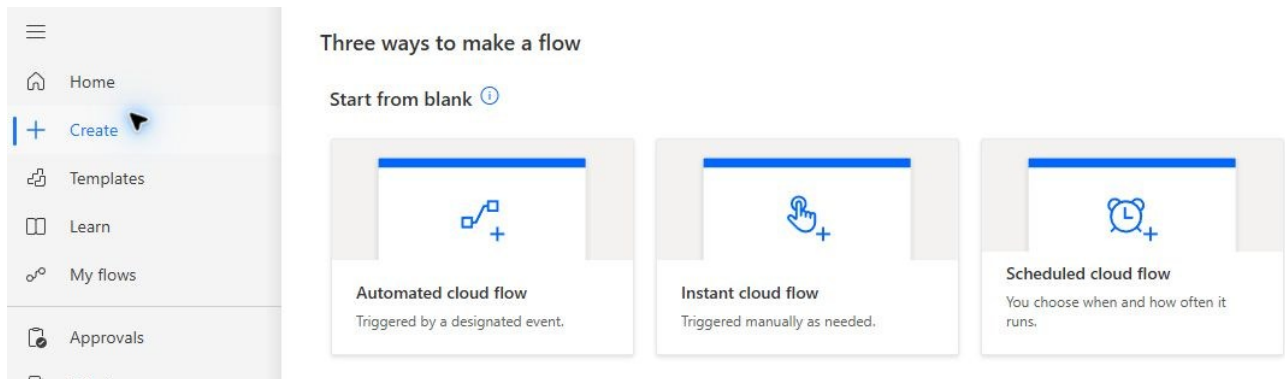


Figure 1.5. Create is selected on the left, and the page displays ways to start a flow.

3. Find the **Start from blank** area. Five tiles appear here. Three of them create cloud flows, which is what this course builds:

Tile	Description shown on the tile	Used in this course
Automated cloud flow	Triggered by a designated event.	Yes
Instant cloud flow	Triggered manually as needed.	Yes
Scheduled cloud flow	You choose when and how often it runs.	Yes
Desktop flow	Automates processes on your desktop environment.	No
Process mining	Evaluate and optimize your existing processes and tasks.	No

Start from blank ⓘ

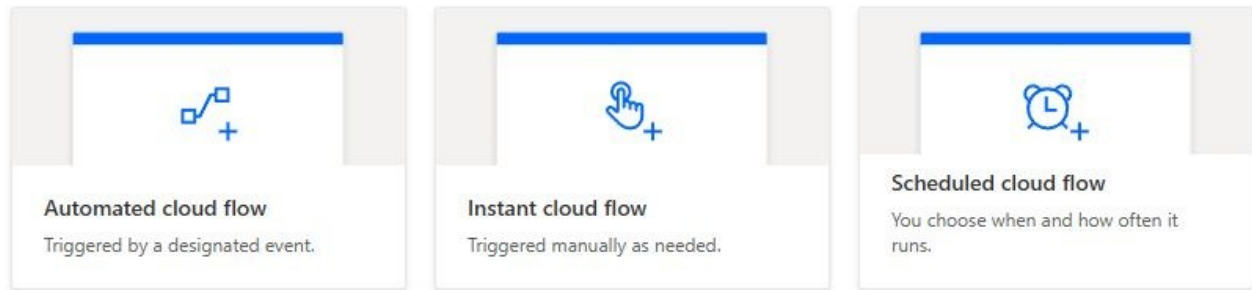


Figure 1.6. Five tiles appear under Start from blank. The first three create cloud flows; Desktop flow and Process mining are outside the scope of this course.

Do not confuse two different sets of three. The page heading reads **Three ways to make a flow**. Those three ways are **Start from blank**, **Start from a template**, and **Start from a connector** - the three sections of the page. They are not the three cloud-flow types. The cloud-flow types are three of the five tiles inside the first section.

4. Locate **Automated cloud flow**. Read its description and connect it to the incoming-email example.

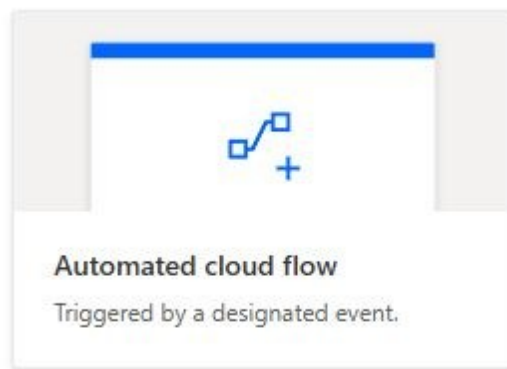


Figure 1.7. An automated cloud flow starts when a designated event occurs.

5. Locate **Instant cloud flow**. Connect it to generating a summary when you choose.

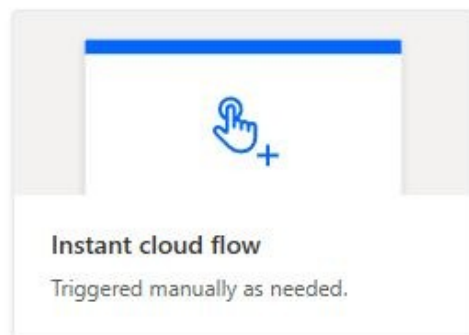


Figure 1.8. An instant cloud flow starts when you manually request it.

6. Locate **Scheduled cloud flow**. Connect it to the weekly reminder example.

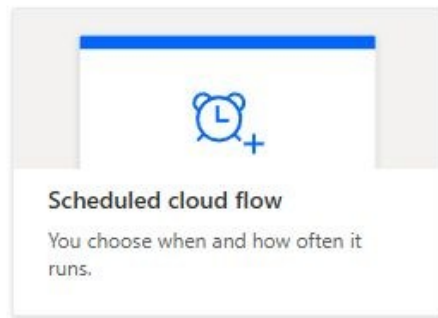


Figure 1.9. A scheduled cloud flow runs at the time and frequency you configure.

7. Look at the three cloud-flow tiles together. Decide which one you would choose for a flow that begins when someone submits a Form, and say why before reading the checkpoint. You do not need to click anything yet.

Start from blank ⓘ

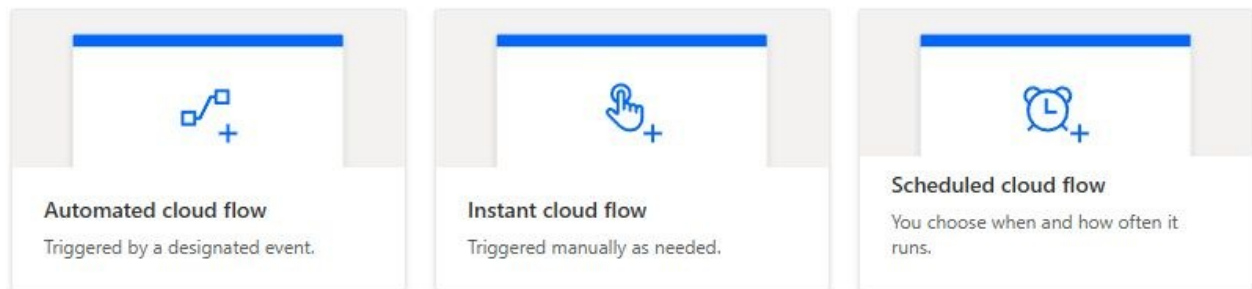


Figure 1.10. The three cloud-flow tiles side by side. Decide which one fits a Form submission before reading the checkpoint.

Checkpoint: A Form submission is an event that happens without you watching for it, so **Automated cloud flow** is the appropriate choice. If you chose Instant, re-read what "triggered manually as needed" means: that describes you starting the flow, not someone else submitting a form. No flow needs to be created in this chapter.

Locate your flows

1. Find **My flows** in the left navigation. This is where you will return to locate flows you create during the course.

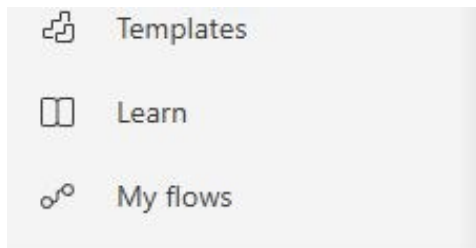


Figure 1.11. Use My flows to find your existing flows; this activity only locates the command.

2. Select **Home** to return to the starting page.

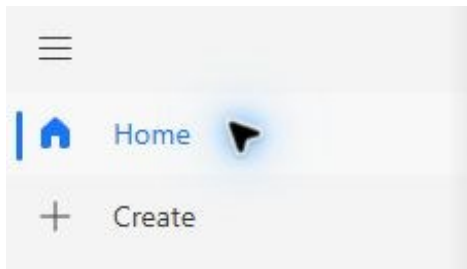


Figure 1.12. The Home command in the left navigation, shown selected. This figure shows the command you select, not the page it opens; compare the page you land on with Figure 1.1.

Discuss: Tell a partner where you would go to create a flow and where you would look for one you already created.

Expected result: You finish on Home and can identify Create and My flows without relying on their position in a screenshot.

1.6 Plan before you build

A short plan makes the designer easier to use. Complete these statements for the attachment scenario.

Planning question	Example answer
What task repeats?	Saving PDF training attachments from email.
What starts it?	An email with the training subject marker arrives.
What information is needed?	Attachment name and contents.
What rule applies?	Save PDF attachments; skip other file types.
Where should the output go?	The training attachments folder in OneDrive for Business.
How will I test it?	Use an email with a PDF and an image, then check the folder.

In Chapter 2, you will prepare the resources this plan refers to. In Chapter 3, you will turn the plan into a working flow.

Independent practice: Plan a weekly reminder

Allow five minutes. Work without copying the attachment example word for word.

Scenario: Every Monday morning, a training coordinator wants to remind people about upcoming sessions. Session details are stored in Excel. People whose training is marked Completed should not receive a reminder.

1. Choose a cloud-flow type.
2. State the starting time and time zone. Use a proposed time for this planning exercise.
3. List three pieces of information the flow needs from Excel.
4. Write one rule that decides whether a reminder should be sent.
5. Describe a test with one person who should receive a reminder and one who should not.

Self-check: A suitable plan uses a scheduled flow, identifies a local time zone, and needs information such as email address, session date, course title, and completion status. Its rule checks that training is upcoming and incomplete. The test checks both an included and an excluded record.

Your wording can differ. The plan should be clear enough for another person to explain who receives a reminder and why.

Tips and troubleshooting

What you notice	What to do
A different Microsoft account opens automatically	Check the account shown in the page. Use the Edge profile intended for training, or follow your trainer's sign-in instructions.
You cannot open Power Automate	Check the address and internet connection, then ask the trainer to verify account access. Record the error message.
The environment name is unfamiliar	Compare it with the class instructions before creating resources.
Create is not visible	Expand the left navigation. If it remains unavailable, ask the trainer to check access.
The page looks different from the screenshot	Look for the named command. Home content and layouts may vary.
A trial or upgrade prompt appears	Check which feature you selected. The core lessons are designed around the Free baseline; ask the trainer to identify the appropriate path.
You are unsure which flow type to choose	Finish the sentence: "This process should begin when..." Then match that starting point to the three types.

Lesson summary

A flow carries out a defined process. Its trigger starts the work, its actions perform steps, and data moves between those steps. Conditions choose a path, and loops repeat work for multiple items.

You can now distinguish automated, instant, and scheduled cloud flows, find the main Power Automate pages, and describe an automation before building it. Next, you will prepare the folders, table, messages, and other resources your flows will use.

Chapter 2: Prepare your Microsoft 365 training workspace

Estimated time: 90 minutes

Your result: A training folder, an Excel workbook with a table, and safe test data for the first two build exercises.

Note: Microsoft Forms and Teams preparation are trainer-led readiness checks, because those destinations depend on your organization's policy and approved training spaces.

What you will learn

By the end of this chapter, you will be able to:

- Create a dedicated training folder structure in OneDrive for Business.
- Build an Excel workbook and format its data as a named table.
- Prepare a safe test email with sample attachments.
- Confirm that Forms, Teams, and your Power Automate environment are ready.

Before you begin

Use your own training account and the environment named by your trainer. Do not place real participant information in the practice resources. Use fictional names and test addresses.

Power Automate Free provides the cloud-flow capability used in this course with standard connectors. It does not provide flow sharing, so each participant creates and owns their own practice flows.

Scenario

You coordinate weekly training. Files arrive by email, session details are tracked in Excel, requests are collected through Forms, and updates are posted to Teams. Preparing these resources first makes the later flow steps predictable and easier to troubleshoot.

2.1 Create the training folders in OneDrive for Business

1. Open OneDrive for Business and select **My files**.

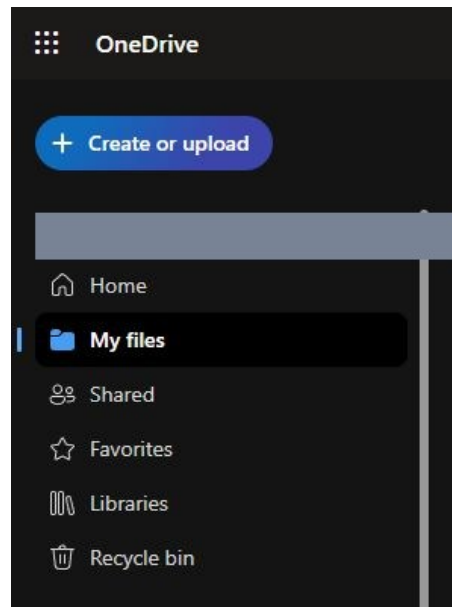


Figure 2.1. The OneDrive navigation with **My files** selected, and the **Create or upload** button above it. This view is cropped to the navigation; your own file list appears to the right of it.

2. Select **Create or upload**, then choose **Folder**.

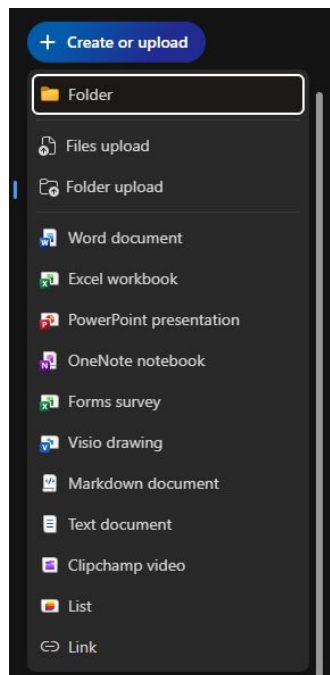


Figure 2.2. The open **Create or upload** menu. **Folder** is the first entry, above the two upload entries and the new-document list. You will return to this same menu in section 2.2 to create the workbook.

3. Enter **Power Automate Training** as the folder name. Leave the default folder color unless your trainer gives a different instruction, then select **Create**.

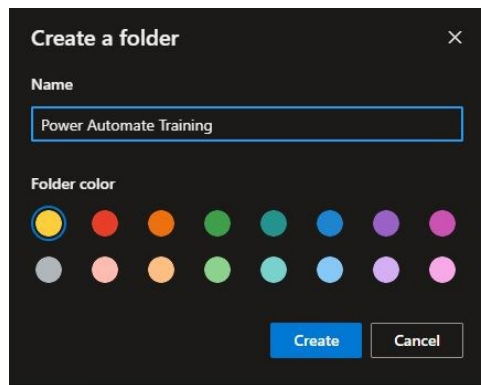


Figure 2.3. Enter the folder name before selecting Create.

4. Open **Power Automate Training**.

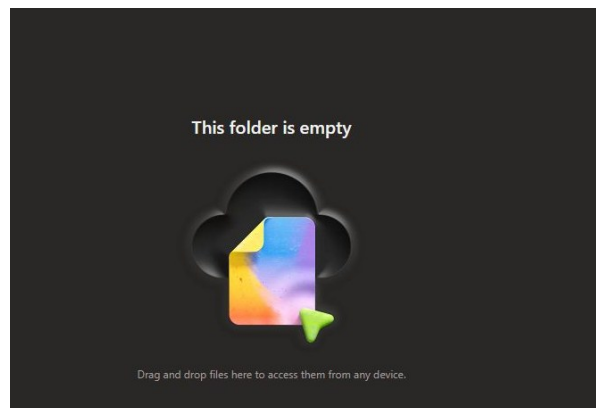


Figure 2.4. The empty-folder message confirms that **Power Automate Training** is ready for the course resources.

5. Inside the folder, create two subfolders named **Attachments** and **Reports**.

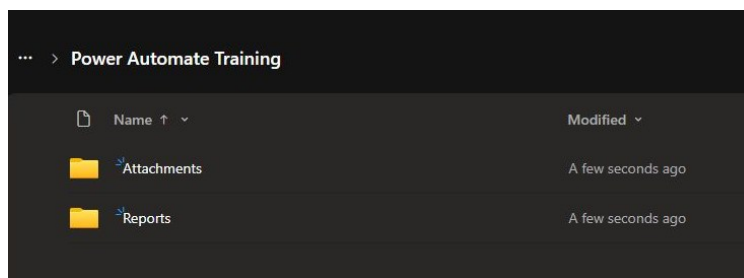


Figure 2.5. Both subfolders inside **Power Automate Training**. **Attachments** receives files saved by the Chapter 3 flow; **Reports** holds summary outputs.

Checkpoint: You can see **Attachments** and **Reports** inside **Power Automate Training**. The later attachment exercise writes to **Attachments**; the summary exercise can use **Reports**.

2.2 Create the Excel workbook

1. In Power Automate Training, select **Create or upload**, then choose **Excel workbook**.

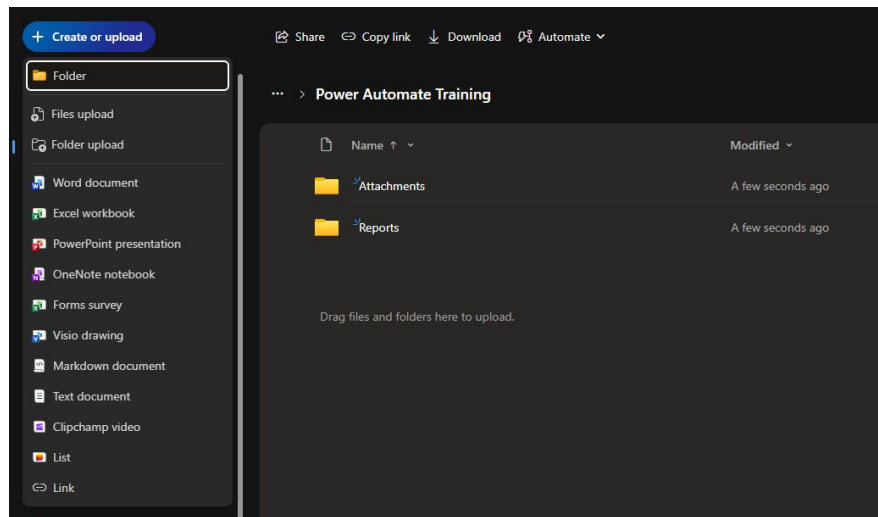


Figure 2.6. The same menu, opened inside **Power Automate Training**. Select **Excel workbook** from the new-document list. Check the breadcrumb first: the new workbook is created in whichever folder you are currently viewing, so **Attachments** and **Reports** should be visible beside the menu.

2. When the workbook opens in Excel for the web, select the file name in the title bar and rename it to **TrainingRegister**. Excel adds the **.xlsx** extension for you.

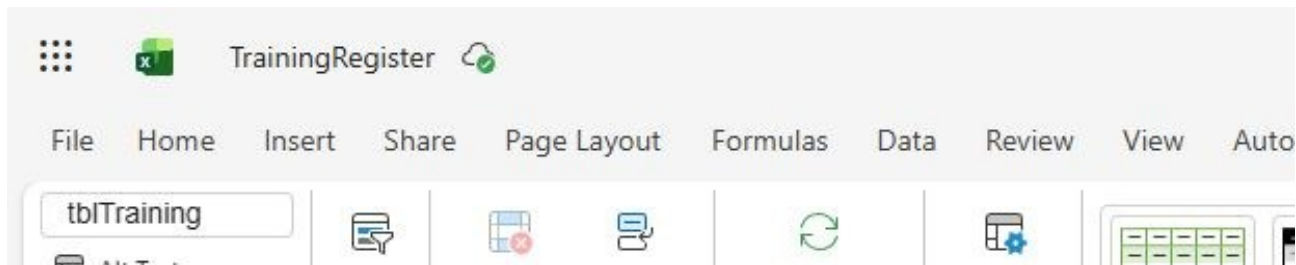


Figure 2.7. The title bar shows the saved name **TrainingRegister**; Excel adds the **.xlsx** extension in OneDrive.

3. In Sheet1, enter these headers in row 1:

TrainingID | ParticipantName | Email | CourseTitle | SessionDate | Status

4. Enter six fictional rows. Use dates that include at least two upcoming sessions, one completed session, and one session outside the training week. Use the trainer's approved test addresses in the Email column.

TrainingID	ParticipantName	Email	CourseTitle	SessionDate	Status
TR-001	Aisha Rahman	learner1@example.com	Power Automate Basics	21-Sep-26	Registered
TR-002	Daniel Lee	learner2@example.com	Excel Reporting	22-Sep-26	Registered
TR-003	Mei Chen	learner3@example.com	Microsoft Forms	23-Sep-26	Completed
TR-004	Ravi Kumar	learner4@example.com	Teams Collaboration	24-Sep-26	Registered
TR-005	Sofia Martinez	learner5@example.com	Approval Workflows	28-Sep-26	Registered
TR-006	Noah Williams	learner6@example.com	Cloud Flow Essentials	5-Oct-26	Completed

Figure 2.8. Six fictional rows provide predictable data for the later flow. The **example.com** addresses are reserved documentation values and cannot receive mail.

5. Select any cell in the range, choose **Insert > Table**, confirm that **My table has headers** is selected, and select **OK**. Then, on the **Table Design** tab, replace the default table name with **tblTraining** and press Enter.

TrainingID	ParticipantName	Email	CourseTitle	SessionDate	Status
TR-001	Aisha Rahman	learner1@example.com	Power Automate Basics	21-Sep-26	Registered
TR-002	Daniel Lee	learner2@example.com	Excel Reporting	22-Sep-26	Registered
TR-003	Mei Chen	learner3@example.com	Microsoft Forms	23-Sep-26	Completed
TR-004	Ravi Kumar	learner4@example.com	Teams Collaboration	24-Sep-26	Registered
TR-005	Sofia Martinez	learner5@example.com	Approval Workflows	28-Sep-26	Registered
TR-006	Noah Williams	learner6@example.com	Cloud Flow Essentials	5-Oct-26	Completed

Figure 2.9. Select a table cell, open **Table Design**, and set the table name to **tblTraining**.

Checkpoint: The workbook is named **TrainingRegister.xlsx**, the data is in a formatted table named **tblTraining**, and the table has six fictional records.

Troubleshooting: If Power Automate cannot find the table later, confirm that the workbook is in OneDrive for Business, that the range was formatted as a table, and that the table name is exactly **tblTraining**.

2.3 Prepare a test email

1. Open Outlook on the web using the training account.
2. Create a message addressed to your own training mailbox, the one you will connect to Power Automate in Chapter 3. Use the subject [PA TRAINING] Attachment test.
3. Attach CourseOutline.pdf and TrainerPhoto.jpg from the supplied test-files folder. Do not send it yet. Close the message so that it is saved in **Drafts**.

Do not attach real course materials or any file whose name contains a person's name: the filename becomes visible in Chapter 3's run history and in your OneDrive folder.

Expected result: The message is saved in Drafts. The Chapter 3 flow reacts only to new messages, so you will send this draft during the test in section 3.6.

2.4 Confirm Forms readiness

Open Microsoft Forms and confirm that the training account can create a form. The Training Request form is built in Chapter 6, immediately before its response-submitted flow, so participants do not create an unused resource on Day 1. The later form will contain Requester name, Course, Preferred date, and Business reason, using fictional responses only.

2.5 Confirm the Teams training destination

Ask the trainer for the designated team and channel or supported training chat, then record its name: ----- . The trainer confirms posting access before the Teams exercises: the optional extension in Chapter 4 and the approval messages in Chapter 6.

The Teams connector is a standard connector, but your organization may restrict posting or require the Workflows app to be enabled. If you cannot post a test message, tell the trainer before building the summary flow.

2.6 Confirm the Power Automate environment

1. Return to Power Automate.
2. Check the environment selector near the top of the page.
3. Confirm that it matches the environment provided by your trainer.

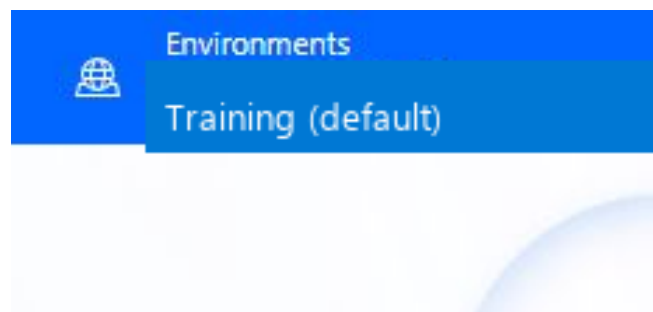


Figure 2.10. Confirm the active environment before creating flows. The account-specific environment name is replaced with a neutral training label.

Important: Resources such as OneDrive files, Excel workbooks, Forms, and Teams destinations are not packaged automatically when a solution is moved. Keep a note of where each resource lives.

Independent practice

Find the row in `tblTraining` whose Status is Completed. Explain which later reminder test should exclude it and why. Then write a second test email without the `[PA TRAINING]` marker, save it as a draft, and predict whether the attachment flow should process it when it is sent.

Lesson summary

You created a dedicated OneDrive training area, prepared the folder structure and Excel table, and assembled safe inputs for the attachment and summary exercises. You also confirmed the environment and recorded where later Forms and Teams work will take place. Next, you will build the automated flow that saves qualifying email attachments.

Chapter 3: Save email attachments automatically

Estimated time: 120 minutes

Your result: An automated cloud flow that examines new training emails and saves only PDF attachments to OneDrive for Business.

Connector baseline: Office 365 Outlook and OneDrive for Business, both standard connectors.

What you will learn

By the end of this chapter, you will be able to:

- Create an automated cloud flow from a business requirement.
- Configure an Outlook trigger with a subject filter.
- Process several attachments with **Apply to each**.
- Use a condition to select PDF files.
- Create a file in a OneDrive folder.
- Test both a matching and a nonmatching case and inspect run history.

Before you begin

Complete Chapter 2. Confirm that the `Attachments` folder exists and that the test email from section 2.3 is saved in Drafts.

The draft uses the subject `[PA TRAINING] Attachment test` and contains one PDF and one JPG from the supplied fictional files in the `test-files` folder:

File	Purpose
<code>CourseOutline.pdf</code>	The attachment that should be saved
<code>TrainerPhoto.jpg</code>	The attachment that should be skipped
<code>SessionNotes.docx</code>	Used only in the independent practice at the end of this chapter

Do not use real participant documents, real course materials, or any file containing a person's name. The flow writes whatever it receives into your OneDrive folder, and the filename becomes visible in run history.

About running the test twice: OneDrive will not silently overwrite a file. If you run the test again with the same attachment, `Create file` either fails with a duplicate-name error or OneDrive appends a counter, producing `CourseOutline (1).pdf`. Both outcomes are correct behavior, not a broken flow. For repeat tests, either delete the file from `Attachments` first, or rename the attachment before sending. Section 3.8 shows how to add a timestamp if you want unique names automatically.

Business scenario

Training coordinators receive course documents by email. They want qualifying PDF attachments saved automatically in OneDrive, while images and unrelated messages are ignored.

The rule is deliberately small:

When a new email arrives with `[PA TRAINING]` in the subject, examine each attachment. If its name ends in `.pdf`, create a file in the `Attachments` folder.

3.1 Create the automated cloud flow

1. Open Power Automate and confirm the correct environment.

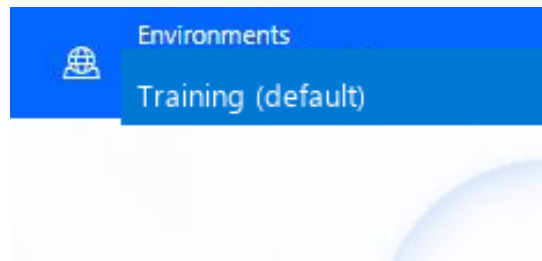


Figure 3.1. Confirm the active environment. The account-specific name is replaced with a neutral training label.

2. Select **Create**, then **Automated cloud flow**.

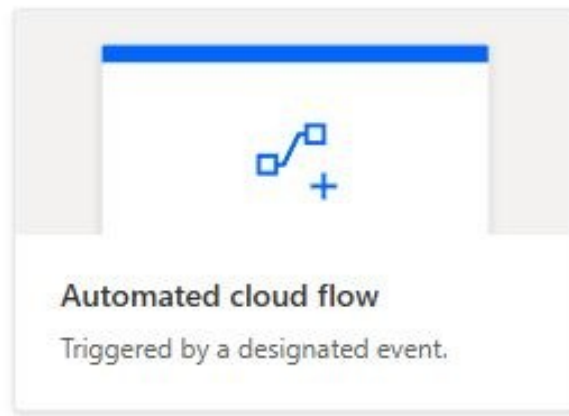


Figure 3.2. Automated cloud flow choice.

3. Enter the flow name PA - Save training PDF attachments.
4. Search for and select the trigger **When a new email arrives (V3)** from **Office 365 Outlook**.

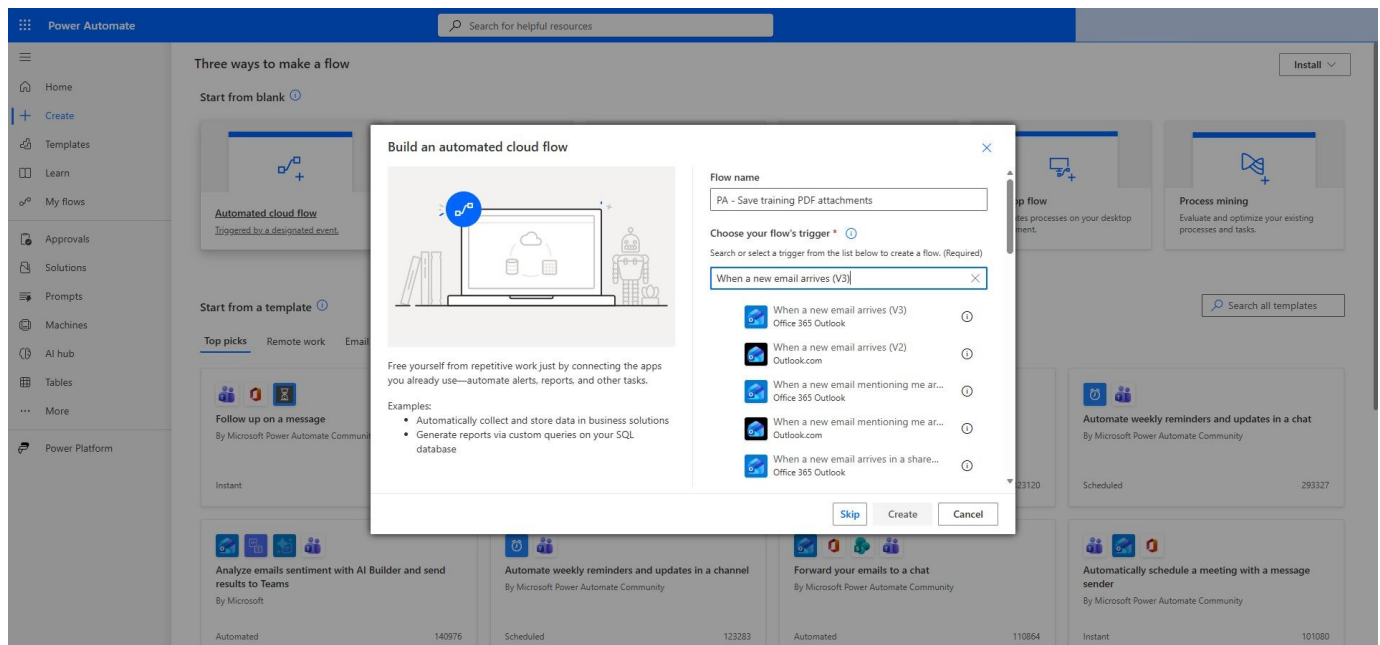


Figure 3.3. Flow name and Outlook trigger.

5. Select **Create**.

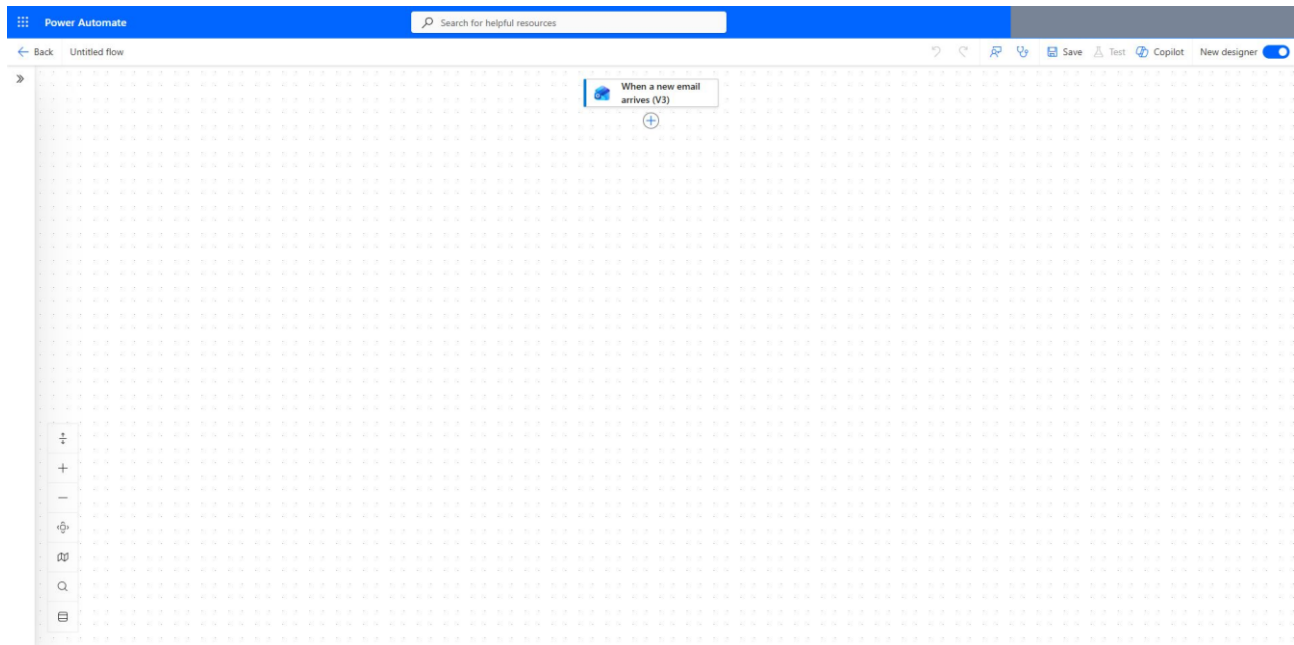


Figure 3.4. The designer opens with only the trigger on the canvas. This capture was taken from a flow created without a name, so its title bar reads "Untitled flow"; yours will show the name you entered in step 3.

3.2 Configure the trigger

1. In the trigger, set **Include Attachments** to **Yes**.
2. Set **Folder** to **Inbox**. This lesson watches the Inbox of the connected mailbox.
3. In **Subject Filter**, enter [PA TRAINING].
4. Set **Only with Attachments** to **Yes**. Use **Show all** under Advanced parameters if the field is hidden.

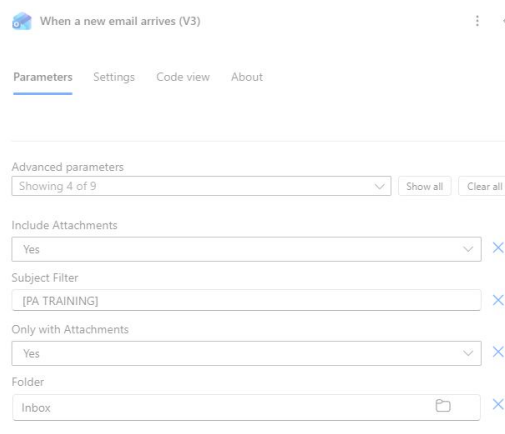


Figure 3.5. The trigger's Parameters tab with all four fields set. Advanced parameters reads "Showing 4 of 9", so use Show all if a field you need is hidden.

5. Select **Save**.

Checkpoint: The trigger is saved without a red validation message. The flow will start only for new matching messages; it will not process old messages simply because they are already in the inbox.

3.3 Add Apply to each

1. Select the + insertion button.
2. Search for **Apply to each** and select the **Control** action.
3. In **Select an output from previous steps ***, choose the trigger's **Attachments** dynamic content.

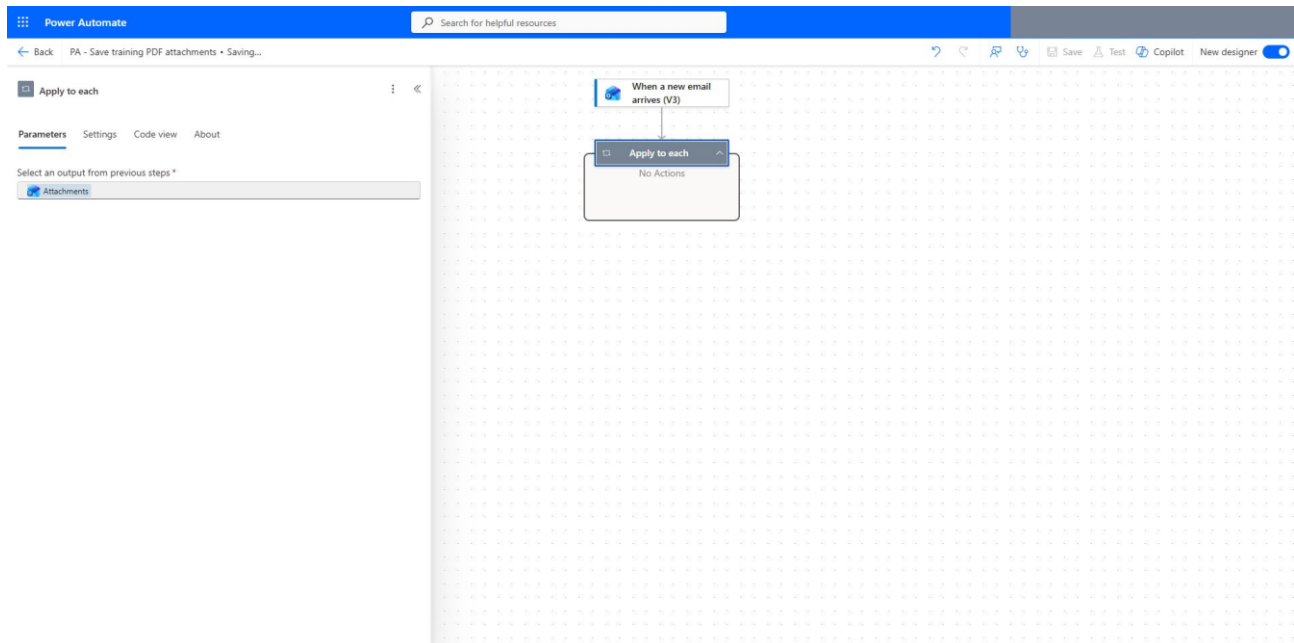


Figure 3.6. Apply to each attachments input.

Checkpoint: The loop contains an empty action area. If the Attachments token does not appear, return to the trigger and confirm that Include Attachments is enabled.

3.4 Add the PDF condition

1. Inside the loop, select **Add an action**.
2. Search for **Condition** and choose the **Control** condition.
3. In the left value field, select **Attachments Name**. If the token picker is difficult to use, type `/`, choose **Insert expression**, and enter `item()?['name']`. Select **Add** to insert it. This reads the current attachment in the loop.
4. Set the comparison to **ends with**.
5. Enter `.pdf` in the right value field.

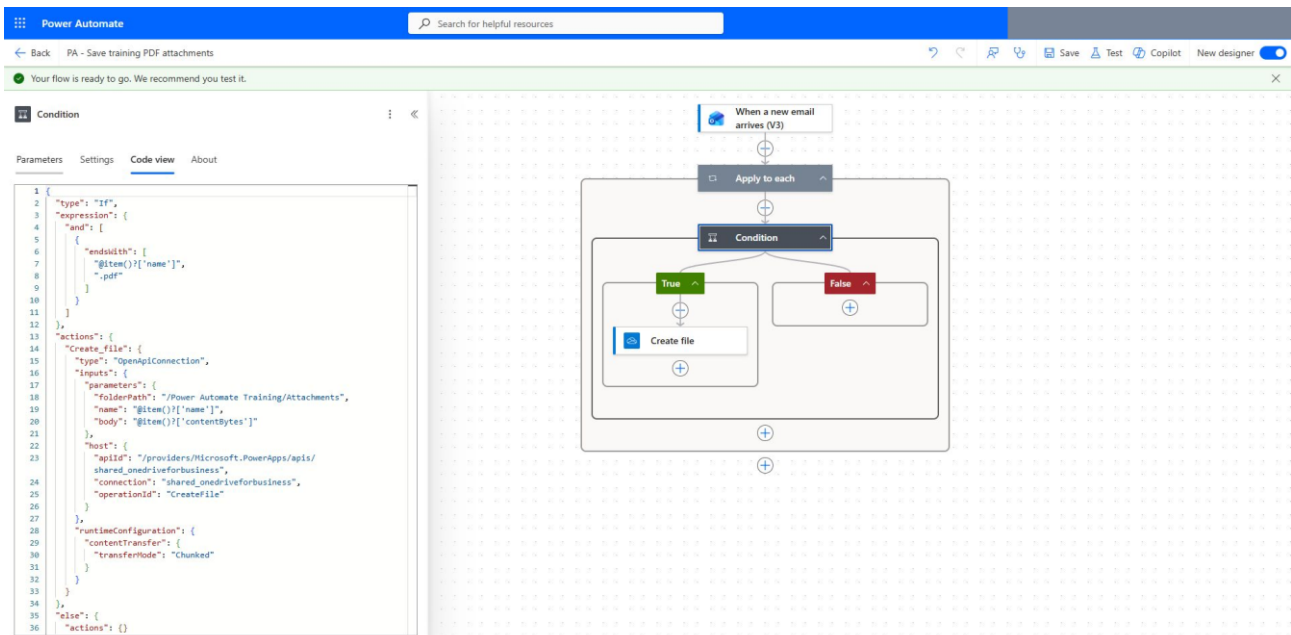


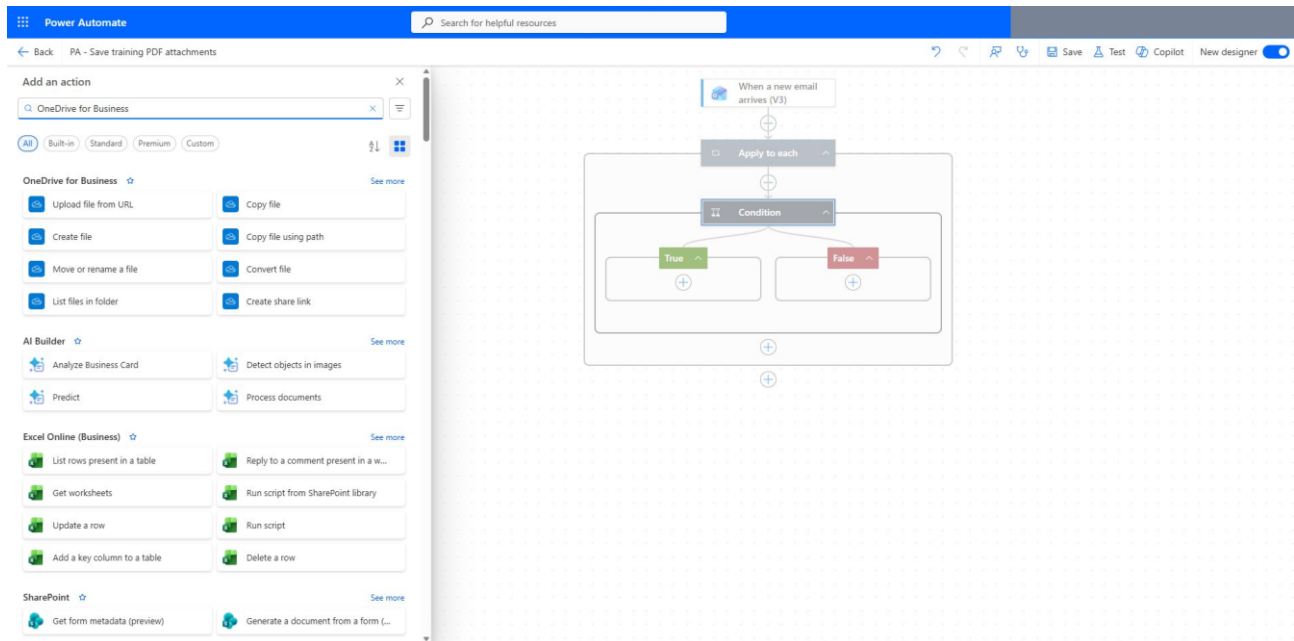
Figure 3.7. The finished condition, shown in Code view rather than the Parameters tab you configure it in. Code view is included here because it states the rule unambiguously: `endsWith` compares `@item()?['name']` with `.pdf`. Configure the condition on the Parameters tab using the three fields in steps 3 to 5, then switch to Code view if you want to confirm your work matches this.

6. In the condition's **True** branch, select **Add an action**.

3.5 Create the OneDrive file

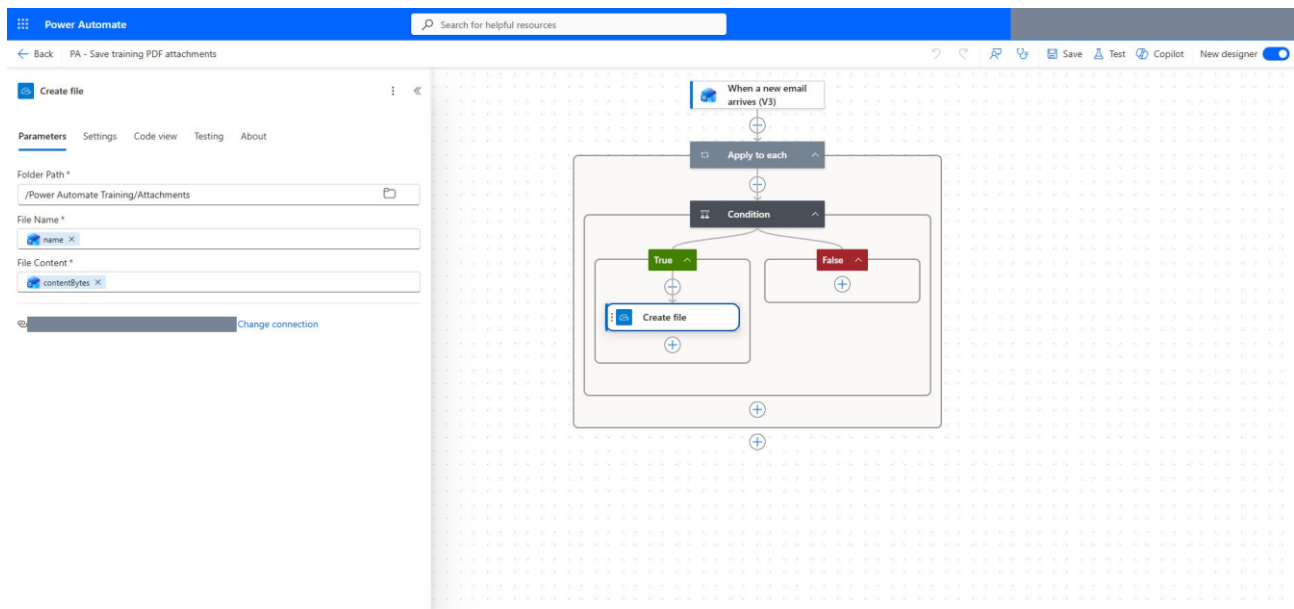
1. In the **Add an action** panel, search for **OneDrive for Business**, then select **Create file** from that connector's group.

Search by connector name rather than by action name. More than one connector offers an action called Create file, and searching the action name alone makes it easy to pick the wrong connector's version.



*Figure 3.8. The Add an action panel searched for **OneDrive for Business**. Results are grouped by connector, and **Create file** appears in the **OneDrive for Business** group. Other connectors are listed below it, which is why searching the connector name is more reliable than searching the action name.*

2. In **Folder Path**, select the Attachments folder under Power Automate Training.
3. In **File name**, select the current attachment's **Name** dynamic content.
4. In **File content**, select the current attachment's **Content** dynamic content. You can also use the expression `item()?['contentBytes']`. After reopening the designer, the tokens may display as **name** and **contentBytes**; these refer to the same attachment properties.



*Figure 3.9. The **Create file** action configured with **Folder Path**, **File Name** and **File Content**. The tokens display as **name** and **contentBytes**, which are the same attachment properties referred to in step 4.*

5. Leave the **False** branch empty. It represents the deliberate decision to skip non-PDF files.
6. Check that **Create file** is inside **True**, and that **False** contains no actions. Keep the default action names to match the screenshots.

7. Select **Save**, then open **Flow checker** and confirm it reports no errors and no warnings.

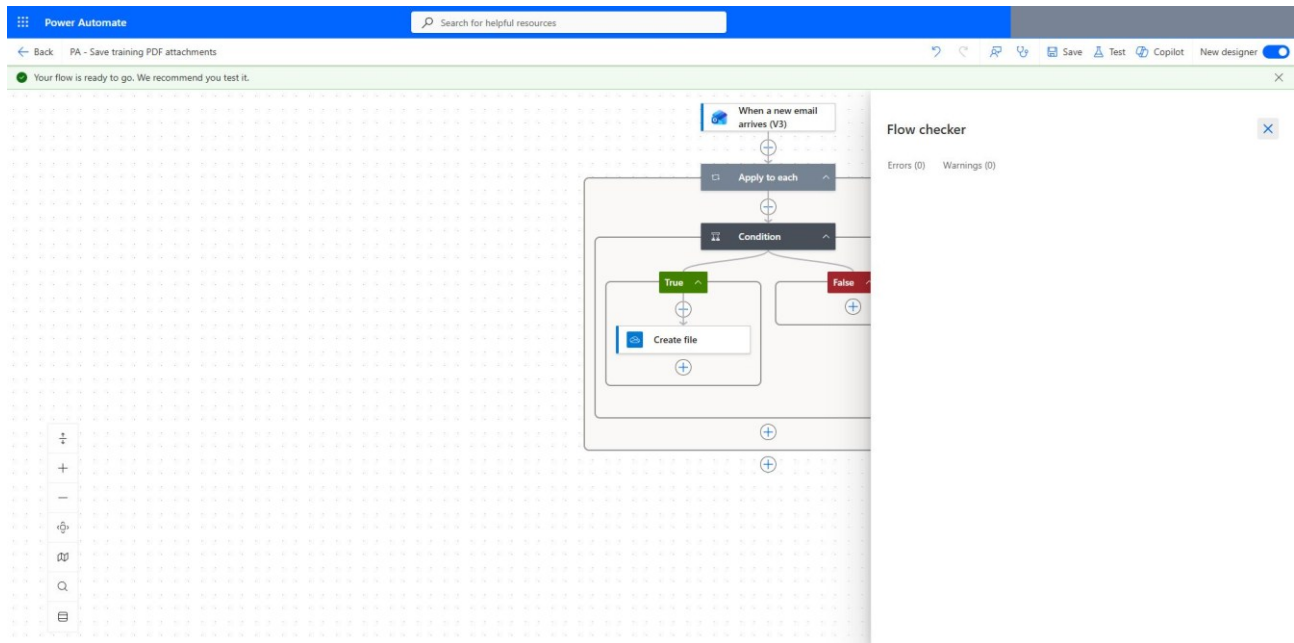


Figure 3.10. The completed flow with Create file inside the True branch, and Flow checker reporting 0 errors and 0 warnings. Note that the open Flow checker panel overlaps the right edge of the False branch in this capture; verify on your own screen that False contains no actions.

3.6 Test a positive case

1. Open the saved flow and select **Test**. Choose **Manually**, then select **Test**.

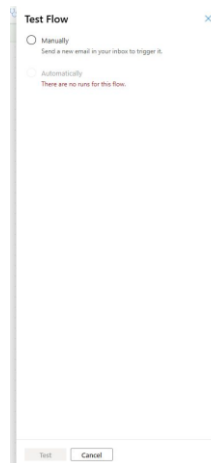


Figure 3.11. The Test Flow panel as it first opens. Neither option is selected yet, which is why Test is dimmed - choose Manually first and the button becomes available. Automatically is unavailable on a new flow, marked "There are no runs for this flow", because it replays a previous run.

2. Wait for the message asking you to send a new email. The flow is now listening and will start on the next matching message.

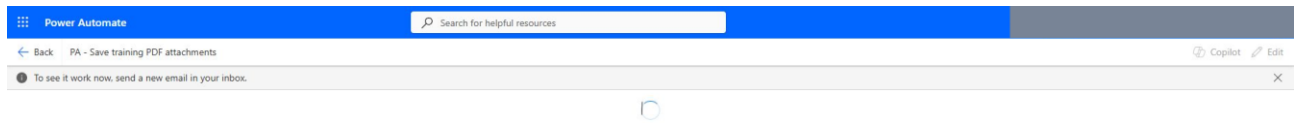


Figure 3.12. The flow is waiting. The banner reads "To see it work now, send a new email in your inbox." Nothing happens until a matching message arrives.

3. Open the draft from section 2.3 and send it. It is addressed to the connected mailbox, uses the subject [PA TRAINING] Attachment test, and contains CourseOutline.pdf and TrainerPhoto.jpg.
4. Wait for the run to finish. Select the completed run or **28-day run history**.

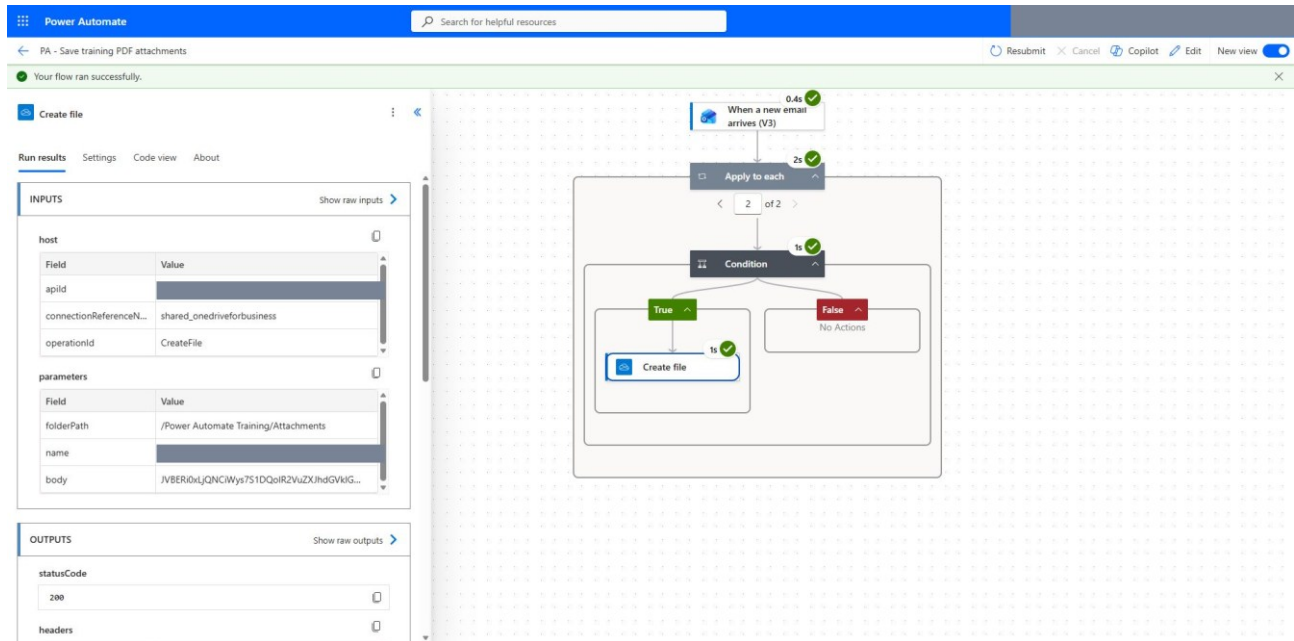


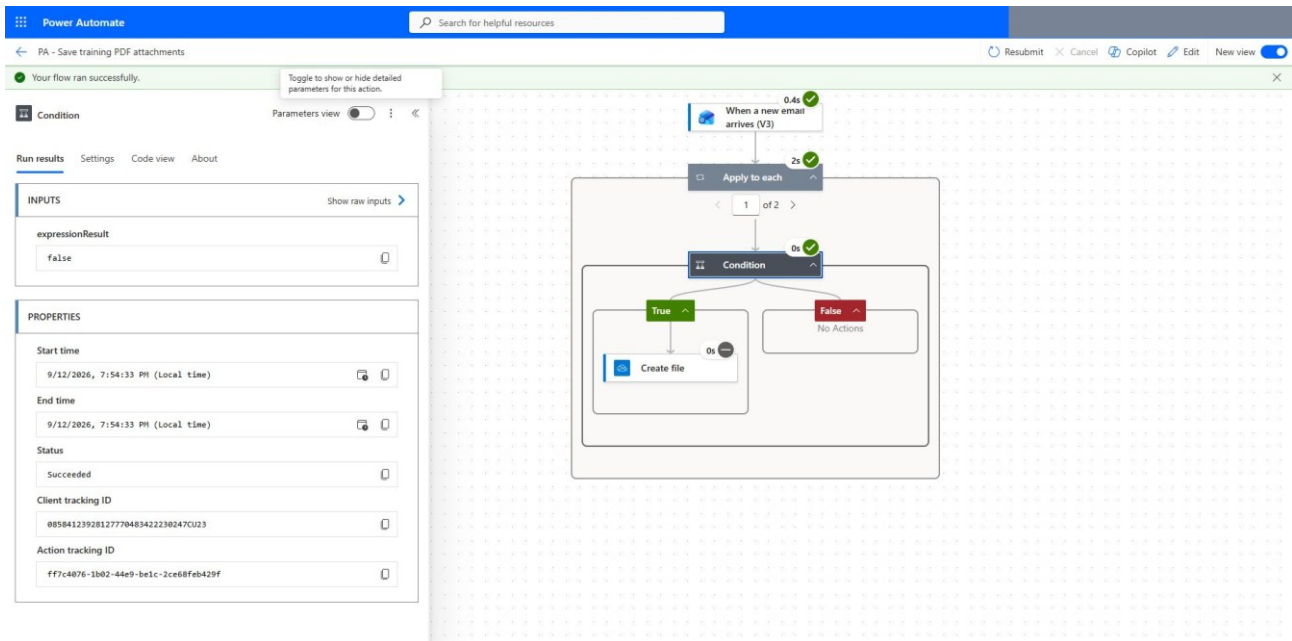
Figure 3.13. The completed run. The loop counter reads 2 of 2, the condition returned true on this iteration, and Create file succeeded with **statusCode 200**. The saved file name and the subscription identifier are masked in this capture.

5. Open the Attachments folder in OneDrive and confirm that the PDF exists and that no image file was created.
- Expected result:** The PDF is created once. The JPG follows the False path and is not created.

3.7 Test a negative case

You do not need to send another email for this. The mixed-attachment test in section 3.6 already produced both outcomes: the run processed two attachments, and one of them was skipped.

1. Open the same run you inspected in section 3.6.
2. In **Apply to each**, change the iteration counter from **2** to **1** to view the other attachment.
3. Select **Condition** and read **expressionResult**.



*Figure 3.14. Iteration 1 of 2 of the same run. **expressionResult** is **false**, **Create file** carries a skip marker rather than a green check, and the **False** branch reports "No Actions". The action status is **Succeeded**: the step completed correctly by doing nothing.*

4. Check the Attachments folder. Confirm that only the PDF is present and that no image file was created.

Optional additional test: if you want a run with no output at all, send a second email with subject [PA TRAINING] Image-only test attaching only `TrainerPhoto.jpg`. The run should succeed and create nothing.

Checkpoint: The flow can run successfully while deliberately producing no file. A successful run means the process completed; it does not mean every branch created an output.

3.8 Decide what happens to duplicate names

Your flow uses the attachment's own name as the file name, so sending the same attachment twice targets the same path. OneDrive does not overwrite it silently. Depending on the action's settings you will see either a duplicate-name error or an automatically numbered file such as `CourseOutline (1).pdf`.

Choose a policy deliberately rather than discovering it mid-class:

Policy	How	When it suits
Keep one copy	Delete the file from Attachments before re-testing	Classroom testing, where you want a clean folder each time
Let OneDrive number them	Change nothing	Quick repeat tests where clutter does not matter
Make every name unique	Put an expression in File name	A real process where two senders may attach files with the same name

To make names unique, set **File name** to this expression instead of the plain token:

```
concat(formatDateTime(utcNow(), 'yyyyMMdd-HH:mm:ss'), '-', item()?['name'])
```

This produces names such as `20260913-094512-CourseOutline.pdf`. Note that `utcNow()` returns UTC, not your local time; Chapter 5 covers converting time zones for values that participants will read.

Checkpoint: You can state what your flow does when the same attachment arrives twice, and you chose that behavior rather than inheriting it.

Independent practice

Change the condition so that the flow saves `.docx` attachments instead of PDFs. Send one test email attaching both `SessionNotes.docx` and `CourseOutline.pdf`. Predict which file should appear before you test, run it, then restore the `.pdf` rule.

Tips and troubleshooting

Problem	Check
The flow does not run	The message must be new, the subject must contain [PA TRAINING], and the message must reach the mailbox used by the Outlook connection.
Attachments token is missing	Enable Include Attachments in the trigger, save the flow, and reopen the loop input.
The loop runs but creates nothing	Inspect the condition value. The ends with operator uses a case-insensitive comparison. Check that the left value is the attachment name and the right value is .pdf.
Create file fails	Confirm the OneDrive connection, folder path, file name, and file content token.
Duplicate file error	Use a unique test filename or add a timestamp to the file name as an extension exercise.
The JPG is saved	Check that Create file is inside True, not outside the condition.

Lesson summary

This flow uses an event trigger, a loop for multiple attachments, a condition for business logic, and a OneDrive action for the result. The tests in this chapter verify both a file-producing path and a successful no-output path. The next chapter will use an instant flow to read the Excel table and distribute a summary.

Chapter 4: Create and distribute an Excel summary

Estimated time: 120 minutes

Your result: An instant cloud flow that reads a training register, selects useful columns, and produces one summary email. A Teams update is an extension after the email is tested.

What you will learn

By the end of this chapter, you will be able to:

- Start a flow on demand with an instant trigger.
- Read rows from an Excel table with List rows present in a table.
- Choose report columns with the Select action.
- Turn an array into one HTML table and send it in a single email.
- Inspect run history to confirm what each action produced.

The business problem

Your coordinator needs a readable list of training sessions. Copying cells into an email every time creates extra work and makes it easy to send an old version. An instant flow lets the coordinator request a fresh summary when needed.

The flow follows this sequence:

Manually trigger a flow > List rows present in a table > Select > Create HTML table > Send an email (V2)

The Excel action returns an array of records. Select shapes each record into the columns readers need. Create HTML table turns the array into one table. The email action runs once, after the table is complete.

Before you begin

Complete Chapter 2, including section 2.2. You need Power Automate Training/TrainingRegister.xlsx in OneDrive for Business, containing a formatted Excel table named tblTraining with these six columns:

Column	Contents	Used in this chapter for
TrainingID	TR-001 to TR-006	The independent practice at the end
ParticipantName	Fictional names	A report column
Email	Placeholder addresses	Read but deliberately left out of the report
CourseTitle	Fictional course names	A report column
SessionDate	Real Excel dates, mixed	A report column, and date formatting in 4.8
Status	Registered or Completed	A report column

Use fictional records only. Placeholder addresses such as learner1@example.com are data in this chapter and are never sent to - example.com is reserved for documentation and cannot receive mail. Chapter 5 is the first exercise that actually emails the people in this column, so substitute approved training addresses before then, not now.

Where the report goes. For this exercise the summary goes to **your own** training mailbox. The flow reads the Email column but does not send one message per participant; producing a single message from many rows is the point of the chapter.

Before each run. Keep the workbook saved and close any active cell edit. Excel for the web holds an uncommitted edit locally, and the connector reads the saved file, so an open edit produces a stale report.

A prepared workbook is available. If your trainer supplies a ready-made `TrainingRegister.xlsx`, it already contains the table named `tblTraining` with six records covering upcoming, completed and out-of-period sessions. Upload it to Power Automate Training instead of building the table by hand.

4.1 Confirm the source table

1. Open `TrainingRegister.xlsx` from the training folder.

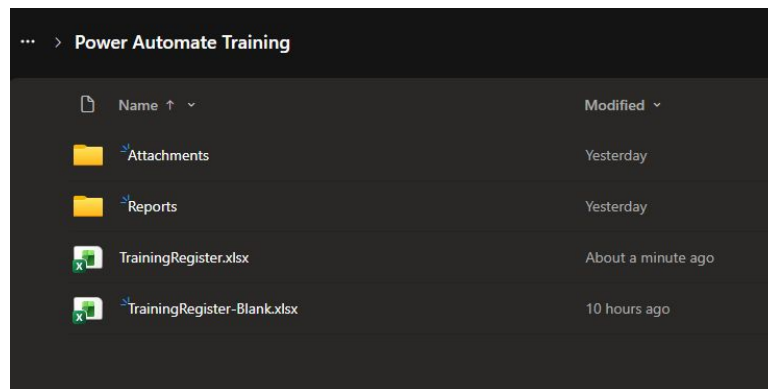


Figure 4.1. TrainingRegister.xlsx in the Power Automate Training folder, alongside Attachments and Reports. Confirm the source is in OneDrive for Business because the Excel connector reads the cloud file.

You built this workbook in Chapter 2, section 2.2. This section only confirms it is ready - you are not creating it again.

2. Check the header row. All six columns must be present and spelled as shown, because the expressions later in this chapter refer to them by name:

TrainingID | ParticipantName | Email | CourseTitle | SessionDate | Status

TrainingID	ParticipantName	Email	CourseTitle	SessionDate	Status
TR-001	Aisha Rahman	learner1@example.com	Power Automate Basics	21-Sep-26	Registered
TR-002	Daniel Lee	learner2@example.com	Excel Reporting	22-Sep-26	Registered
TR-003	Mei Chen	learner3@example.com	Microsoft Forms	23-Sep-26	Completed
TR-004	Ravi Kumar	learner4@example.com	Teams Collaboration	24-Sep-26	Registered
TR-005	Sofia Martinez	learner5@example.com	Approval Workflows	28-Sep-26	Registered
TR-006	Noah Williams	learner6@example.com	Cloud Flow Essentials	5-Oct-26	Completed

Figure 4.2. The prepared source contains six fictional records. The **example.com** addresses are safe documentation values and cannot receive mail.

3. Check that the range is a **formatted table** and that its name is exactly **tblTraining**. Select any cell inside the data, then look at the table name in the table controls.

TrainingID	ParticipantName	Email	CourseTitle	SessionDate	Status
TR-001	Aisha Rahman	learner1@example.com	Power Automate Basics	21-Sep-26	Registered
TR-002	Daniel Lee	learner2@example.com	Excel Reporting	22-Sep-26	Registered
TR-003	Mei Chen	learner3@example.com	Microsoft Forms	23-Sep-26	Completed
TR-004	Ravi Kumar	learner4@example.com	Teams Collaboration	24-Sep-26	Registered
TR-005	Sofia Martinez	learner5@example.com	Approval Workflows	28-Sep-26	Registered
TR-006	Noah Williams	learner6@example.com	Cloud Flow Essentials	5-Oct-26	Completed

Figure 4.3. The formatted Excel table is named **tblTraining**. The worksheet name alone is not sufficient for the connector.

4. Confirm that **SessionDate** holds real dates rather than text, and that **Status** uses only **Registered** or **Completed**.

If anything is missing, return to Chapter 2 section 2.2 and complete steps 3 to 5 before continuing. Do not build the flow against an incomplete table; the Excel action will either return nothing or fail to list the table at all.

Checkpoint: A worksheet name and a table name are different things. The Excel connector requires a formatted table, not merely cells with colored headings. If the later Table dropdown is empty, this is almost always the cause.

A note on dates. `SessionDate` is stored as an Excel date, which the connector may return as a serial number such as 46286 rather than a readable date. Section 4.3 step 5 sets the output format to avoid this. Inspect the actual run output before adding any date formatting - see section 4.8.

4.2 Create an instant cloud flow

1. In Power Automate, confirm the training environment and select **Create > Instant cloud flow**.

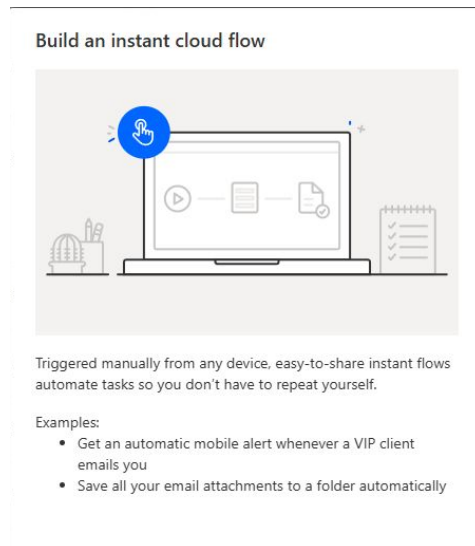


Figure 4.4. The resulting dialog confirms that you selected an instant cloud flow.

2. Name the flow PA – Training register summary and choose **Manually trigger a flow**.

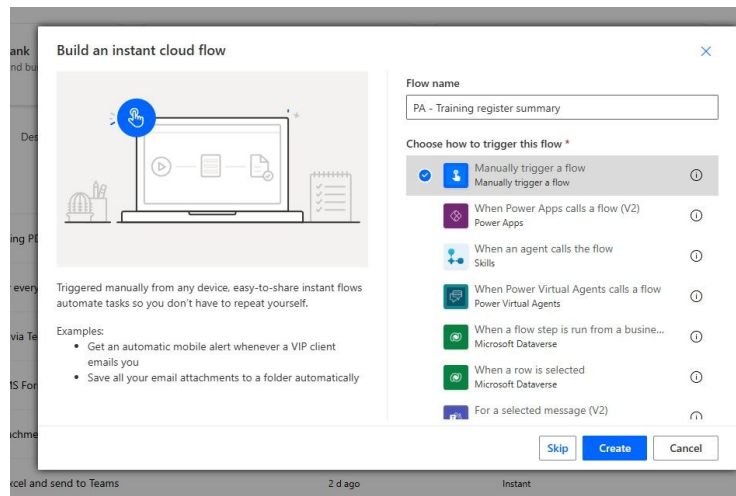


Figure 4.5. Name the flow and select the manual trigger.

3. Select **Create** and locate the plus button below the trigger.

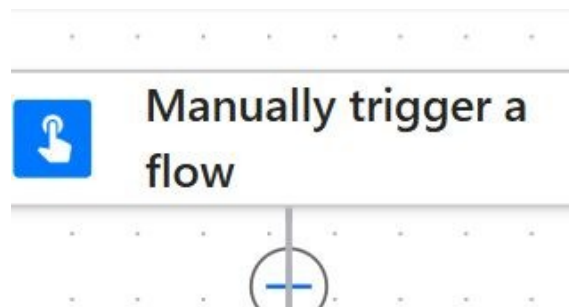


Figure 4.6. The initial canvas contains the manual trigger and an insertion control.

Checkpoint: This trigger starts when you run it. A new email or a changed spreadsheet does not start this flow.

4.3 Read the Excel records

1. Add Excel Online (Business) > List rows present in a table.

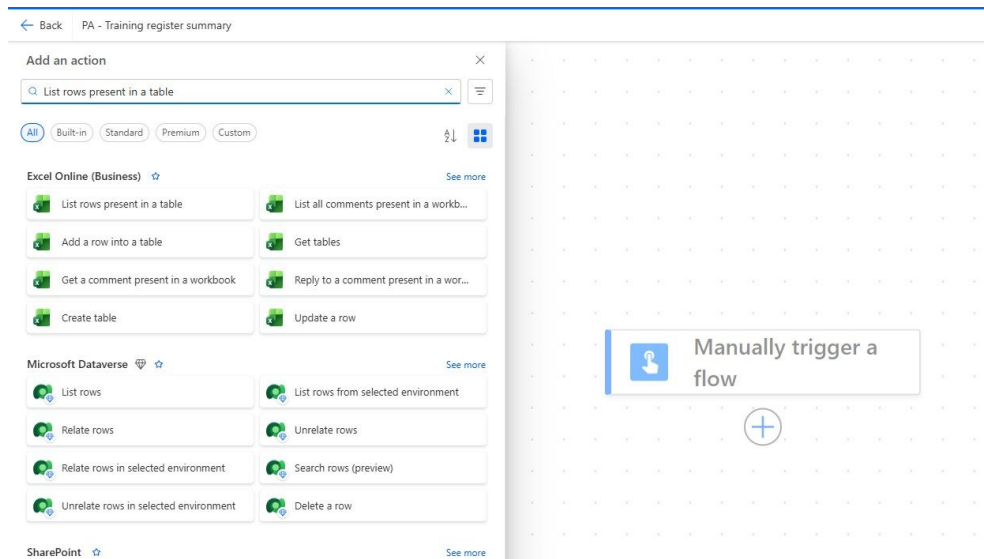


Figure 4.7. Add List rows present in a table from Excel Online (Business).

2. Set Location to OneDrive for Business.

The Location list also offers every Microsoft 365 group and SharePoint site you belong to. Pick OneDrive for Business for this exercise.

3. Set Document Library to OneDrive.

Three libraries are offered - KARuntime, OneDrive and PersonalCacheLibrary. Only **OneDrive** holds your files; the other two are internal libraries and will not contain the training workbook.

 List rows present in a table

<<

Parameters

Settings

Code view

Testing

About

Location *

OneDrive for Business

Document Library *

OneDrive

File *

Select an Excel file through File Browse.

Table *

Select a table from the drop-down.

Extract Sensitivity Label

A boolean whether to extract sensitivity label Id for associated artefact.

Sensitivity Label Metadata

A boolean whether to fetch sensitivity label Metadata for associated LabelId.

Advanced parameters

Showing 0 of 6

Show all

Clear all

Figure 4.8. The source location uses OneDrive for Business and the OneDrive document library.

4. In **File**, use the folder browse control and navigate to Power Automate Training, then select the training workbook. Then choose tblTraining in **Table**.

If Table shows "No items", stop here. The workbook has no formatted Excel table, and no amount of re-selecting the file will fix it. Return to Chapter 2 section 2.2 and create the table.

 List rows present in a table

⋮

⏪

Parameters

Settings

Code view

Testing

About

Location *

OneDrive for Business

▼

Document Library *

OneDrive

▼

File *

/Power Automate Training/TrainingRegister.xlsx



Table *

tblTraining

▼

Extract Sensitivity Label

A boolean whether to extract sensitivity label Id for associated artefact.

▼

Sensitivity Label Metadata

A boolean whether to fetch sensitivity label Metadata for associated LabelId.

▼

Advanced parameters

Showing 0 of 6

▼

Show all

Clear all

Figure 4.9. Browse to **TrainingRegister.xlsx**, then choose **tblTraining**.

5. If **DateTime Format** is available under advanced parameters, choose **ISO 8601** to make later date formatting predictable. The advanced parameter list for this action is long and includes sensitivity-label options that this course does not use.

✔ Your flow is ready to go. We recommend you test it.

List rows present in a table

⋮ ⏪

Parameters

Settings

Code view

Testing

About

Location *

OneDrive for Business

▼

Document Library *

OneDrive

▼

File *

/Power Automate Training/TrainingRegister.xlsx

📁

Table *

tblTraining

▼

Extract Sensitivity Label

A boolean whether to extract sensitivity label Id for associated artefact.

▼

Sensitivity Label Metadata

A boolean whether to fetch sensitivity label Metadata for associated LabelId.

▼

Advanced parameters

Showing 1 of 6

▼

Show all

Clear all

DateTime Format

ISO 8601

▼

✕

Figure 4.10. Keep only the DateTime Format advanced parameter and set it to ISO 8601.

Checkpoint: If the table list is empty, save Excel, confirm that the range is a table, and reselect the workbook. Do not replace the table name with a worksheet name.

4.4 Choose the report columns

1. Add **Data Operation > Select** after the Excel action.

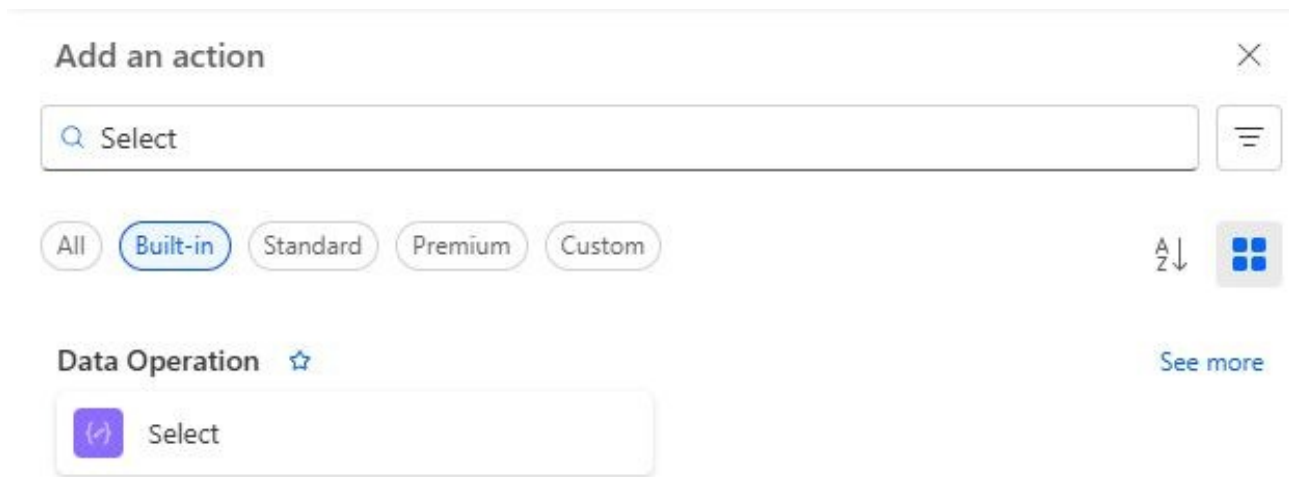


Figure 4.11. Choose the built-in Data Operation **Select** action.

2. In **From**, insert the Excel action's **value** output. This is the array containing all returned rows.
3. Add the following keys and values to the mapping. Use the Excel row fields or the supplied expressions in the expression editor.

Report heading	Value expression
Participant	item()?['ParticipantName']
Course	item()?['CourseTitle']
Session date	item()?['SessionDate']
Status	item()?['Status']

✓ Your flow is ready to go. We recommend you test it.

Select



Parameters Settings Code view About

From *

List rows presen... X

Map *

Participant	ParticipantName X	🗑️ 🔄
Course	CourseTitle X	🗑️
Session date	SessionDate X	🗑️
Status	Status X	

Figure 4.12. **From** uses the Excel **value** array. The four mappings define the report headings and omit Email and internal row metadata.

Why Select helps: Readers see four useful columns. Internal row metadata and the Email column are omitted from the report. Select transforms every row without a manually added Apply to each.

4.5 Build one HTML table

1. Add **Data Operation > Create HTML table** after Select.

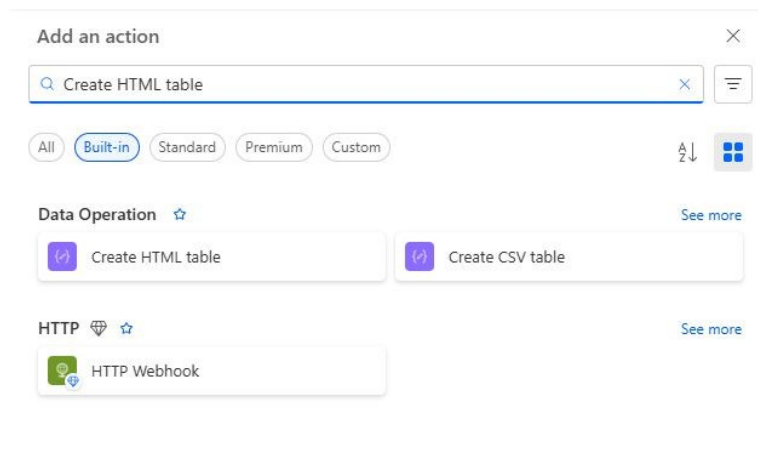


Figure 4.13. Add the built-in **Create HTML table** action.

2. In **From**, select **Output** from Select. Keep the table's column mode automatic because Select already defines the headings.

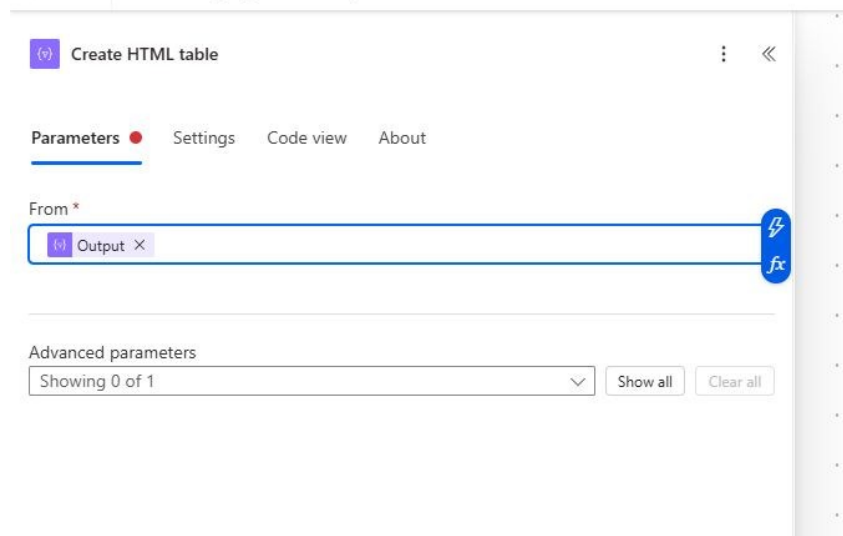


Figure 4.14. **Create HTML table** receives the complete **Output** array from **Select**.

Checkpoint: The HTML table action receives the whole array. If an **Apply to each** appears around this action or the email, check whether you chose a single row field instead of the array or table output.

4.6 Send the coordinator summary

1. Add **Office 365 Outlook > Send an email (V2)** after Create HTML table.

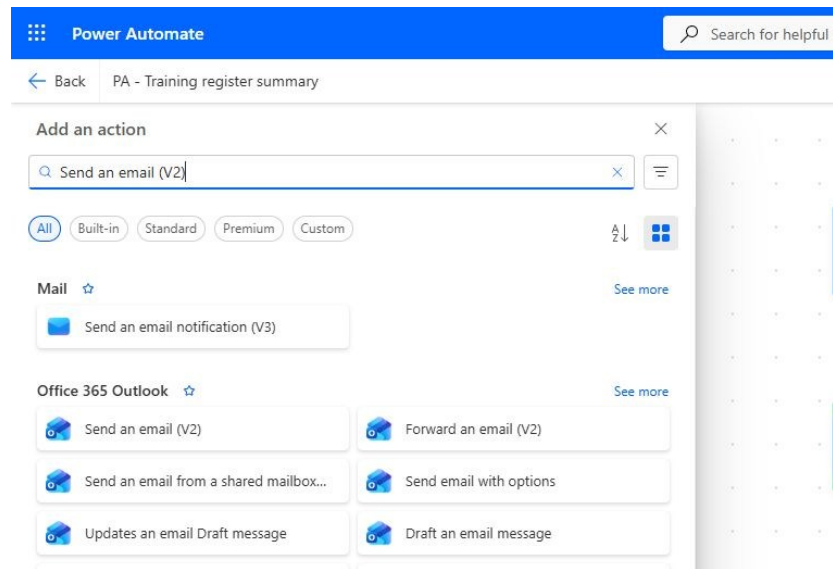


Figure 4.15. Choose **Send an email (V2)** from Office 365 Outlook.

2. Set **To** to your own approved training mailbox and **Subject** to [PA TRAINING] Training register summary.
3. In the body, enter a short introduction and insert **Output** from Create HTML table below it. Use the email editor's HTML/code mode if necessary so the output renders as a table rather than literal tags.

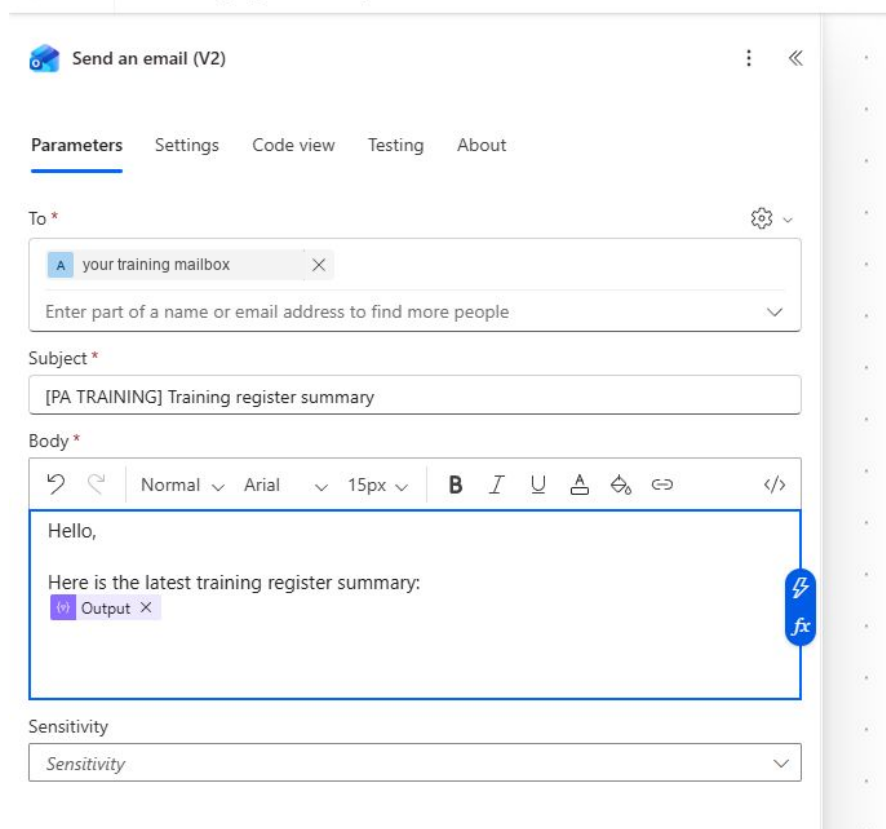


Figure 4.16. The action sends one marked training message to the connected user's own mailbox and inserts the generated HTML table below the introduction.

4. Select **Save**, then run **Flow checker** and resolve any errors.



Figure 4.17. Flow checker reports no errors or warnings.

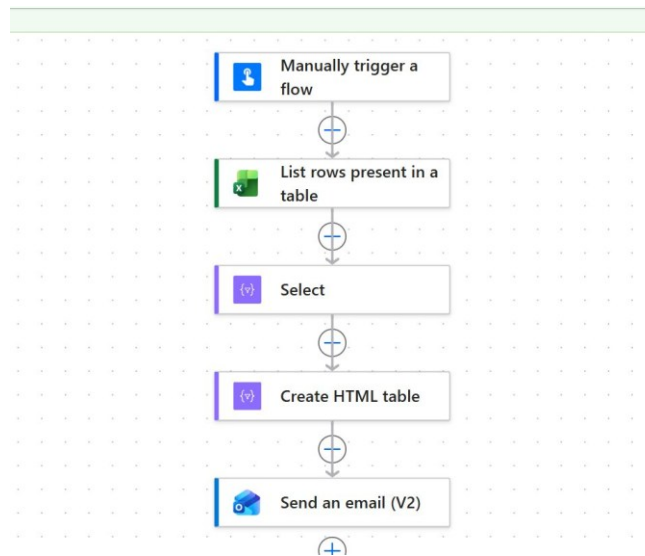


Figure 4.18. The saved five-action sequence produces one report from all workbook rows.

4.7 Test and inspect

1. Select **Test > Manually > Test**. Running the flow sends an actual training email, so follow the trainer's instruction before selecting **Run flow**. After the run completes, open the run history and confirm **Test succeeded**.



Figure 4.19. The manual test completed successfully in three seconds.

2. Open the completed run. Confirm that every action has a green check and that **Send an email (V2)** ran once.

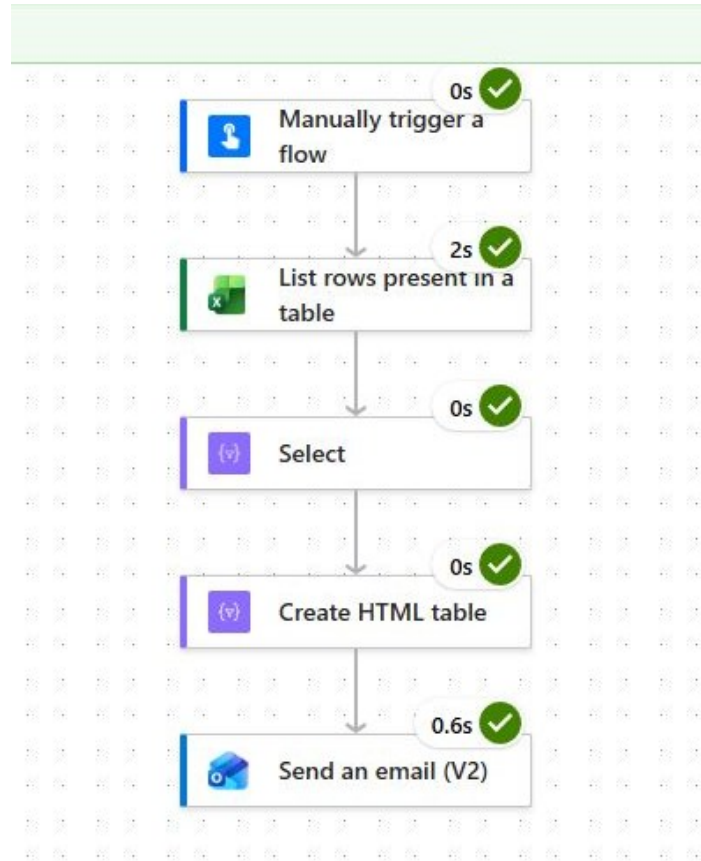


Figure 4.20. All five actions succeeded. The single email action runs after the complete table is created.

3. Inspect the Excel output and confirm that it returned six records. Then open **Show raw outputs** for Select and verify that the six records contain real participant values rather than formula text.



Figure 4.21. Select returned six objects with the four required report fields. This proves the expressions were inserted as dynamic values.

4. Open the received email. Confirm four headings, six data rows, and one email message.

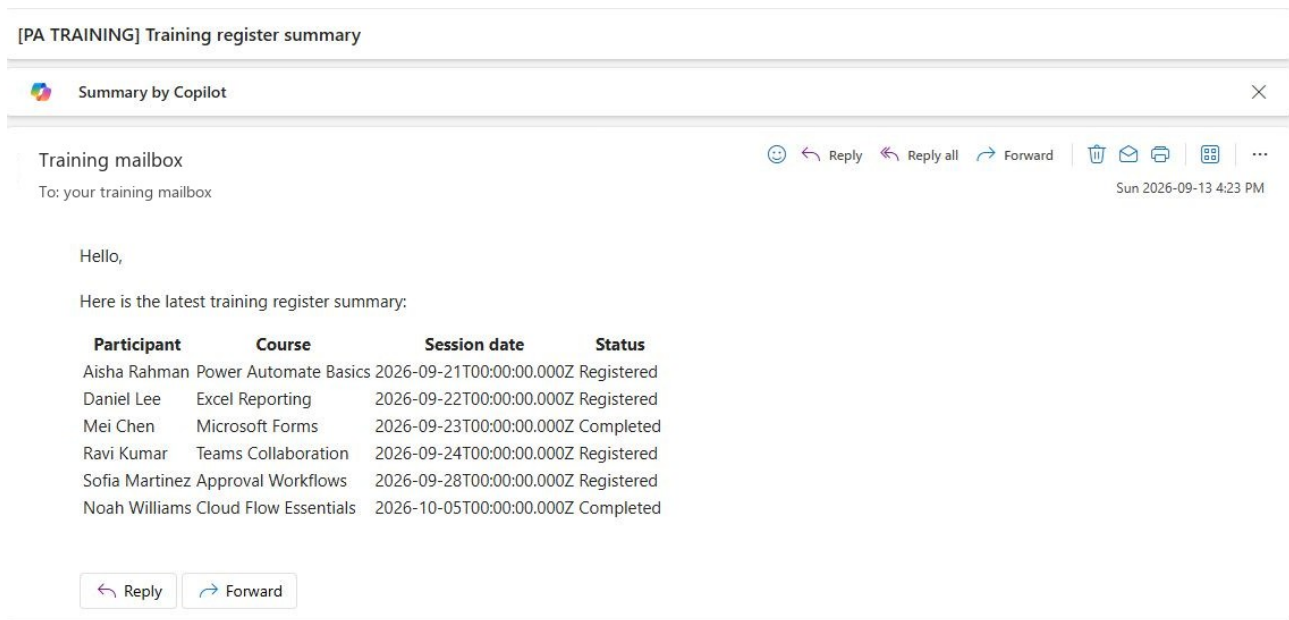


Figure 4.22. Outlook received one HTML table containing all six fictional training records. The connected mailbox identity is masked for distribution.

- For optional practice, change one fictional Status value in Excel, wait for save, and run the flow again only when the trainer authorizes another message. Compare the new email with the earlier report.

Expected result: Each manual run sends one current report. Previous email messages remain historical snapshots.

4.8 Optional formatting practice

Start with the working report. Use the HTML/code editor to wrap the output in a short introduction and add modest table styling. Keep a readable font, strong contrast, a header row, and sufficient cell padding. Email clients can interpret CSS differently, so inspect the received message after each change.

For an optional row-count line, use `length(body('Select'))` if the action is still named Select. If you renamed it, insert its output from dynamic content and confirm the expression reference matches the action name.

Practice: Format Session date as `dd MMM yyyy` only after checking the Excel output. If it is a valid ISO date, use `formatDateTime(item()?['SessionDate'], 'dd MMM yyyy')` in Select. Empty dates need an explicit blank-value check before formatting.

4.9 Optional Teams update

Use this trainer-led extension only after a permitted destination is identified. Add the available **Microsoft Teams > Post message in a chat or channel** action after the email, choose the approved posting identity and destination, and enter a concise record count plus a OneDrive link to the workbook. Check the link's access before posting because pasting a link does not grant permission. Save and test only when the trainer confirms the destination. Teams may render the content differently from Outlook, so use a short message rather than reusing the email's HTML table.

Troubleshooting

Symptom	What to check
No Excel table is listed	Workbook location, saved table, table name, and connector account.
One email arrives for every row	Move the summary email outside any row loop; use Create HTML table output.
The report includes internal columns	Use Select to define report headings and values.
The email displays HTML tags	Inspect the email body editor mode and selected output token.
A date appears as a number	Inspect Excel's DateTime Format option and the actual run output before adding formatDateTime.
The report is stale	Finish the Excel edit, wait for save, and run the flow again.
Teams posting fails	Destination membership, posting identity, organizational policy, and Workflows app availability.

Independent practice

Add Training ID to the report without adding Email. Predict where the change belongs, make it, and verify the received message. Explain why a table containing six records should still produce only one summary email.

Lesson summary

An instant trigger gives the coordinator control over timing. Excel supplies the records, Select defines the report columns, and Create HTML table produces one report. In the next chapter, a recurrence trigger will start a reminder process on a schedule.

Reference notes

The Excel connector exposes List rows present in a table and a DateTime Format parameter. See [Excel Online (Business) connector reference](<https://learn.microsoft.com/en-us/connectors/excelonlinebusiness/>).

Select reshapes an array and Create HTML table converts an array into table markup. See [Microsoft data operations guidance](<https://learn.microsoft.com/en-us/power-automate/data-operations>).

Chapter 5: Send weekly training reminders

Estimated time: 75 minutes

Your result: A scheduled cloud flow that runs every Monday at 9:00 AM Singapore time, reads the training register, and prepares one reminder email for each row.

What you will learn

By the end of this chapter, you will be able to:

- Schedule a flow with the Recurrence trigger and a local time zone.
- Convert a UTC timestamp into readable local time.
- Loop through Excel rows with Apply to each.
- Build a personalized email from the values in the current row.

The business problem

Training coordinators often repeat the same reminder task each week: open a register, read every row, and prepare a message for each participant. A scheduled cloud flow can perform that work at a predictable local time.

The flow in this chapter follows this sequence:

Recurrence > Current time > Convert time zone > List rows present in a table > Apply to each > Send an email (V2)

The Recurrence trigger controls **when the flow starts**. Current time supplies a timestamp in UTC. Convert time zone produces a readable Singapore-time value for the message. The Excel action returns an array of rows, and Apply to each processes those rows one at a time.

Before you begin

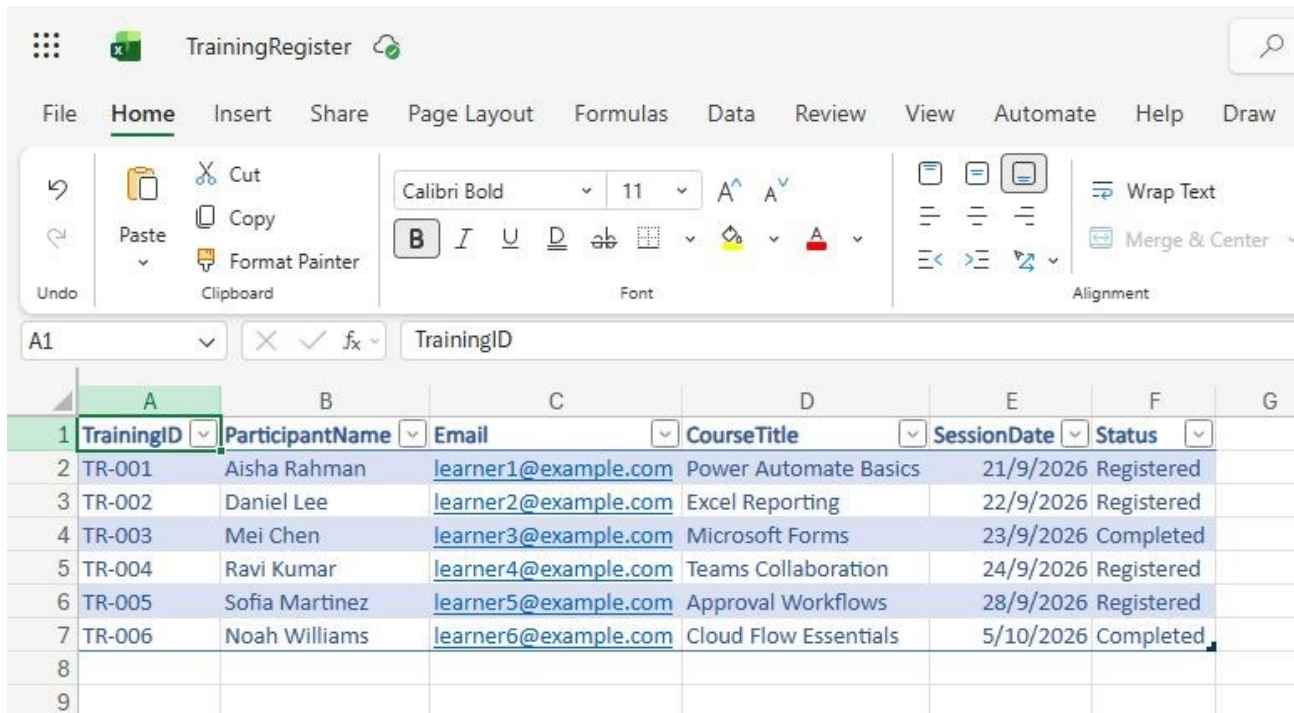
You need Power Automate Training/TrainingRegister.xlsx in OneDrive for Business. The workbook must contain a formatted table named tblTraining with these columns:

Column	Used for
TrainingID	A unique training record reference
ParticipantName	The greeting in the reminder
Email	The recipient address
CourseTitle	The email subject and body
SessionDate	The session information in the body
Status	The current registration state

The prepared workbook uses addresses such as learner1@example.com. These are reserved documentation values and do not receive email. Replace them with trainer-approved test addresses before any live test. Keep the documentation values while learning the build steps.

5.1 Confirm the training register

1. Open `TrainingRegister.xlsx`, confirm that all six columns are present, and check that the data is formatted as a table named `tblTraining`.



TrainingID	ParticipantName	Email	CourseTitle	SessionDate	Status
TR-001	Aisha Rahman	learner1@example.com	Power Automate Basics	21/9/2026	Registered
TR-002	Daniel Lee	learner2@example.com	Excel Reporting	22/9/2026	Registered
TR-003	Mei Chen	learner3@example.com	Microsoft Forms	23/9/2026	Completed
TR-004	Ravi Kumar	learner4@example.com	Teams Collaboration	24/9/2026	Registered
TR-005	Sofia Martinez	learner5@example.com	Approval Workflows	28/9/2026	Registered
TR-006	Noah Williams	learner6@example.com	Cloud Flow Essentials	5/10/2026	Completed

Figure 5.1. The prepared register contains six fictional rows and the six required columns. The `example.com` addresses are safe placeholders and must be replaced before a live email test.

Checkpoint: Save the workbook and finish any active cell edit. The connector reads the saved cloud file.

5.2 Create the weekly schedule

1. In Power Automate, select **Create > Scheduled cloud flow**. Name the flow `PA - Weekly training reminders`, set it to repeat every **1 Week**, select **Monday**, and choose **9:00 AM**. Your starting date will differ from the example.

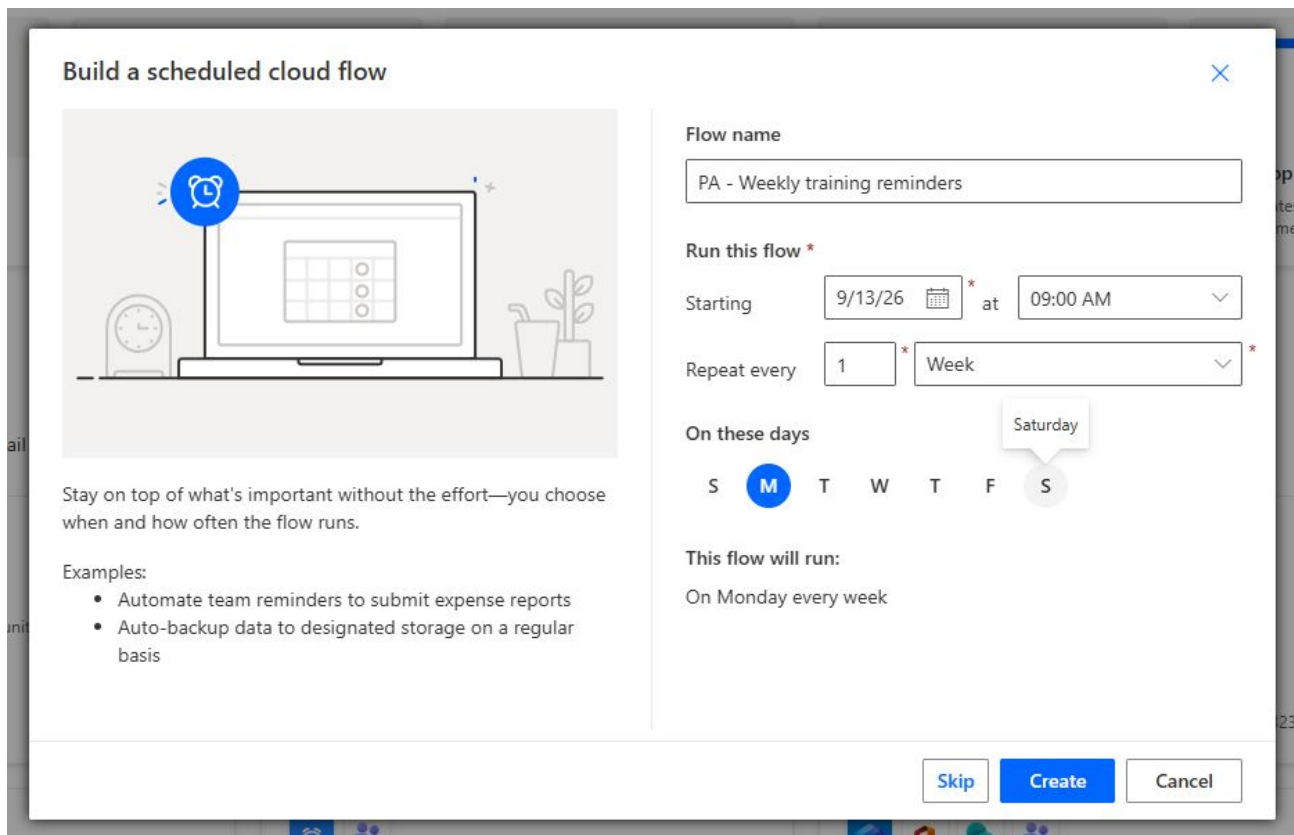


Figure 5.2. The scheduled-flow dialog creates the Recurrence trigger and the first weekly schedule.

2. Open **Recurrence** and confirm these values:

Setting	Value
Interval	1
Frequency	Week
Time zone	(UTC+08:00) Kuala Lumpur, Singapore
On these days	Monday
At these hours	9
At these minutes	0

 Recurrence

Parameters

Settings

Code view

About

Recurrence *

Interval *

1

Frequency *

Week

Time zone

(UTC+08:00) Kuala Lumpur, Singapore

Start time

2026-09-13T01:00:00.000Z

On these days

Monday

At these hours

9

At these minutes

0

Preview

Runs at 9:00 on Monday every week

Figure 5.3. The preview states that the flow runs at 9:00 on Monday every week. The Start time value is stored in UTC, while the Time zone setting controls the local schedule.

Checkpoint: A weekly frequency without a day and local time is incomplete for this scenario. Read the preview before continuing.

5.3 Produce a readable local timestamp

1. Add **Date Time > Current time** below Recurrence. This action needs no parameters.

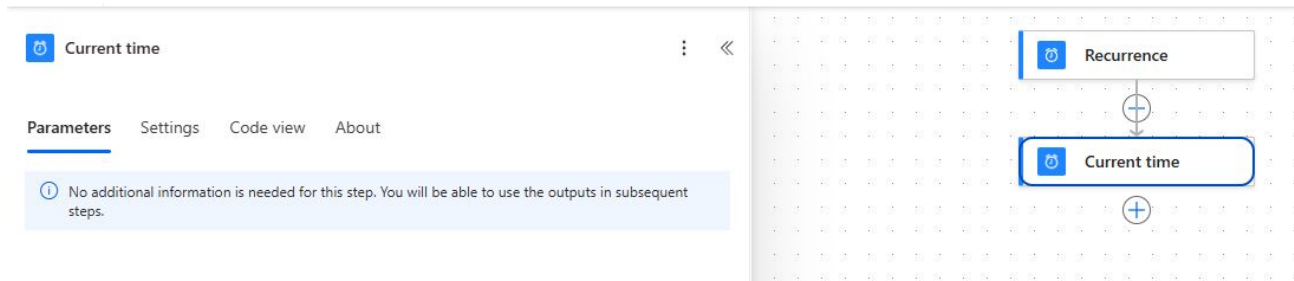


Figure 5.4. Current time provides a UTC timestamp that later actions can use.

2. Add **Date Time > Convert time zone** and configure it as follows:

Setting	Value
Base time	Current time from the previous action
Source time zone	(UTC) Coordinated Universal Time
Destination time zone	(UTC+08:00) Kuala Lumpur, Singapore
Format string	Full date/time pattern (short time)

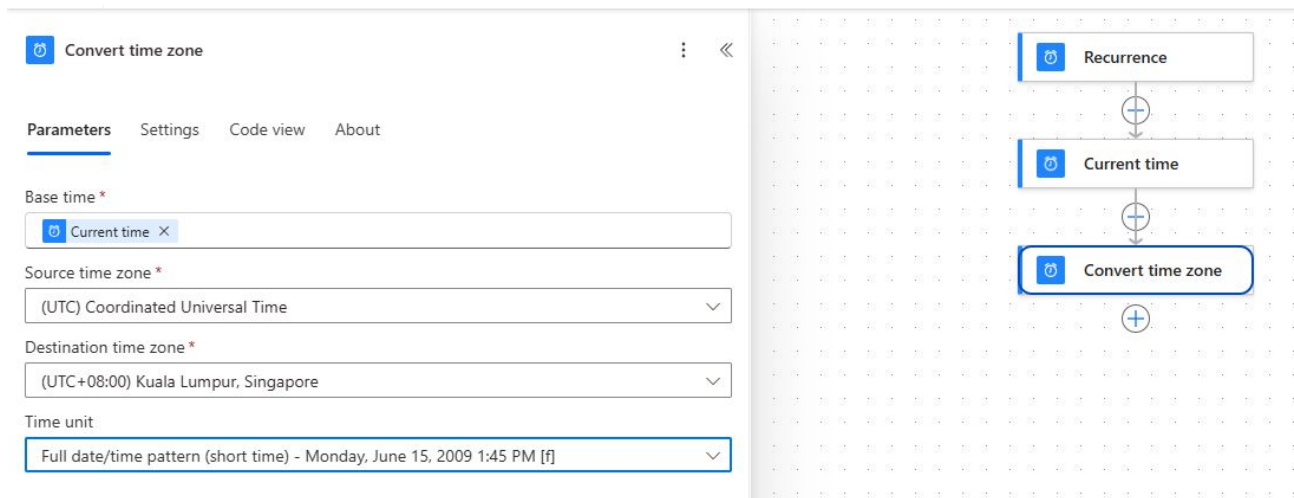


Figure 5.5. The action converts the UTC timestamp into a readable local value for the reminder body.

The two time-zone settings have different jobs. Recurrence determines when the flow starts. Convert time zone changes a timestamp that appears in the message; it does not change the schedule.

5.4 Read the Excel rows

1. Add **Excel Online (Business)** > **List rows present in a table** and set:

Setting	Value
Location	OneDrive for Business
Document Library	OneDrive
File	/Power Automate Training/TrainingRegister.xlsx
Table	tblTraining

✓ Your flow is ready to go. We recommend you test it.

List rows present in a table

Parameters Settings Code view Testing About

Location *
OneDrive for Business

Document Library *
OneDrive

File *
/Power Automate Training/TrainingRegister.xlsx

Table *
tblTraining

Extract Sensitivity Label
A boolean whether to extract sensitivity label Id for associated artefact.

Sensitivity Label Metadata
A boolean whether to fetch sensitivity label Metadata for associated LabelId.

Recurrence

Current time

Convert time zone

List rows present in a table

Figure 5.6. The Excel action points to the saved workbook and its formatted table.

If the table list is empty: save the workbook, confirm the range is a formatted table, confirm its name is `tblTraining`, and then reselect the file.

5.5 Process one row at a time

1. Add **Control > Apply to each**. In **Select an output from previous steps**, insert **value** from List rows present in a table. If you use the expression editor, the same array is:
`outputs('List_rows_present_in_a_table')['body/value']`

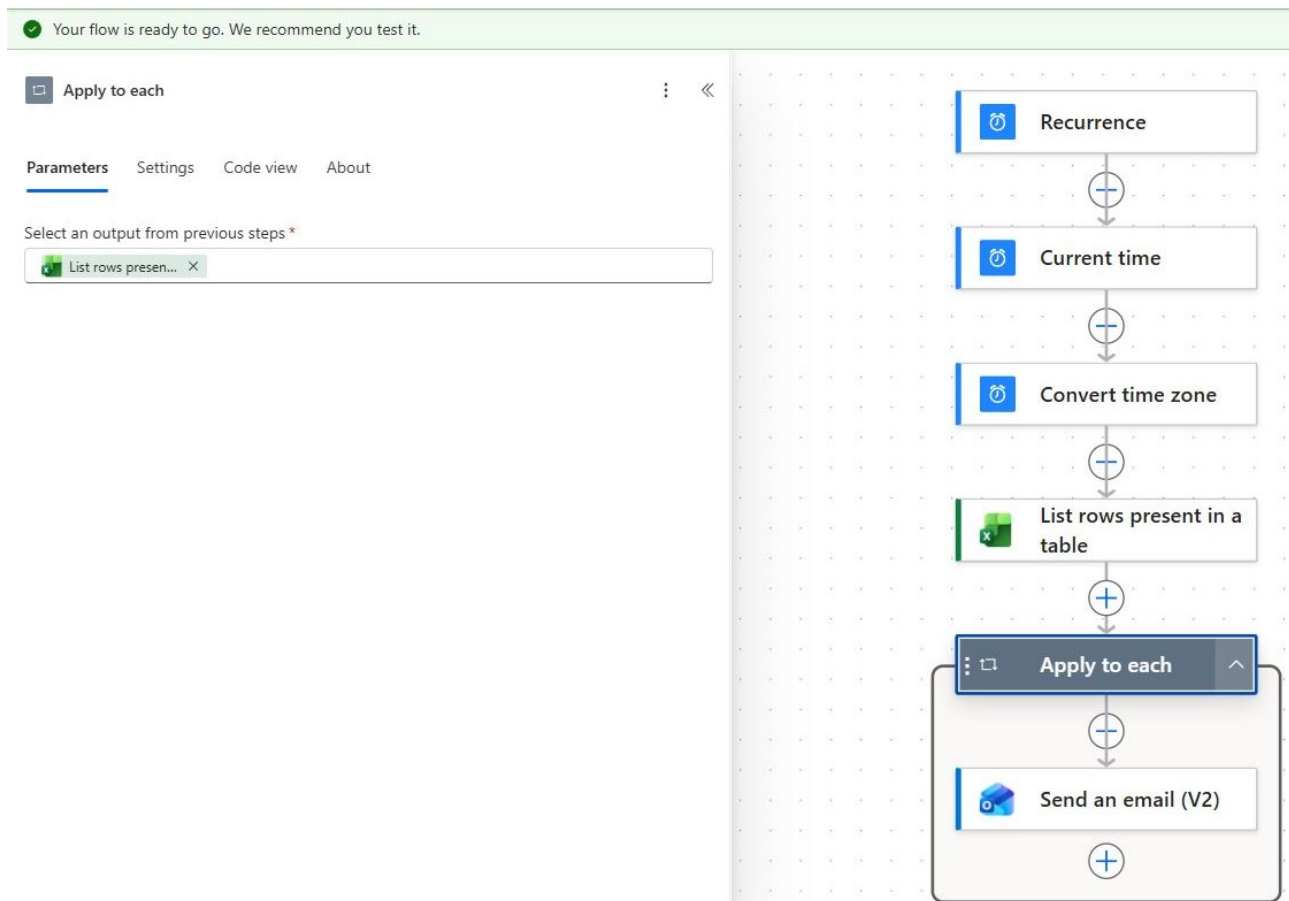


Figure 5.7. Apply to each receives the complete Excel row array. The email action is placed inside the loop so it can use the current row.

Checkpoint: Apply to each must receive the Excel **value** array. A single field such as Email or CourseTitle is not the array.

5.6 Prepare the reminder email

1. Inside Apply to each, add **Office 365 Outlook > Send an email (V2)**. Configure the three required fields with dynamic values from the current row:

Field	Value
To	item()?['Email']
Subject	concat(['PA TRAINING] Reminder: ', item()?['CourseTitle'])
Body	Use the expression below

```
concat('<p>Hello ', item()?['ParticipantName'],
',</p><p>This is your weekly training reminder.</p><p><strong>Course:</strong> ',
item()?['CourseTitle'], '<br><strong>Session date:</strong> ',
item()?['SessionDate'], '<br><strong>Status:</strong> ', item()?['Status'],
'<br><strong>Reminder generated:</strong> ', body('Convert_time_zone'),
'</p><p>Please contact the training coordinator if your plans change.</p>')
```

✓ Your flow is ready to go. We recommend you test it.

Send an email (V2)

Parameters Settings Code view Testing About

To *

Email X

Subject *

concat(...) X

Body *

Normal Arial 15px B I U A E G

concat(...) X

Sensitivity

Sensitivity

Advanced parameters

Showing 1 of 6 Show all Clear all

Importance

Normal X

Recurrence

Current time

Convert time zone

List rows present in a table

Apply to each

Send an email (V2)

Figure 5.8. The recipient comes from the current row. The subject and body each contain one expression token.

Checkpoint: The email action belongs inside Apply to each. In this version it will prepare one message for every row, including rows whose Status is Completed. Chapter 7 will add filtering so only relevant upcoming records remain.

5.7 Save and check the flow

1. Select **Save**, then open **Flow checker**. Resolve any reported error before testing.

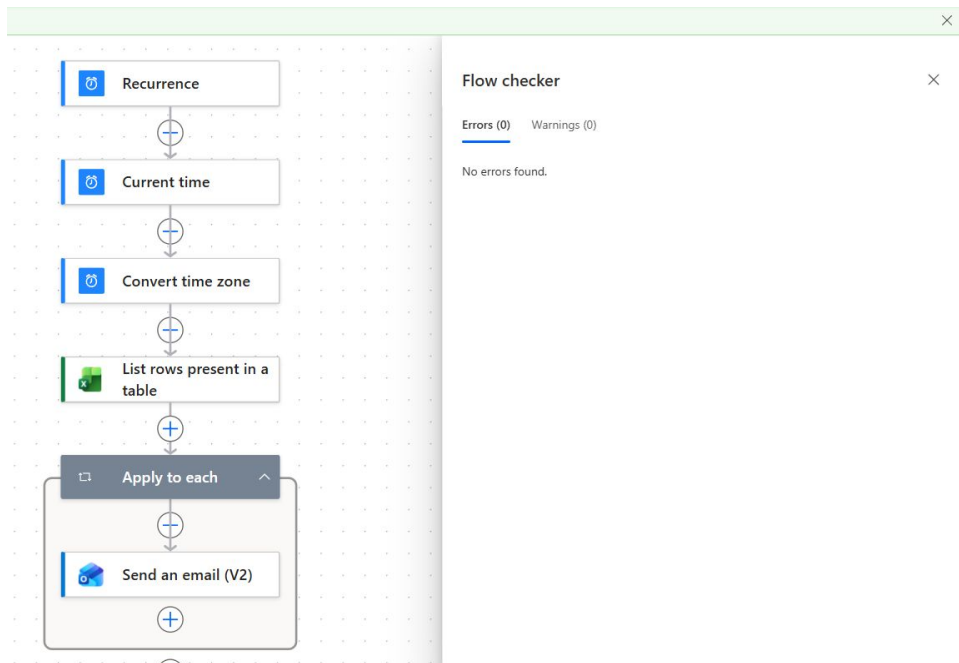


Figure 5.9. The saved flow passes Flow checker with zero errors and zero warnings.

2. Close Flow checker and confirm the complete sequence on the canvas.

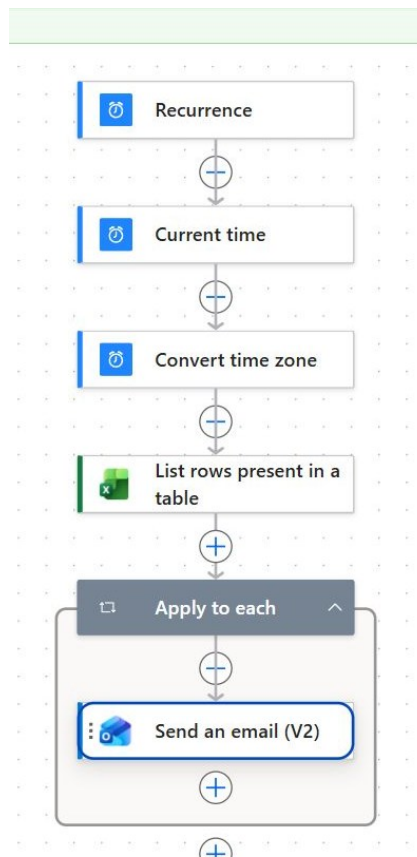


Figure 5.10. The complete saved flow contains five top-level steps and one email action inside Apply to each.

Trainer-controlled test

Do not select **Test** while the workbook contains `example.com` placeholders. Before a live run, replace every Email value with an explicitly approved training destination and save the workbook.

A manual test of this chapter's flow attempts **one email for every Excel row**. With the six-row sample, that means six email actions. The trainer must confirm the recipients and authorize the run immediately before **Run flow** is selected.

After an authorized test, inspect the run history and check that:

- Recurrence, Current time, Convert time zone, and List rows present in a table succeeded.
- Apply to each shows the expected number of iterations.
- Each Send an email (V2) action used the row's Email, CourseTitle, SessionDate, and Status.
- The displayed reminder timestamp is in Singapore time.

Troubleshooting

Symptom	What to check
The flow starts at the wrong local time	Recurrence Time zone, day, hour, minute, and preview.
The message timestamp is eight hours early	Convert time zone source and destination settings.
No Excel rows are returned	OneDrive location, workbook save state, formatted table, and table name.
Apply to each is empty	Use the Excel action's <code>value</code> array.
A recipient or course is blank	Check the workbook header spelling and the current-row expressions.
Completed records receive reminders	This is expected in Chapter 5; Chapter 7 adds the filter.
Multiple real messages could be sent	Stop before testing, verify every Email value, and obtain trainer authorization.

Independent practice

Change the scheduled hour to a trainer-specified value, read the Recurrence preview, and then restore 9:00. Explain why changing Convert time zone would not change when the flow starts.

Lesson summary

A scheduled trigger starts the flow at a predictable local time. Date Time actions produce a readable timestamp, Excel supplies an array of records, and Apply to each makes the current row available to the email action. The saved flow is structurally valid. The next chapter adds a human approval decision, and Chapter 7 extends this reminder flow with filtering and a coordinator summary.

Chapter 6: Process a training approval

Estimated time: 75 minutes

Your result: An automated cloud flow that reads a Microsoft Forms response, asks a person to approve or reject it, and prepares a different Teams message for each outcome.

What you will learn

By the end of this chapter, you will be able to:

- Start a flow when a Microsoft Forms response is submitted.
- Retrieve the answers with Get response details.
- Request a decision with Start and wait for an approval.
- Branch on the approval outcome and post a different Teams message for each result.

The business problem

A training coordinator needs enough information to review a request before confirming it. The requester should complete one consistent form, an approver should make a clear decision, and the flow should follow the correct branch automatically.

The flow in this chapter follows this sequence:

When a new response is submitted > Get response details > Start and wait for an approval > Condition > Approved or rejected Teams message

The Forms trigger supplies a response identifier. Get response details uses that identifier to retrieve the answers. Start and wait for an approval pauses until a decision is recorded. The Condition reads the approval Outcome and directs the flow into either the True or False branch.

Before you begin

You need access to Microsoft Forms, Approvals, and Microsoft Teams through your work or school account. Your trainer must identify an approved test approver and Teams destination before a live test.

The screenshots use `approver@example.com`. This is a documentation placeholder. Replace it with a trainer-approved account before testing. Do not submit the form or run the flow while the placeholder remains.

6.1 Prepare the request form

1. In Microsoft Forms, create a form named `Training Request`. Add the description `Submit a fictional training request for the Power Automate approval exercise`. Create these four required questions:

Question	Type	Options or setting
Requester name	Text	Single line
Course	Choice	Power Automate Basics; Approval Workflows; Excel Reporting
Preferred date	Date	Required
Business reason	Text	Long answer

Training Request

Submit a fictional training request for the Power Automate approval exercise.

* Required

1. Requester name *

Enter your answer

2. Course *

☐ Power Automate Basics

☐ Approval Workflows

☐ Excel Reporting

3. Preferred date *

Please input date (M/d/yyyy)

4. Business reason *

Figure 6.1. The form collects the minimum information needed for an approval decision. The signed-in account greeting has been masked.

Checkpoint: Confirm that all four questions show the red required marker. A missing answer can produce an incomplete approval card.

6.2 Create the automated flow

1. In Power Automate, select **Create > Automated cloud flow**. Name the flow PA – Training request approval, search for Forms, select **When a new response is submitted**, and select **Create**.

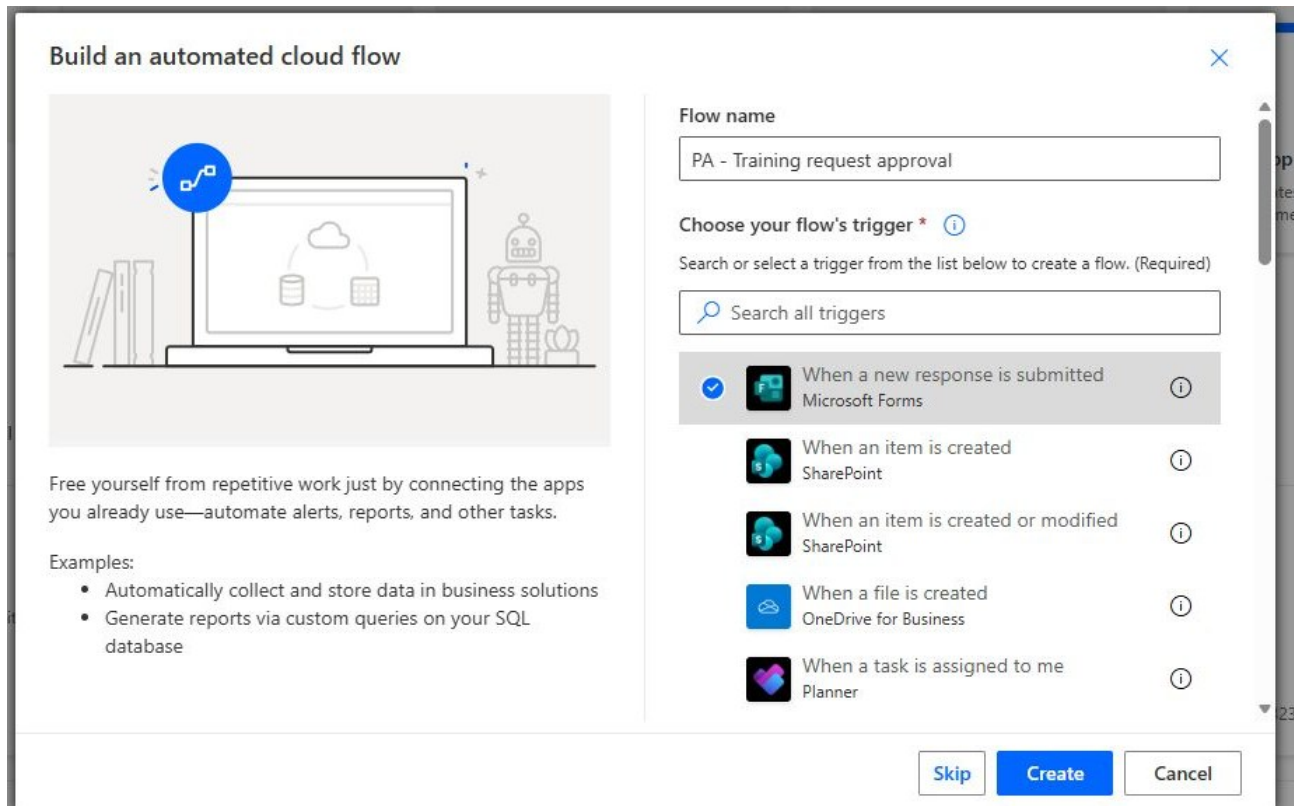


Figure 6.2. The Forms event creates an automated flow that starts whenever the selected form receives a response.

6.3 Select the form

1. Open **When a new response is submitted** and set **Form Id** to **Training Request**.

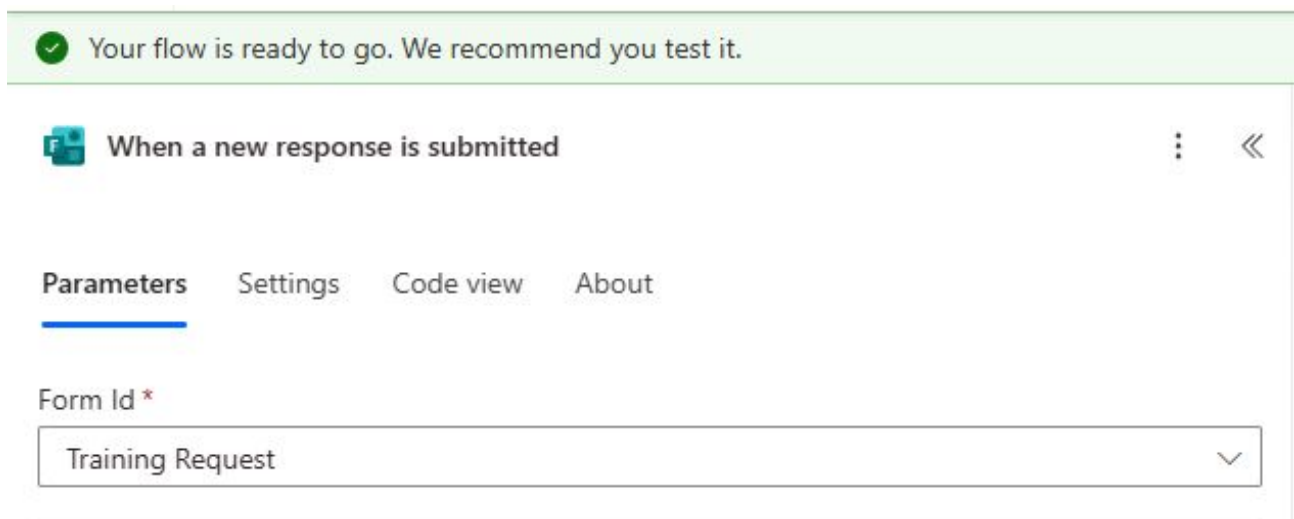
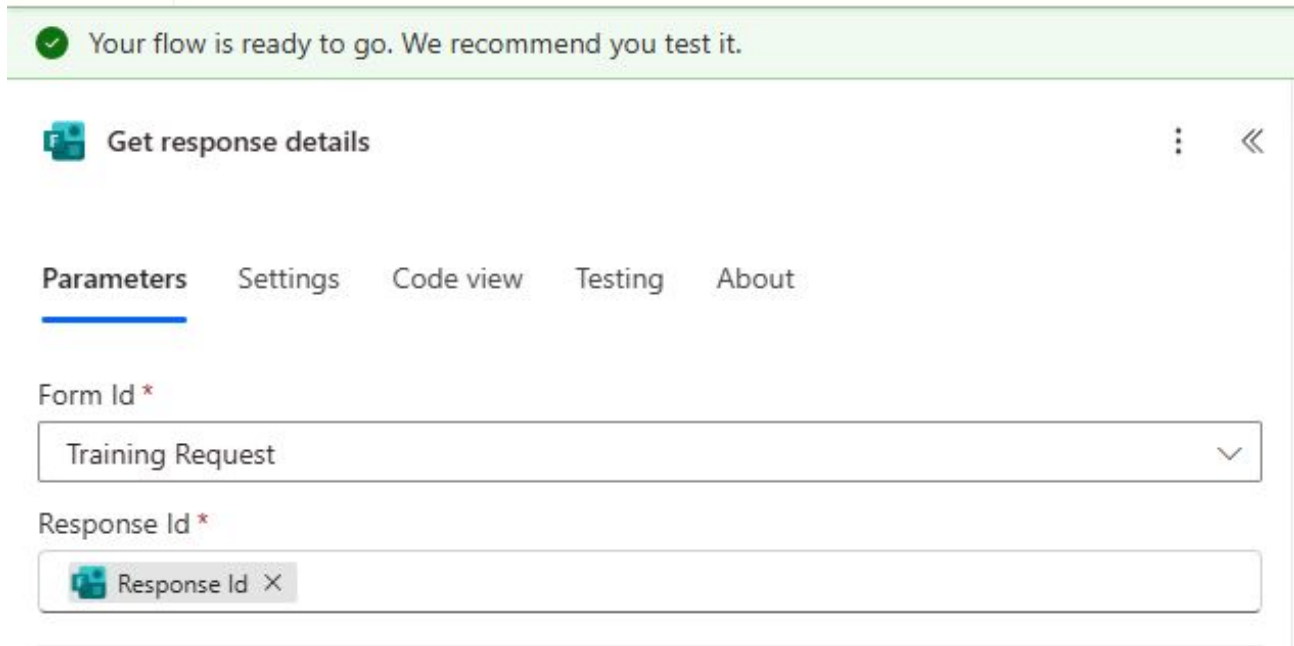


Figure 6.3. The trigger watches only the Training Request form.

If the form does not appear: confirm that you own or can access it, refresh the list, and verify that the Microsoft Forms connection uses the intended work or school account.

6.4 Retrieve the answers

1. Add **Microsoft Forms > Get response details**. Set **Form Id** to **Training Request**. In **Response Id**, insert **Response Id** from When a new response is submitted.



The screenshot shows the configuration for the 'Get response details' action in Microsoft Power Automate. At the top, a green banner with a checkmark icon states: 'Your flow is ready to go. We recommend you test it.' Below this, the action is titled 'Get response details' with a Microsoft Forms icon. To the right of the title are three vertical dots and a double-left arrow icon. Below the title is a tabbed interface with five tabs: 'Parameters' (which is selected and underlined with a blue line), 'Settings', 'Code view', 'Testing', and 'About'. Under the 'Parameters' tab, there are two required fields, each marked with a red asterisk. The first field is 'Form Id', which contains the text 'Training Request' and a dropdown arrow on the right. The second field is 'Response Id', which contains a dynamic content link represented by a small Microsoft Forms icon, the text 'Response Id', and a close 'X' button on the right.

Figure 6.4. The dynamic Response Id links this action to the exact submission that started the flow.

The trigger tells the flow that a response exists. Get response details returns the individual question answers so later actions can use them.

6.5 Request the decision

1. Add **Approvals > Start and wait for an approval** and configure:

Field	Value
Approval type	Approve/Reject - First to respond
Title	[PA TRAINING] Training request
Assigned to	Trainer-approved approver
Details	Requester name, Course, Preferred date, and Business reason from Get response details

Arrange the Details field so each label is followed by its matching dynamic value.

Start and wait for an approval

⋮

⏪

Parameters
Settings
Code view
Testing
About

Approval type *

Approve/Reject - First to respond

▼

Title *

[PA TRAINING] Training request

Assigned to *

⚙️ ▼

A
approver@example.com
X

Enter part of a name or email address to find more people

▼

Details

Requester:
Requester name
X

Course:
Course
X

Preferred date:
Preferred date
X

Business reason:
Business reason
X

Item link

Add a link to the item to approve

Item link description

Describe the link to the item

Figure 6.5. The approver receives the four answers needed to make the decision. The screenshot uses a reserved **example.com** placeholder.

Checkpoint: Use the dynamic values from **Get response details**, not similarly named values from another action.

6.6 Branch on the outcome

1. Add **Control > Condition**. In the completed condition row, select **Outcome** from Start and wait for an approval, choose **is equal to**, and enter **Approve**.

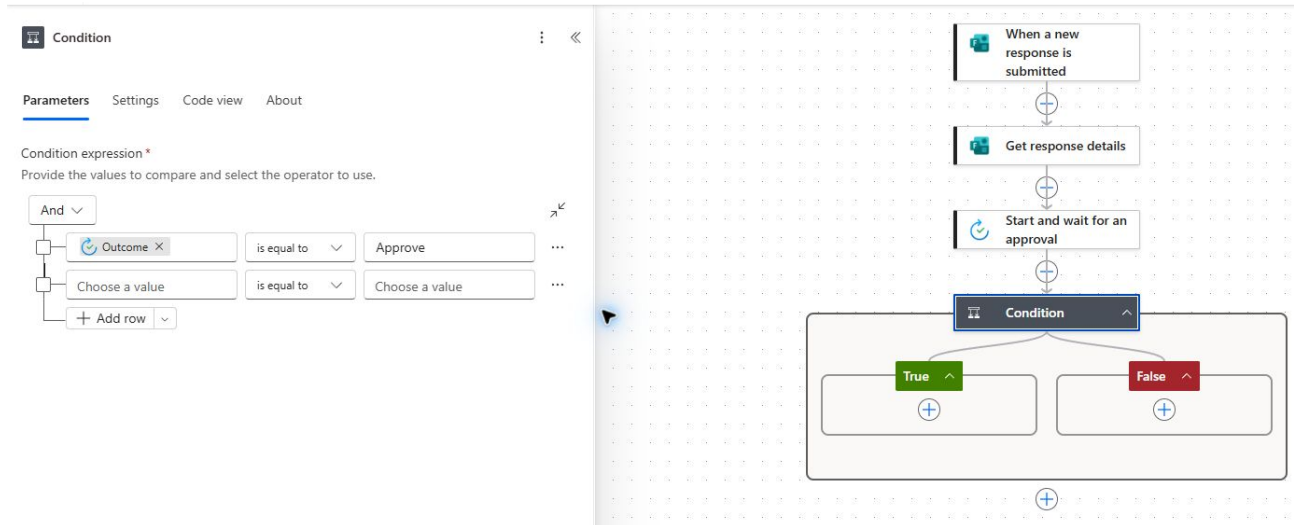


Figure 6.6. An Approve outcome follows the True branch; every other outcome follows the False branch. The designer may display a trailing blank row for adding another rule; only the completed Outcome row is part of this decision.

Checkpoint: The comparison text must be **Approve**. A misspelling or extra space sends an approval down the False branch.

6.7 Prepare the approved message

1. In the **True** branch, add **Microsoft Teams > Post message in a chat or channel**. Configure:

Field	Value
Post as	Flow bot
Post in	Chat with Flow bot
Recipient	Trainer-approved Teams account
Message	Training request approved for followed by Course from Get response details

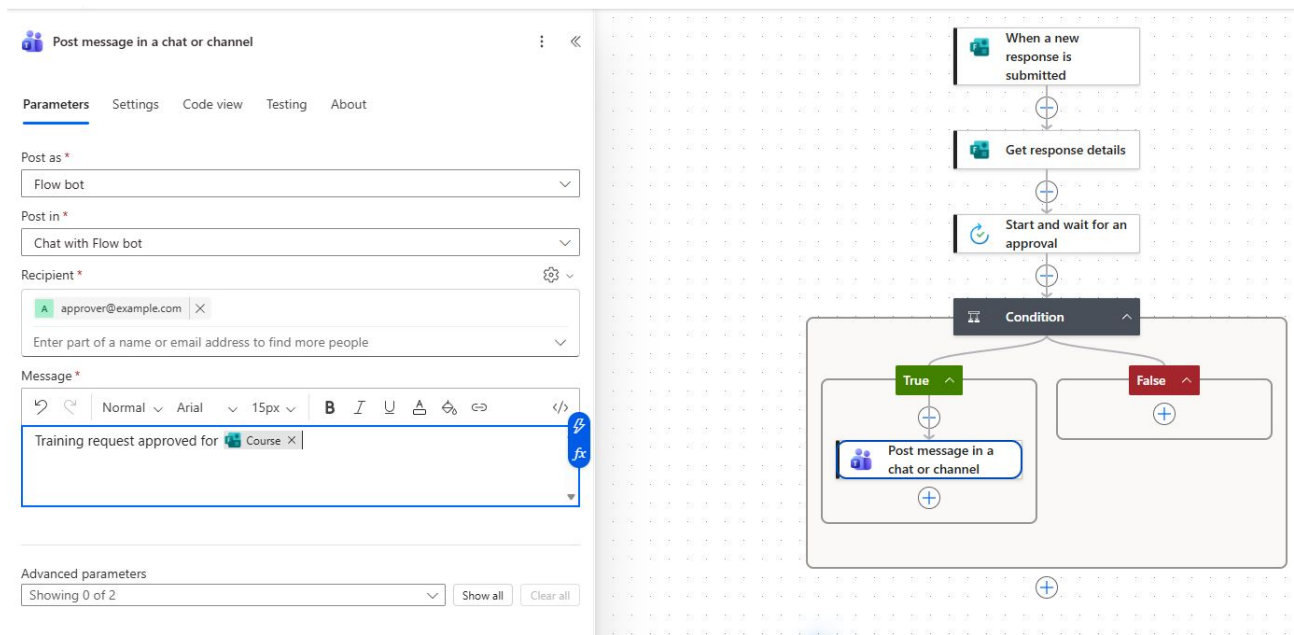


Figure 6.7. The True branch prepares the approval message and includes the requested course.

6.8 Prepare the rejected message

1. In the **False** branch, add a second **Microsoft Teams > Post message in a chat or channel** action. Use the same Post as, Post in, and trainer-approved recipient settings. Set **Message** to Training request rejected for followed by **Course** from Get response details.

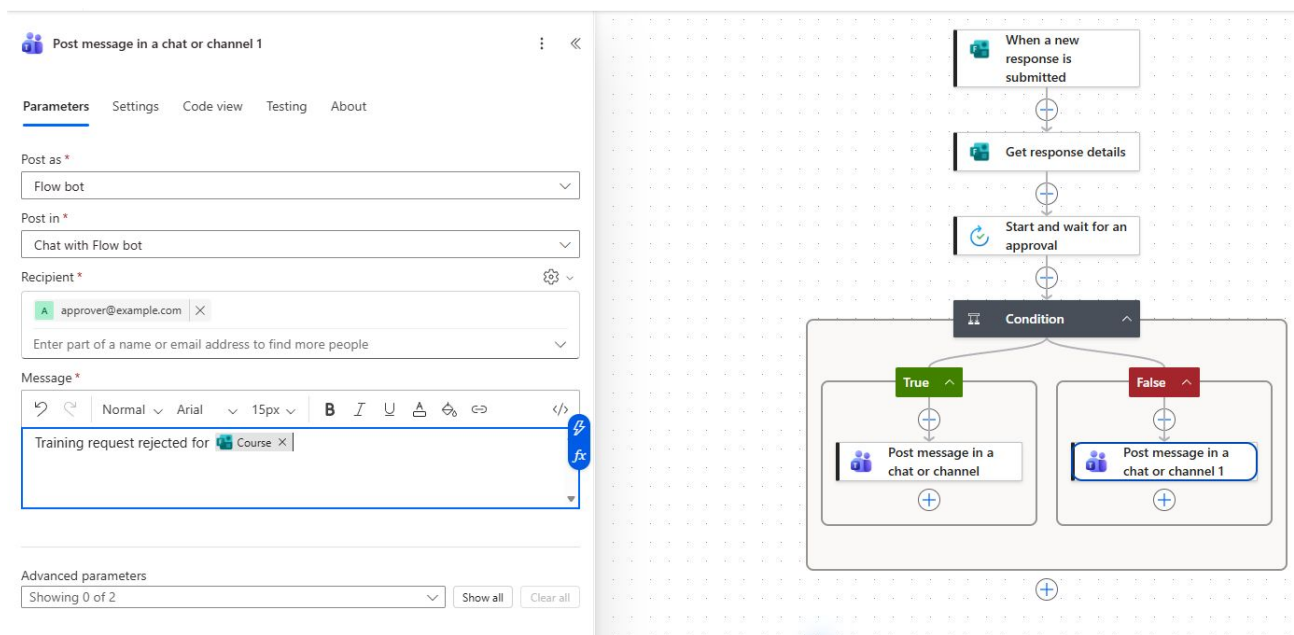


Figure 6.8. The False branch prepares a different message while reusing the course answer.

For a production process with more outcomes, add explicit handling for values such as Cancelled or timed-out approvals. This beginner exercise uses the two standard Approve/Reject buttons.

6.9 Save and check the flow

1. Select **Save**, then open **Flow checker**. Resolve every error before testing.

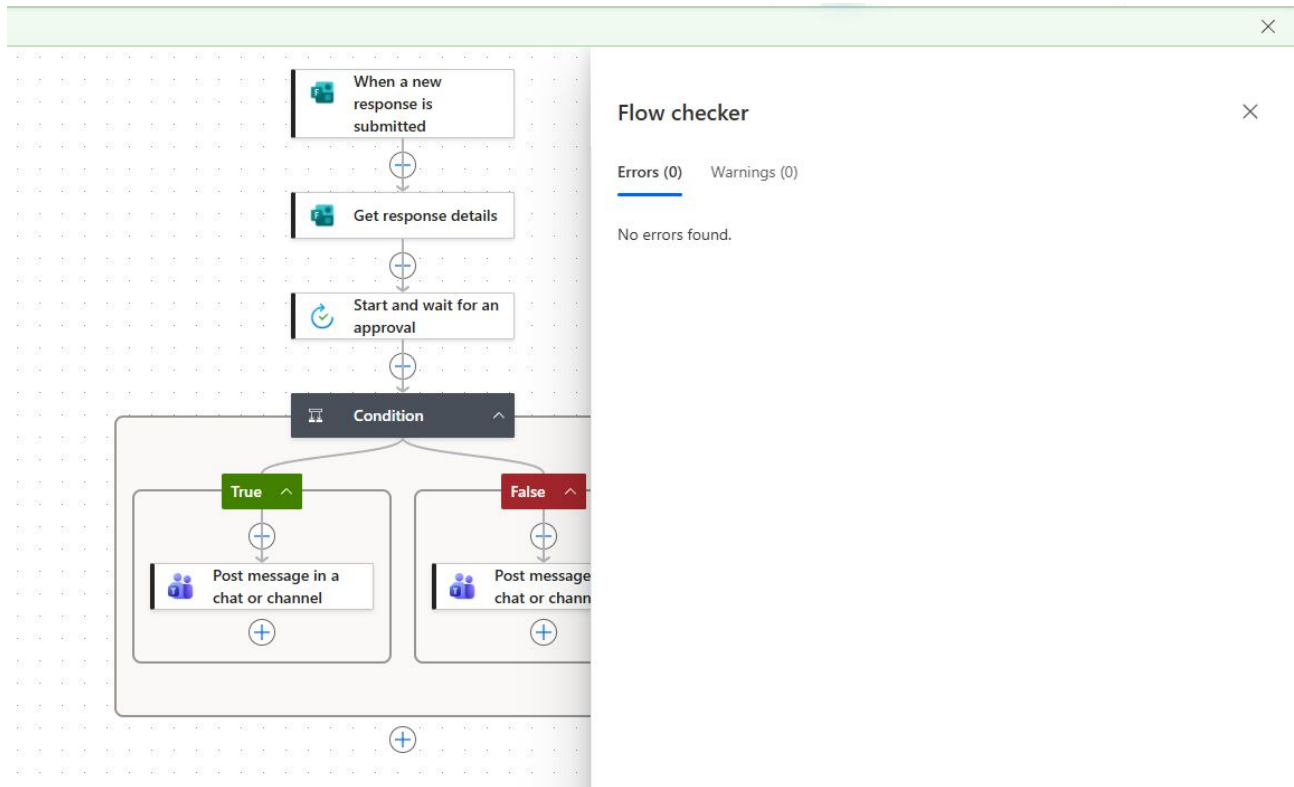


Figure 6.9. The saved flow passes Flow checker with zero errors and zero warnings.

2. Close Flow checker and confirm the full sequence on the canvas.

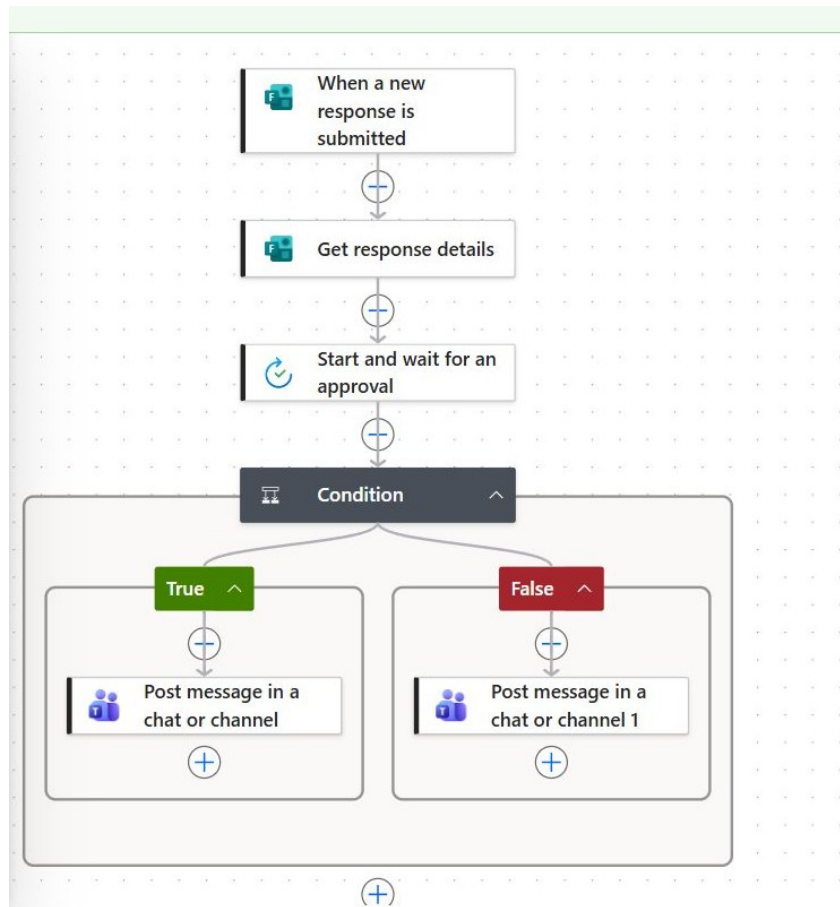


Figure 6.10. The complete saved flow contains three steps before the Condition and one Teams action in each branch.

Trainer-controlled test

Testing this flow creates an approval request and may post a Teams message. Before a live test, replace both `approver@example.com` placeholders with an explicitly approved account and confirm that the recipient is ready to receive the approval and Teams notification.

The trainer should then authorize these actions immediately before they occur:

1. Submit one fictional Training Request response.
2. Approve the request and inspect the True branch and approved Teams message.
3. Submit a second fictional response.
4. Reject it and inspect the False branch and rejected Teams message.

After each authorized test, open the run history and confirm that the correct branch succeeded while the other branch was skipped.

Troubleshooting

Symptom	What to check
Training Request is absent from Form Id	Form ownership or access, the Forms connection account, and a refreshed form list.
The approval has no request information	Get response details Form Id, Response Id token, and Details tokens.
The flow waits indefinitely	The assigned approver, approval notification, and whether a decision was submitted.
An approval follows the False branch	Outcome token, <code>is equal to</code> , and the exact text <code>Approve</code> .
The Teams action cannot resolve the recipient	Replace the documentation placeholder with the trainer-approved work or school account.
No Teams message appears	Check the selected branch, Teams connection, Post in setting, recipient, and action result in run history.
Flow checker is clean but the flow cannot be tested	Flow checker validates structure; it does not prove that placeholder recipients or organization policies permit a live run.

Independent practice

Without running the flow, change the approval title so it also includes the Course dynamic value. Explain which action supplies that value and why the trigger alone cannot supply it.

Lesson summary

A Forms trigger starts the flow and provides a response identifier. Get response details turns that identifier into usable answers. Start and wait for an approval records a human decision, and a Condition sends the request through the matching branch. The saved flow is structurally valid and ready for a trainer-controlled test after approved recipients replace the placeholders.

Chapter 7: Capstone - weekly training reminders and summary

Estimated time: 90 minutes

Your result: A copy of the weekly reminder flow that selects upcoming incomplete training records, counts the prepared reminders, and follows a clear summary or no-records branch.

What you will learn

By the end of this chapter, you will be able to:

- Filter an array before a loop so that only relevant rows are processed.
- Normalize text and compare dates inside an expression.
- Count processed items with an integer variable.
- Branch after a loop, including a week with no matching records.

The business problem

Chapter 5 prepared one reminder for every row in the register. A coordinator needs a more useful process: ignore completed training, limit the reminder window, count the records that were processed, and handle the week when nothing matches.

The capstone follows this sequence:

Recurrence > Current time > Convert time zone > List rows > Filter array > Initialize variable > Apply to each > Send reminder > Increment variable > Condition > Summary or no-records message

The filter is placed before the loop so Apply to each receives only relevant rows. The counter starts at zero and increases once for every reminder prepared. After the loop, a Condition checks the counter and selects the matching result branch.

Before you begin

Complete Chapter 5 and keep `Power Automate Training/TrainingRegister.xlsx` in OneDrive for Business. Its table must be named `tblTraining` and include `Email`, `CourseTitle`, `ParticipantName`, `SessionDate`, and `Status`.

The sample register uses `example.com` addresses. These are reserved documentation values that do not receive email. Keep the capstone turned off and do not test it until the trainer has replaced every recipient with an approved destination.

7.1 Make a safe capstone copy

1. Open `PA - Weekly training reminders`, select **Save As**, enter `PA - Weekly training capstone`, and save the copy. Power Automate creates the copy turned off by default.

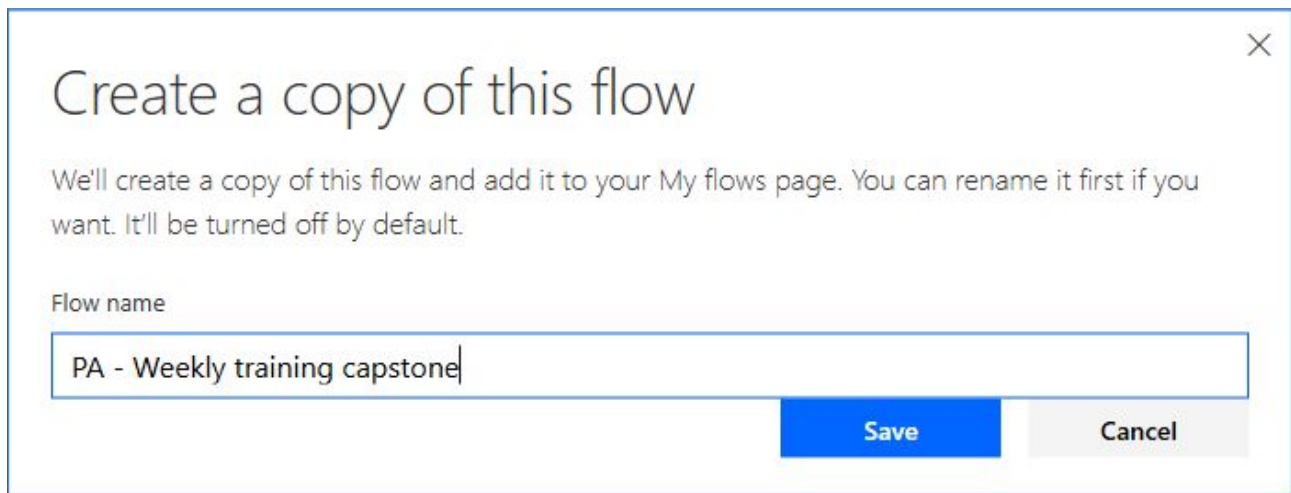


Figure 7.1. The capstone begins as a separate copy, preserving the Chapter 5 flow.

Checkpoint: Confirm that you are editing PA – Weekly training capstone before making the remaining changes.

7.2 Return Excel dates in a consistent format

1. Open **List rows present in a table**. Under **Advanced parameters**, add **DateTime Format** and select **ISO 8601**.

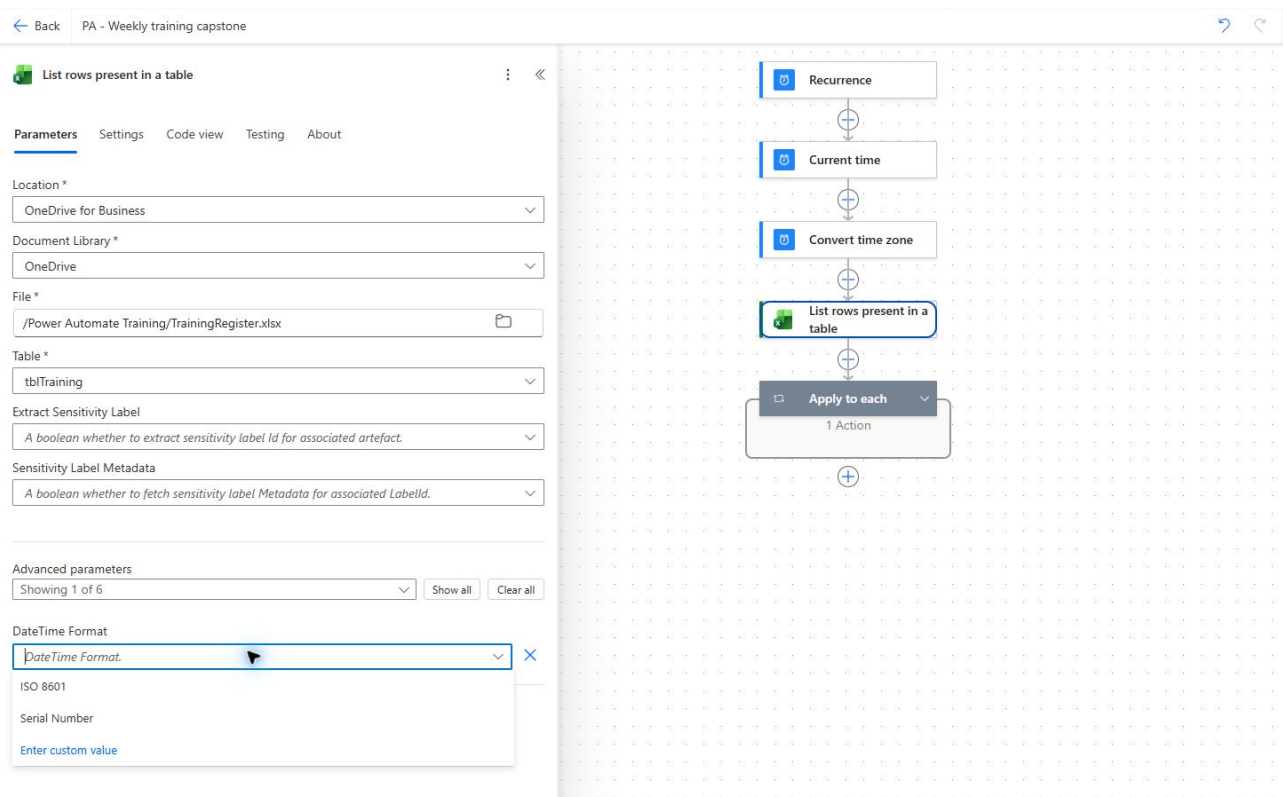


Figure 7.2. ISO 8601 gives the filter a predictable SessionDate value for date comparisons.

7.3 Select upcoming incomplete records

1. Add **Data Operation > Filter array** after List rows present in a table. Rename it **Filter upcoming incomplete training**. Set **From** to **value** from List rows present in a table, choose **Edit in advanced mode**, and enter:

```
@and(  
  not(equals(toLower(trim(item()?['Status'])), 'completed')),  
  greaterOrEquals(ticks(item()?['SessionDate']), ticks(startOfDay(utcNow()))),  
  lessOrEquals(ticks(item()?['SessionDate']), ticks(addDays(startOfDay(utcNow()), 14)))  
)
```

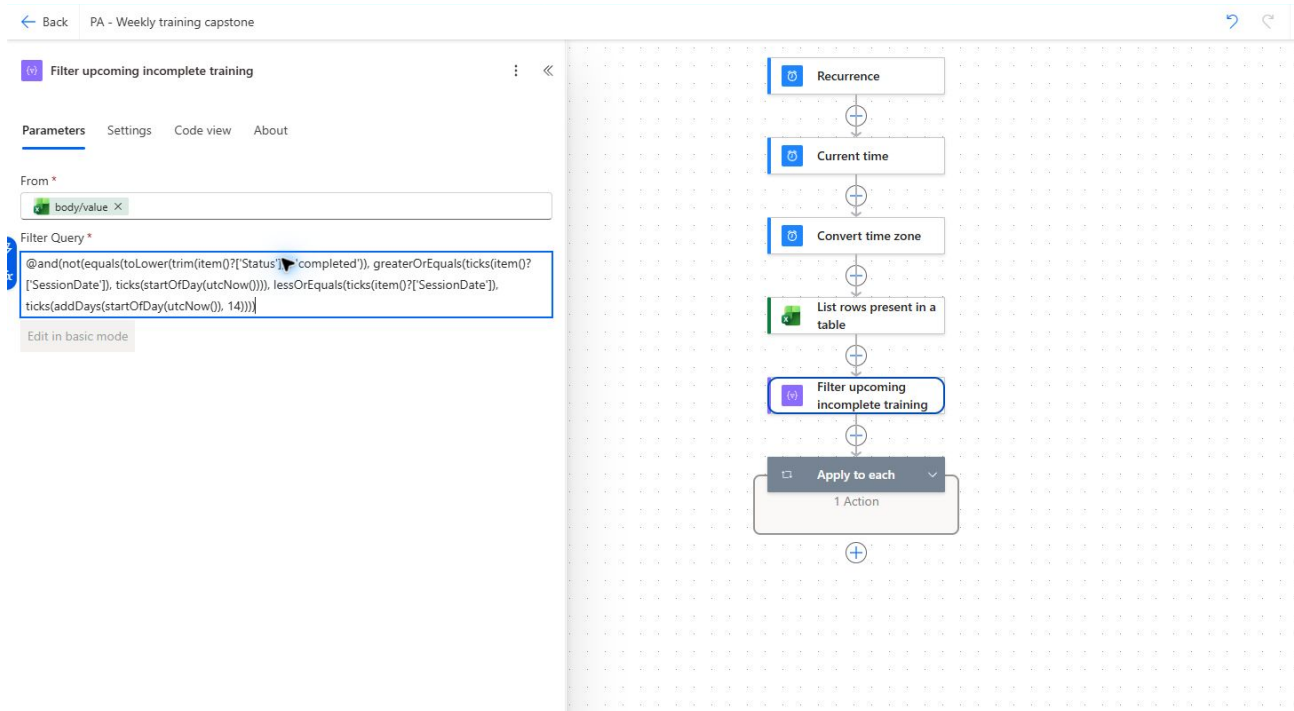


Figure 7.3. The expression normalizes *Status*, excludes completed rows, and keeps dates from today through the next 14 days.

`trim()` removes accidental spaces and `toLower()` makes the status comparison independent of capitalization. `ticks()` converts each date to a number that can be compared reliably. The start and end comparisons are inclusive.

Checkpoint: With the prepared six-row sample and a current date of 17 September 2026, four rows match: 21, 22, 24, and 28 September. The completed rows do not match, including the one on 23 September.

7.4 Start the reminder counter

1. Add **Variables > Initialize variable** after Filter array. Rename the action **Initialize reminder count** and configure:

Field	Value
Name	ReminderCount
Type	Integer
Value	0

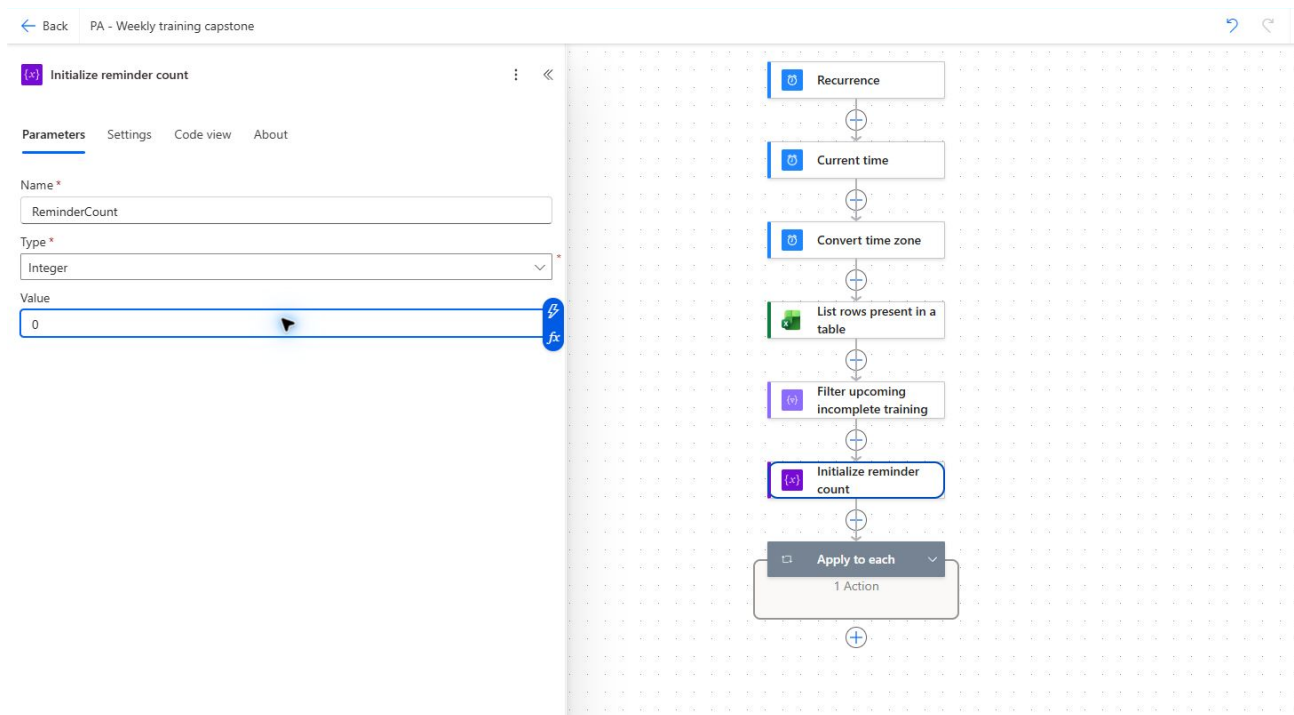


Figure 7.4. The counter begins at zero before any row enters the loop.

7.5 Loop through the filtered rows

1. Open **Apply to each** and replace its input with **Body** from Filter upcoming incomplete training. If you use the expression editor, the equivalent array is:

`body('Filter_upcoming_incomplete_training')`

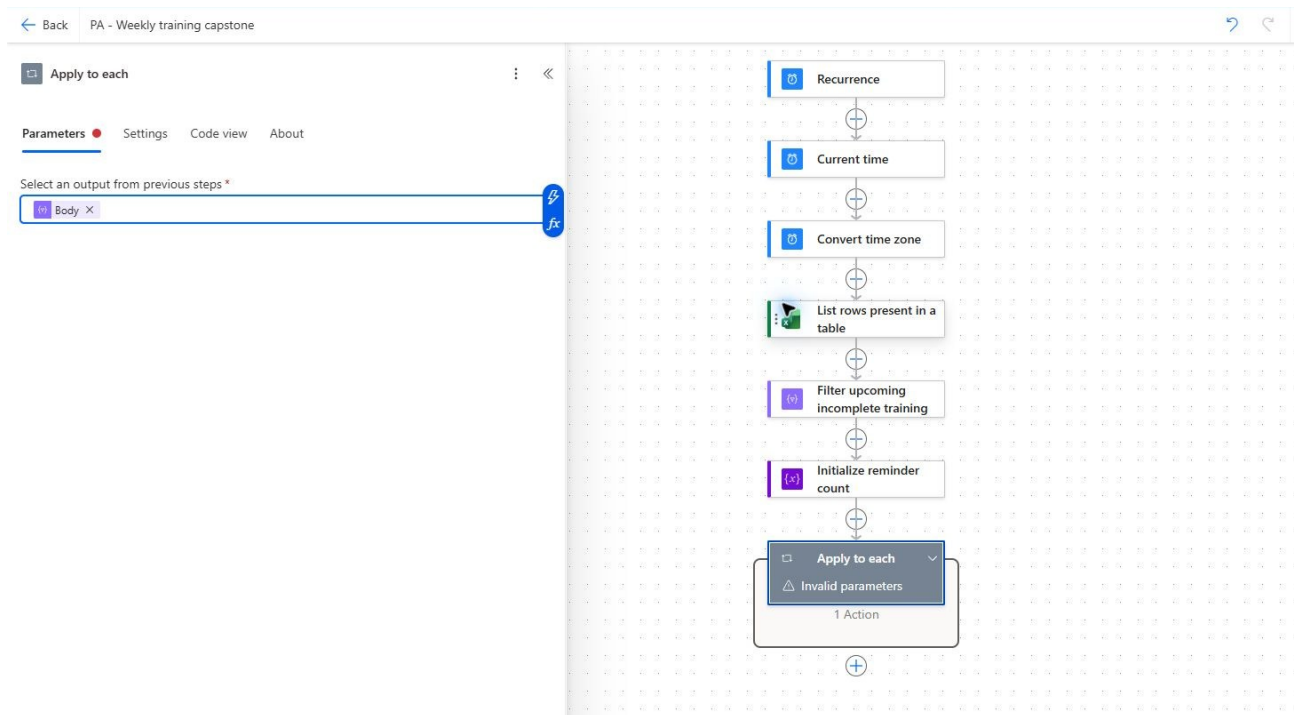


Figure 7.5. The loop now processes only the rows that passed the capstone filter.

Checkpoint: Do not use the original Excel value array here. That would restore the Chapter 5 behavior and include completed or out-of-window rows.

7.6 Count each prepared reminder

1. Inside Apply to each, after **Send an email (V2)**, add **Variables > Increment variable**. Select ReminderCount and enter 1 for Value.

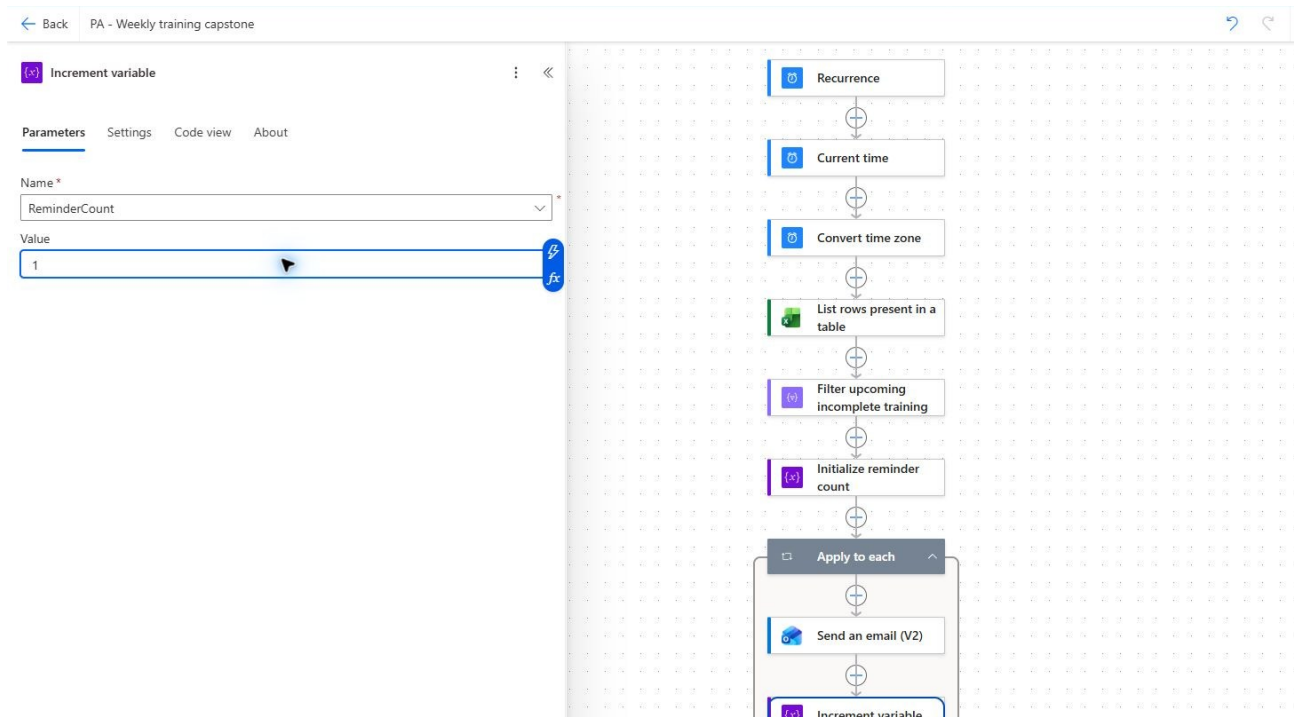


Figure 7.6. The counter increases once after each reminder action in the loop.

Keep the counter update after the email action. In a later authorized test, this ordering makes the count represent reminders whose email action reached the increment step.

7.7 Review the learner reminder

1. Confirm that **Send an email (V2)** remains inside Apply to each and uses the current filtered row:

Field	Value
To	item()?['Email']
Subject	concat(['PA TRAINING] Reminder: ', item()?['CourseTitle'])
Body	The Chapter 5 HTML expression using ParticipantName, CourseTitle, SessionDate, Status, and Convert time zone

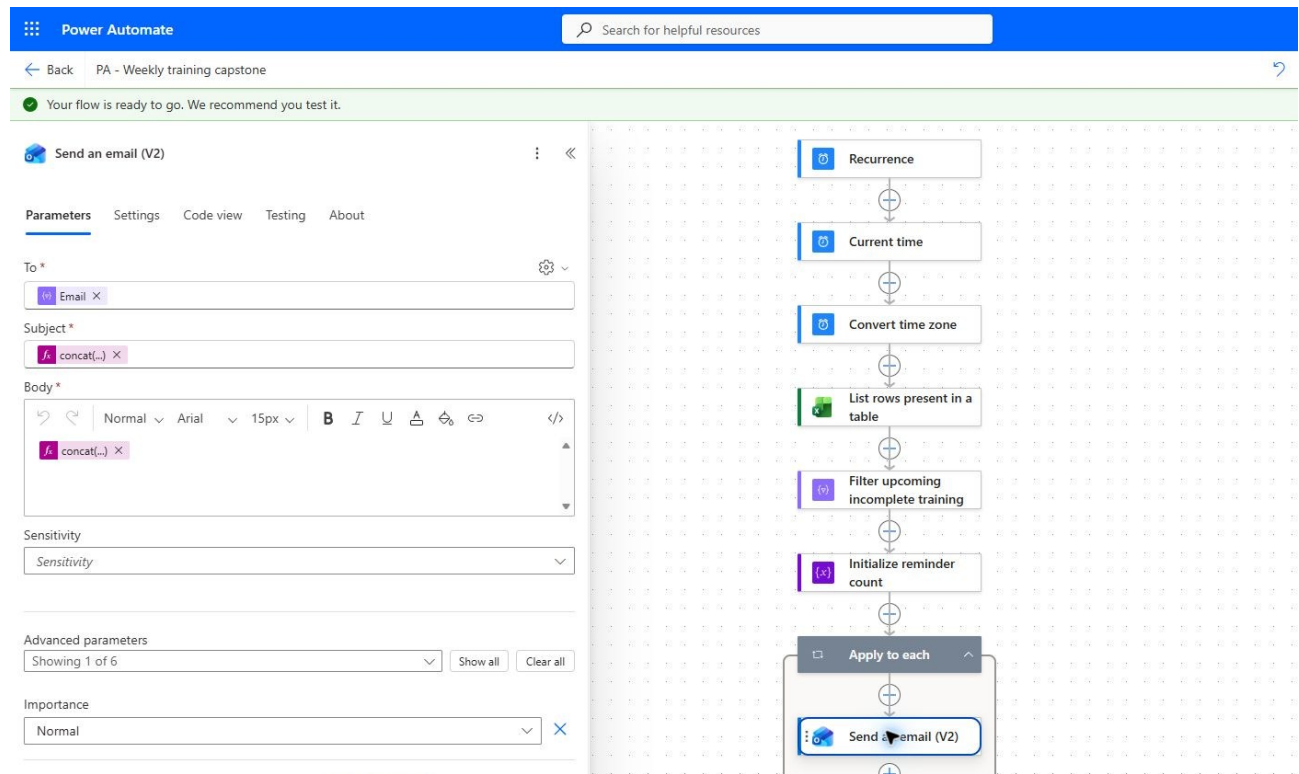


Figure 7.7. The email still uses the current row, but the revised loop supplies only upcoming incomplete records.

7.8 Decide whether a summary is needed

1. Add **Control > Condition** after Apply to each. Rename it **If reminders were prepared** and configure the completed row as:

Left value	Operator	Right value
ReminderCount	is greater than	0

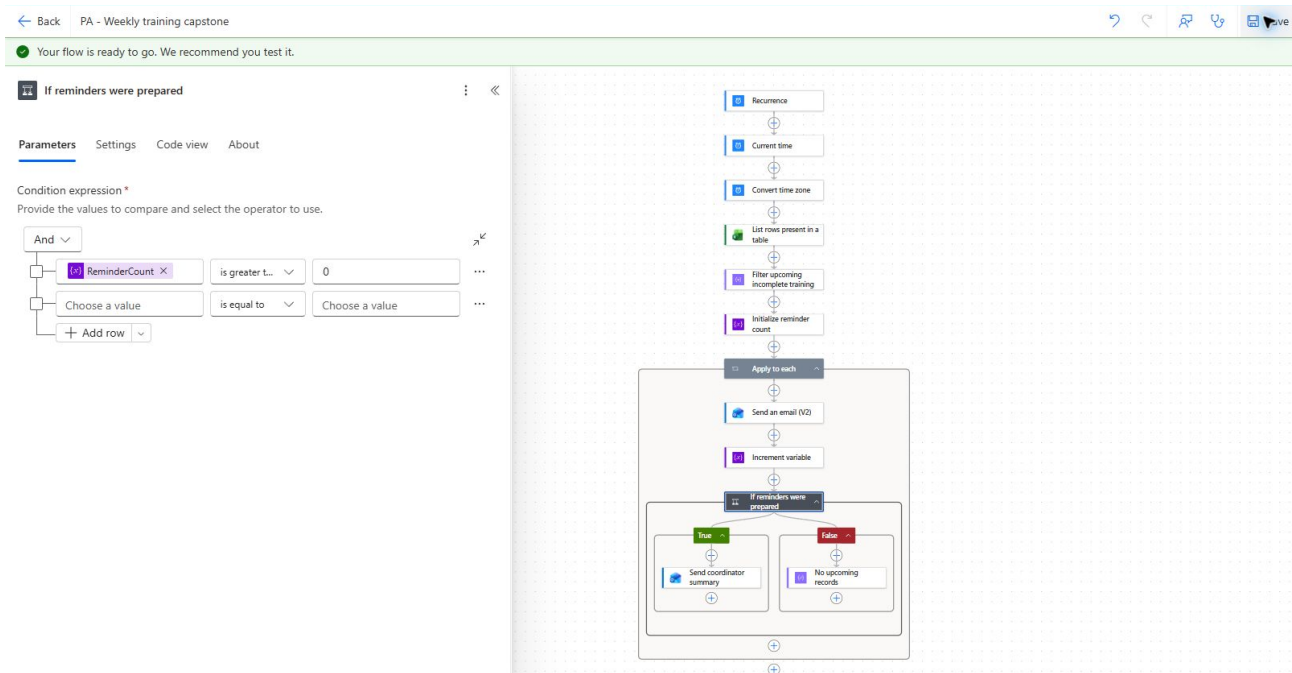


Figure 7.8. A positive count follows True; zero follows False. The designer displays one blank row as the control for adding another rule.

7.9 Prepare the coordinator summary

1. In the **True** branch, add **Office 365 Outlook > Send an email (V2)** and rename it **Send coordinator summary**. Configure:

Field	Value
To	coordinator@example.com
Subject	[PA TRAINING] Weekly reminder summary
Body	Type The weekly training reminder flow prepared , insert ReminderCount from dynamic content, thentype reminder(s) for upcoming incomplete sessions in the next 14 days.

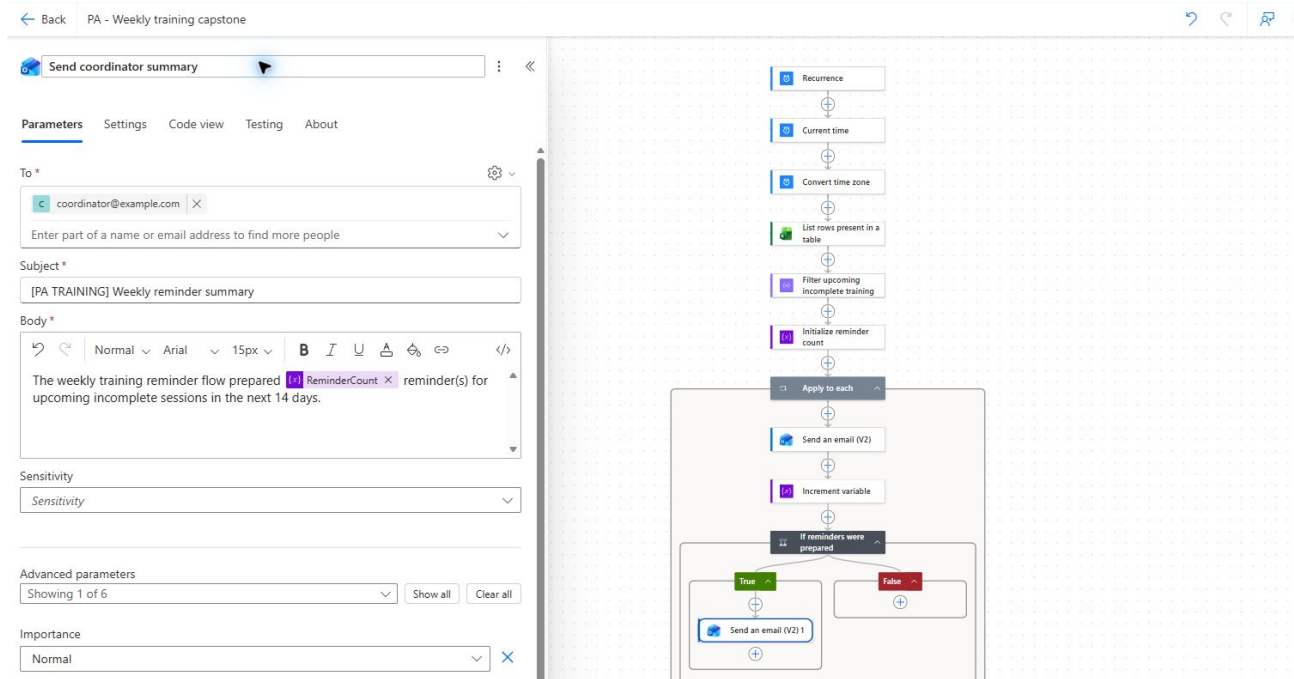


Figure 7.9. The True branch prepares a concise count for the coordinator. The reserved example.com address is a documentation placeholder.

7.10 Handle a week with no matching rows

1. In the **False** branch, add **Data Operation > Compose**, rename it **No upcoming records**, and enter: No upcoming incomplete training records were found for the next 14 days.

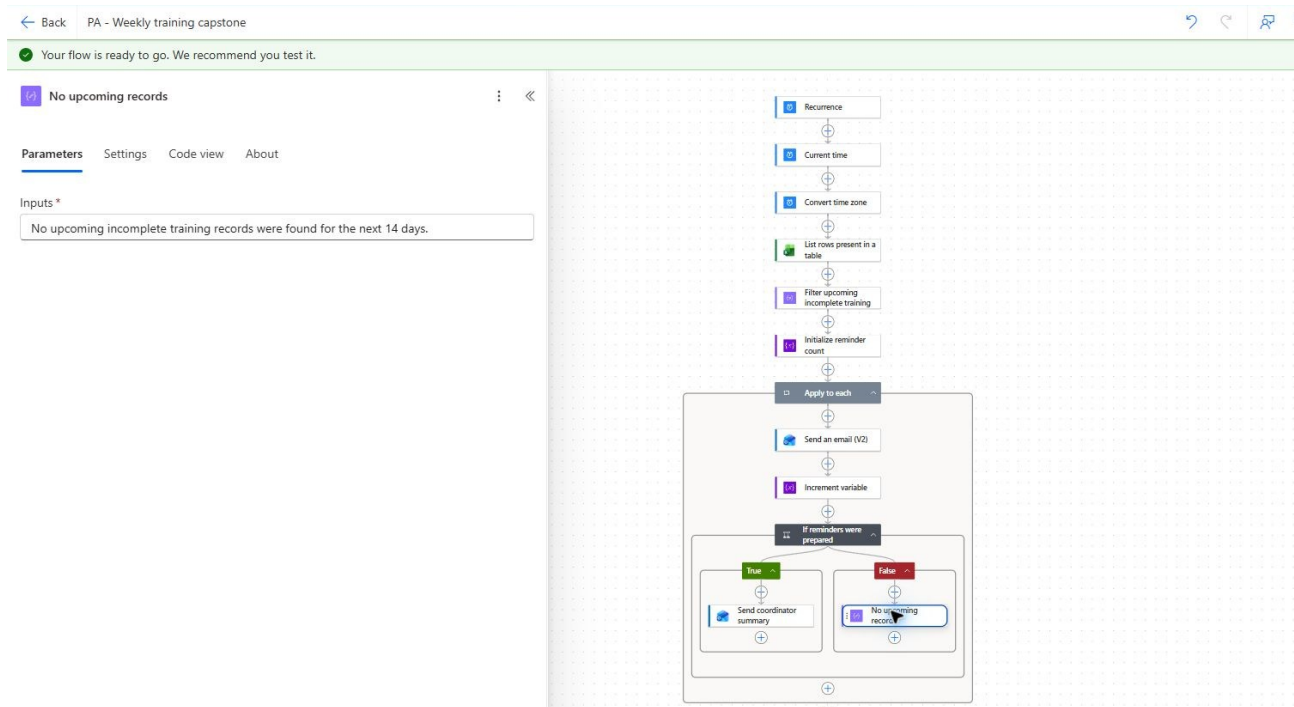


Figure 7.10. The False branch records an explicit outcome without preparing an unnecessary coordinator email.

7.11 Save and check the capstone

1. Select **Save**, then open **Flow checker**. Confirm that both **Errors** and **Warnings** show zero.

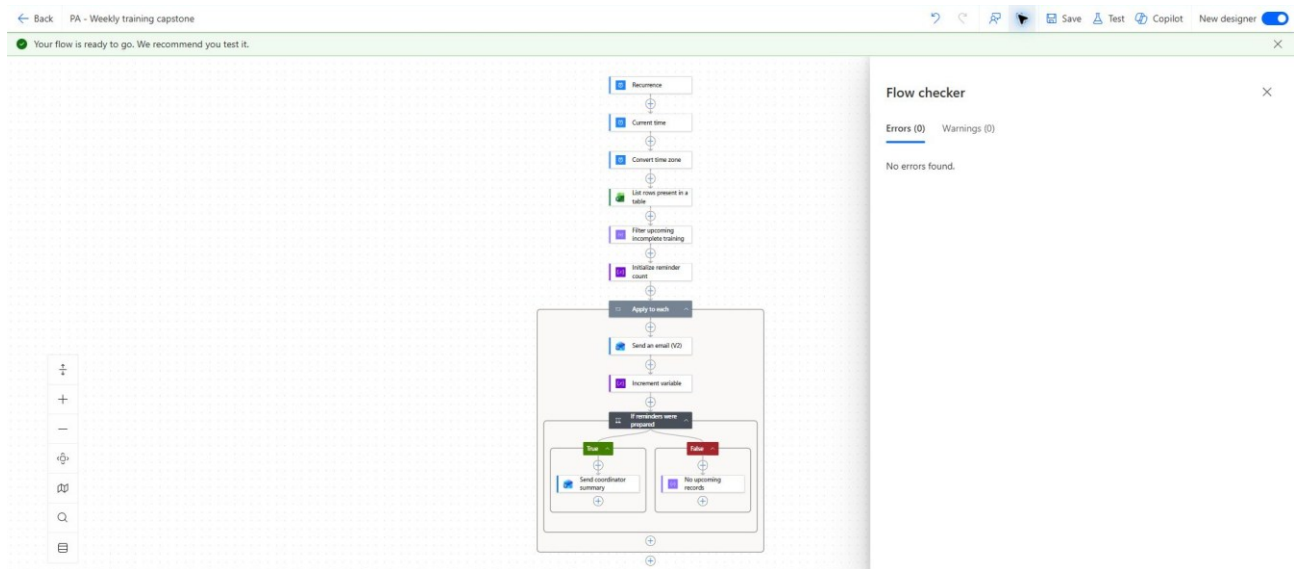


Figure 7.11. The saved capstone passes Flow checker with zero errors and zero warnings.

2. Close Flow checker, select **Zoom view to fit**, and inspect the complete sequence and both branches.

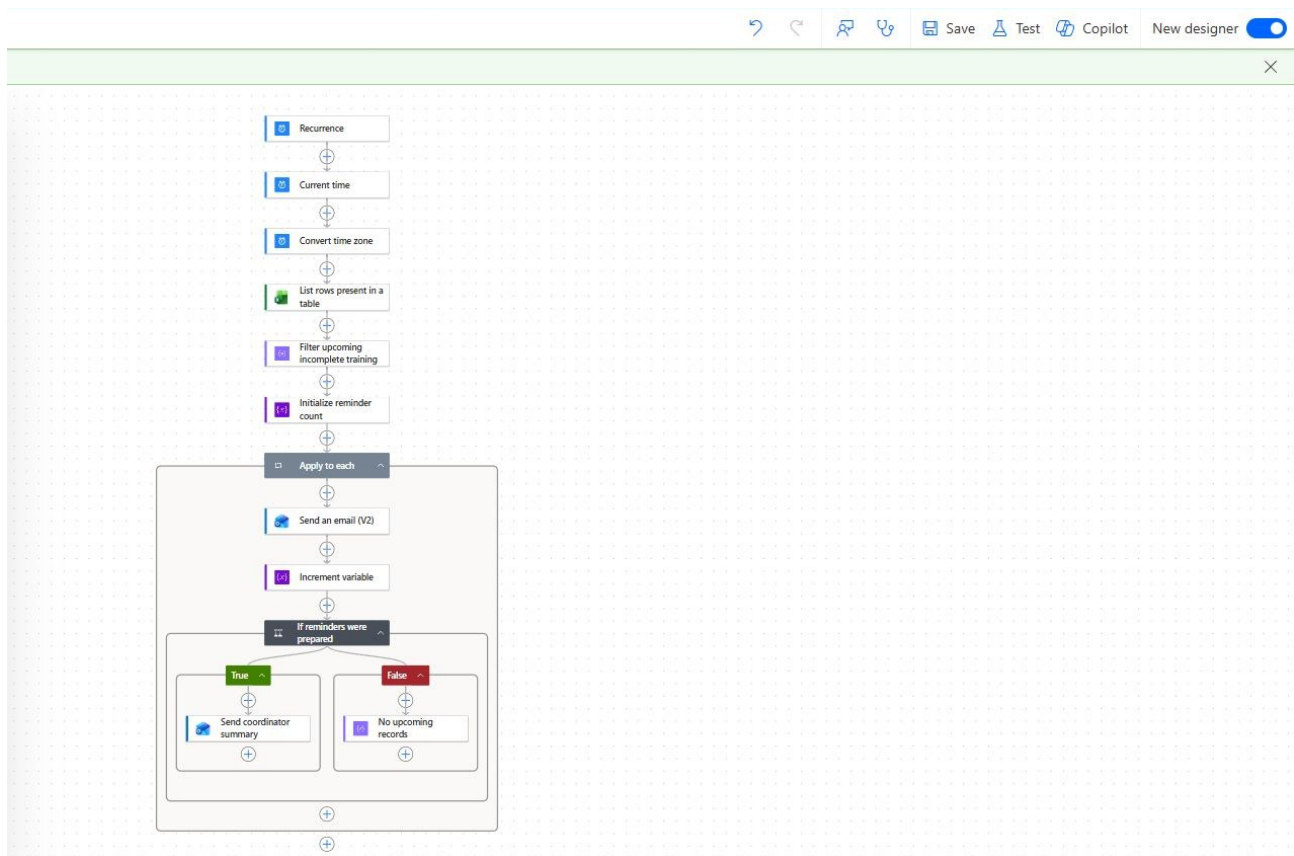


Figure 7.12. The complete saved capstone filters before the loop, counts inside the loop, and decides what to do after the loop.

Trainer-controlled test

Testing can send one learner reminder for every matching row and, when the count is positive, a coordinator summary. With the prepared sample on 17 September 2026, the filter is expected to select four rows.

Before a live test, the trainer must replace every `example.com` address with explicitly approved destinations, save the workbook, confirm the coordinator address, and authorize the run immediately before **Test** or **Run flow** is selected.

After an authorized test, inspect the run history and confirm that:

- Filter upcoming incomplete training returned the expected rows.
- Apply to each completed one iteration for every filtered row.
- ReminderCount increased once per completed reminder iteration.
- A positive count followed the True branch and prepared the coordinator summary.
- A zero-count test followed the False branch and produced the no-records Compose output.

Troubleshooting

Symptom	What to check
A completed row passes the filter	Status spelling, <code>trim()</code> , <code>toLowerCase()</code> , and the exact text completed .
A date comparison fails	DateTime Format is ISO 8601 and SessionDate contains valid dates.
Apply to each still includes every row	Use Filter upcoming incomplete training Body , not the Excel <code>value</code> array.
ReminderCount remains zero	The variable was initialized before the loop and incremented inside the loop.
The summary runs before reminders finish	Place the Condition after Apply to each, outside its container.
The False branch contains an email action	Use Compose for the no-records result in this exercise.
Flow checker is clean but a test is unsafe	Flow checker validates structure; it does not approve recipients or prevent messages from being sent.

Independent practice

Without running the flow, change the reminder window from 14 days to 7 days by editing only the final date expression. Identify which sample rows would match on 17 September 2026, then restore 14 days.

Lesson summary

The capstone filters an array before looping, normalizes text inside an expression, compares dates numerically, maintains an integer counter, and branches on the final count. These steps turn a broad weekly reminder into a controlled process with a clear coordinator outcome.

Chapter 8: Plan an automation with Copilot

Estimated time: 30 minutes

Your result: A reviewed, improved flow plan produced with Power Automate Copilot, without creating or running a cloud flow.

What you will learn

By the end of this chapter, you will be able to:

- Describe an automation to Copilot with a clear, bounded prompt.
- Review an AI-generated flow plan against the business rule.
- Refine a suggestion by naming the exact change you need.
- Leave the planning experience without creating a flow.

The business problem

A natural-language description can produce a useful starting point, but a suggested flow is not a finished design. Copilot may omit a calculation, choose an unnecessary action, or interpret a broad instruction differently from the intended business rule.

In this chapter, you will ask Copilot to plan a weekly coordinator summary. You will review the proposed trigger and actions, give precise corrections, and stop before Power Automate creates the flow.

The intended plan is:

Recurrence > List rows present in a table > Filter array > Compose count > Send one coordinator email

Before you begin

Copilot availability can differ by organization, environment, region, and administrator settings. If **Create with Copilot** or **Describe it to design it** is unavailable, use the screenshots in this chapter as a supplied example and complete the review questions without entering a prompt.

The exercise uses only fictional training details and `coordinator@example.com`, a reserved documentation address. Do not include confidential information, personal data, passwords, or real recipients in a training prompt.

8.1 Open the Copilot planning experience

1. On the Power Automate home page, select **Create with Copilot**. The Create page may label the same experience **Describe it to design it**.

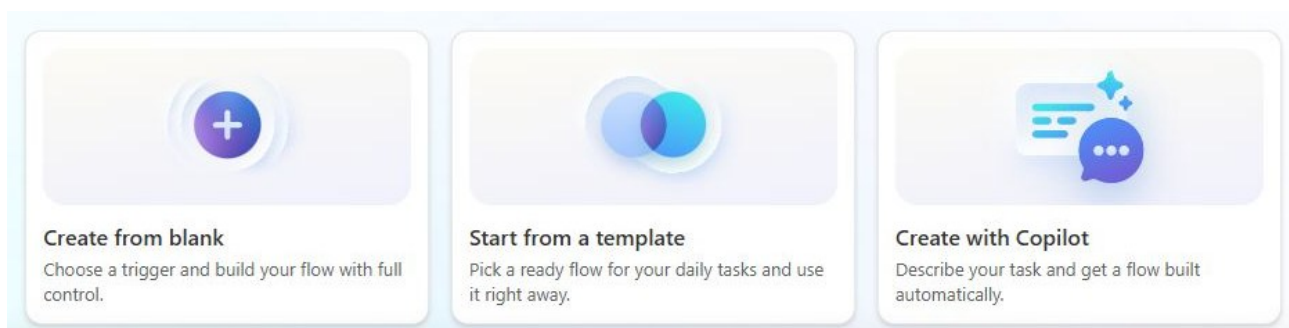


Figure 8.1. Create with Copilot opens the natural-language flow planning experience.

8.2 Describe the complete business outcome

1. In **What will your flow do?**, enter the following prompt, then select **Submit**:

Every Friday at 4:00 PM Singapore time, read rows from the tblTraining table in TrainingRegister.xlsx in OneDrive for Business. Keep rows whose Status is Registered and SessionDate is within the next 14 days. Email the training coordinator a summary count. Do not send participant emails.

Describe it to design it

Step 1 of 2

What will your flow do?

Describe what you want to automate and AI will help you make it happen. [How it works](#)

Every Friday at 4:00 PM Singapore time, read rows from the tblTraining table in TrainingRegister.xlsx in OneDrive for Business. Keep rows whose Status is Registered and SessionDate is within the next 14 days. Email the training coordinator a summary count. Do not send participant emails.

Or try one of these examples to get started

- Every time I receive an email, move it to a folder
- Send an automatic message to a Teams channel once every week
- Send an email reminder every week

[View more examples](#)

Cancel

Figure 8.2. The prompt names the schedule, source, table, filter, output, and an explicit action to avoid.

A strong prompt states the event, data source, rule, output, and boundaries. It still does not replace configuration: Copilot does not know the approved recipient, exact workbook location, organizational policies, or how the result should be tested.

8.3 Review the first suggestion

1. Read the suggested trigger and every proposed action before selecting **Keep it and continue**. In the captured first version, Copilot suggested:

- Recurrence
- List rows present in a table
- FilteredRows
- Initialise Count
- Send an email

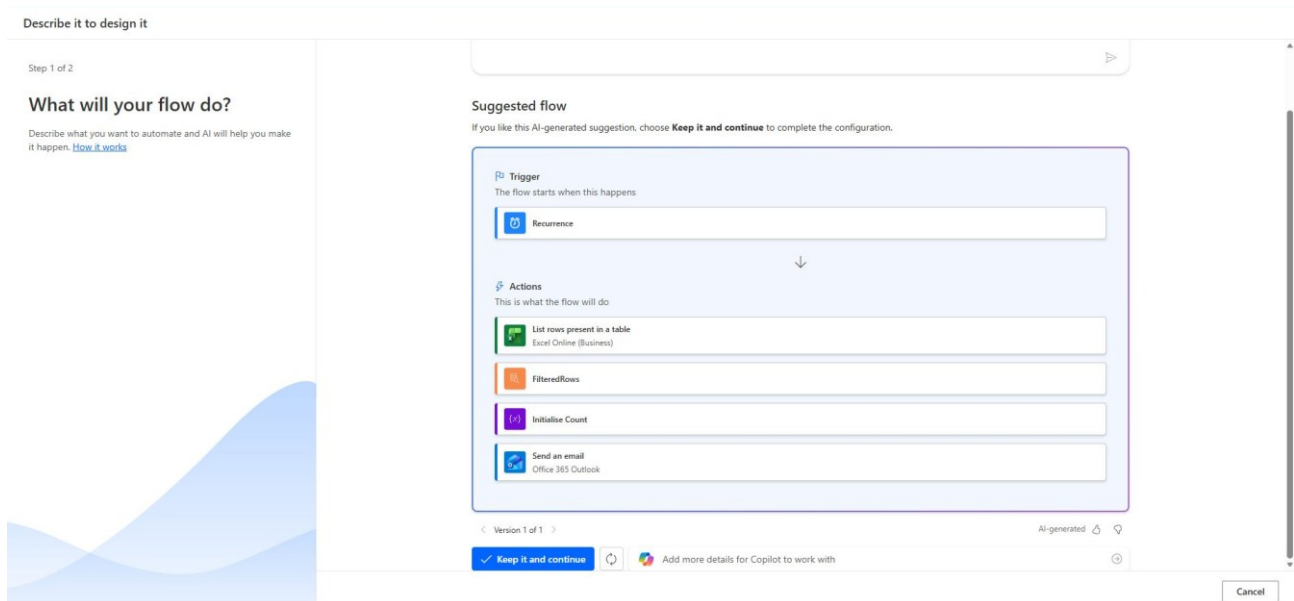


Figure 8.3. The initial structure is plausible, but *Initialise Count* does not show how the value will be calculated.

Use these questions whenever you review an AI-generated flow plan:

Review area	Question
Trigger	Does it start at the intended event, time, and time zone?
Data	Does it use the correct connector, file, table, list, form, or mailbox?
Logic	Are filtering, looping, conditions, and calculations explicit?
Output	Is the destination approved, and could the action contact real people?
Licensing	Are all proposed connectors and actions available to the participants?
Testing	Can the plan be tested safely with fictional data and approved destinations?

8.4 Clarify the filter and count

1. In **Add more details to improve the flow**, enter this refinement and select **Send**:

Use Filter array to keep rows where Status is Registered and SessionDate is between today and 14 days from today. Calculate the count with `length(body('FilteredRows'))`. Send one summary email only to `coordinator@example.com`. Do not add an Apply to each or any participant email.

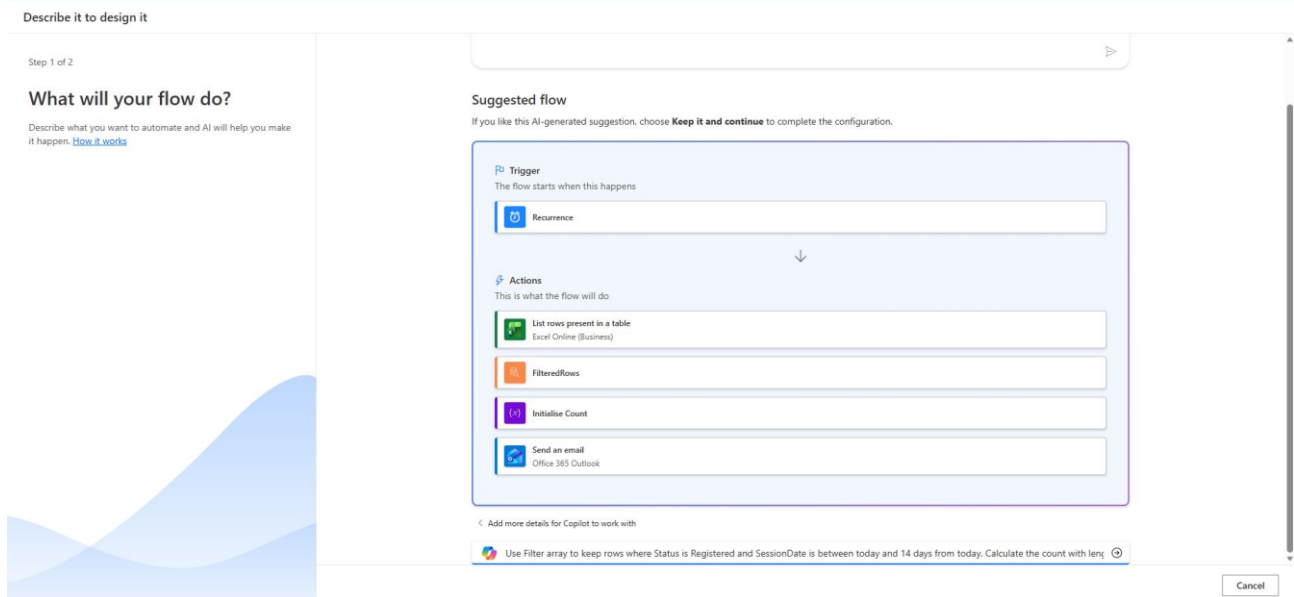


Figure 8.4. The refinement converts vague intent into named actions, a counting expression, and a clear messaging boundary.

8.5 Identify an unnecessary action

1. Review the second suggestion. It now includes **Count** and still has one email action, but it also adds **Get file metadata using path** before **List rows present in a table**.

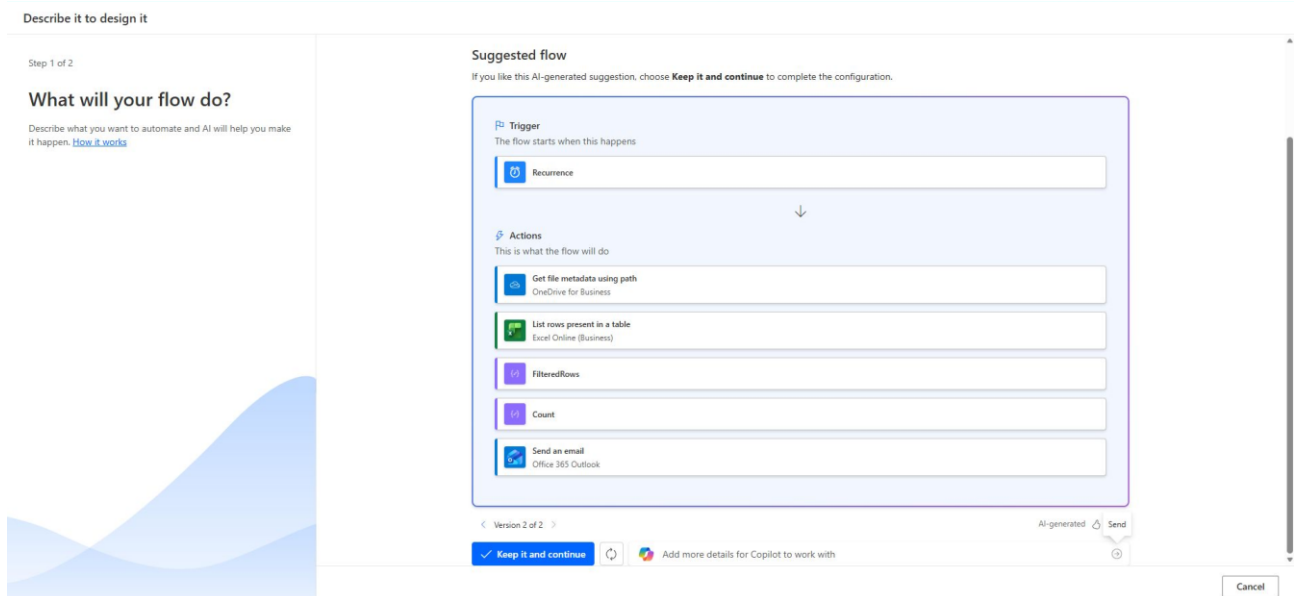


Figure 8.5. The second version improves the count but introduces an additional OneDrive action that is unnecessary for this plan.

List rows present in a table can select the workbook directly during configuration. An extra metadata action adds another dependency without contributing to the intended output.

8.6 Request a streamlined design

1. Enter the following refinement and select **Send**:

Remove Get file metadata using path because List rows present in a table can select the workbook directly. Keep only Recurrence, List rows present in a table, Filter array named FilteredRows, Compose named Count using

```
length(body('FilteredRows'))), and one Send an email action to  
coordinator@example.com.
```

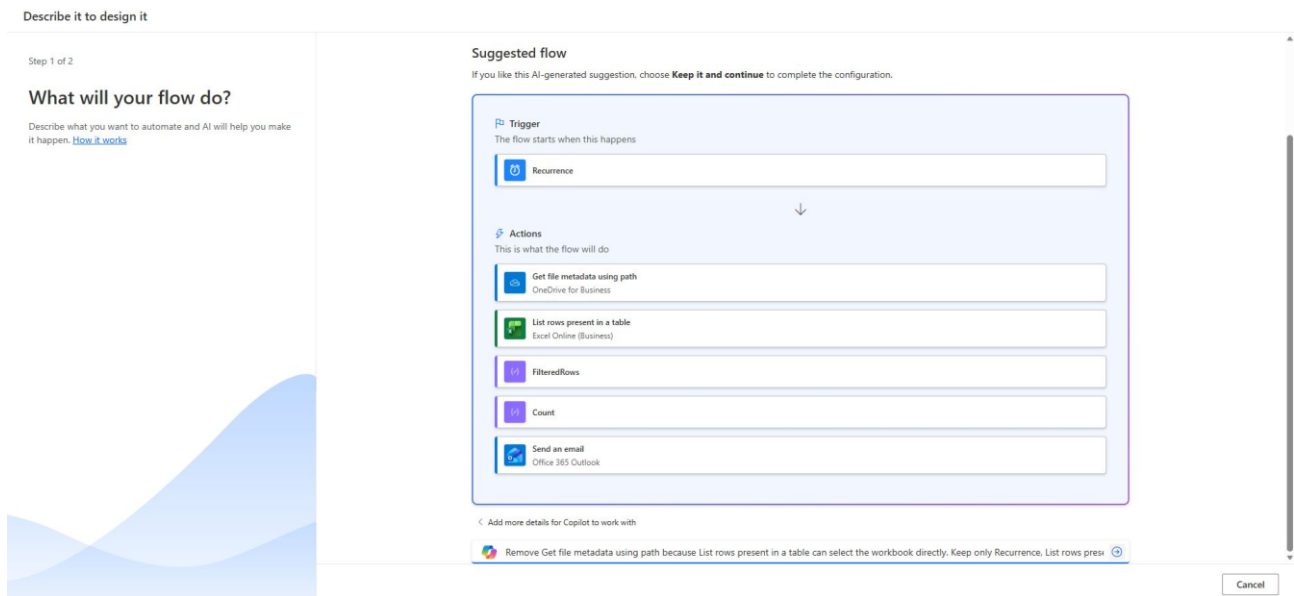


Figure 8.6. The request explains what to remove, why it is unnecessary, and the exact structure to retain.

8.7 Validate the final suggestion

1. Confirm that the third suggestion contains:

Order	Proposed step
1	Recurrence
2	List rows present in a table
3	FilteredRows
4	Compose Count
5	Send an email

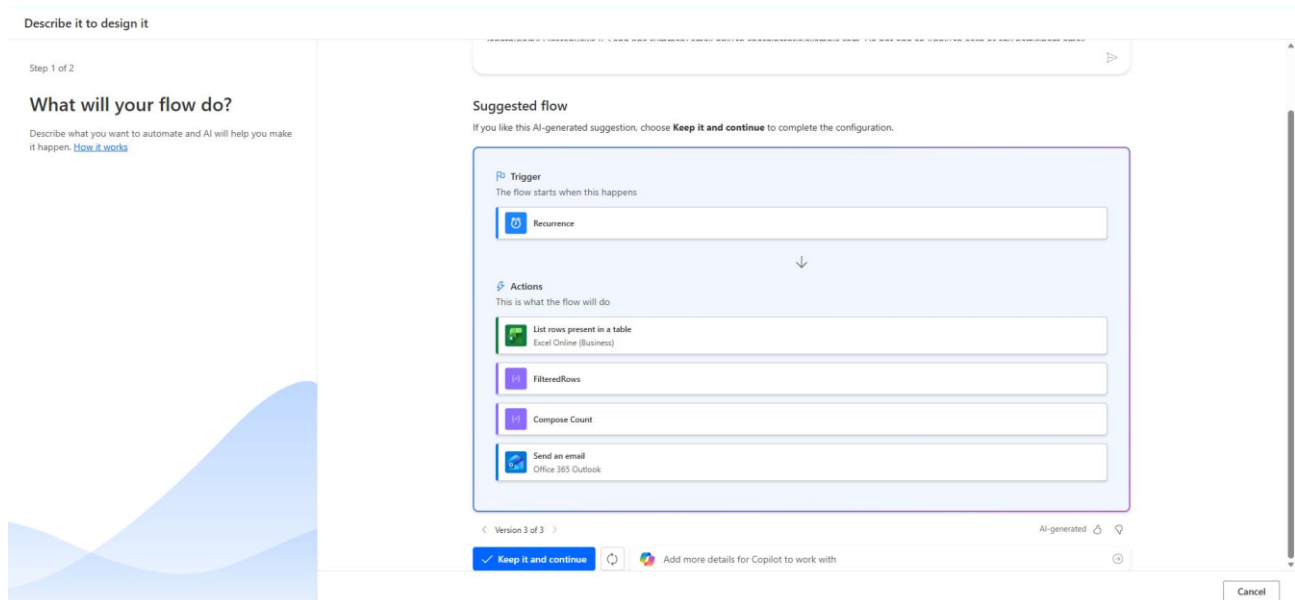


Figure 8.7. The final suggestion matches the intended five-step structure and remains marked AI-generated.

This screen verifies the outline only. If you continued, you would still need to configure the Friday schedule and time zone, workbook and table, filter expression, count expression, recipient, subject, body, error handling, and test data.

8.8 Exit without creating the flow

1. Select **Cancel** instead of **Keep it and continue**. Confirm that Power Automate returns to the Create page.

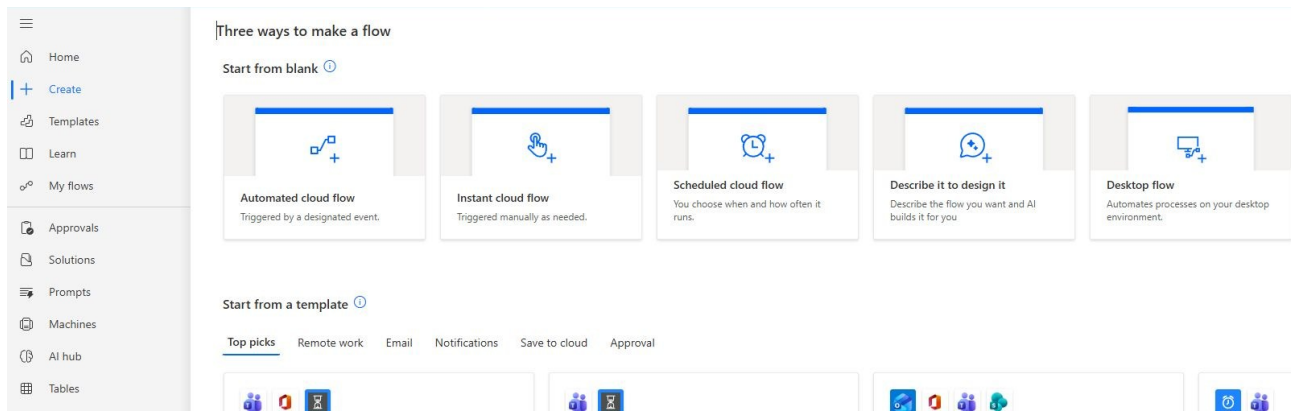


Figure 8.8. Returning to the Create page confirms that the exercise remained a planning activity.

Troubleshooting

Symptom	What to check
Create with Copilot is missing	Organization settings, environment, region, account entitlement, and the supplied-example fallback.
Copilot suggests the wrong trigger	State the exact event or schedule, frequency, day, time, and time zone.
The plan contains a participant email loop	State that only one coordinator summary is required and explicitly exclude participant emails.
The count is unclear	Name the filtered array and request <code>length(body('FilteredRows'))</code> .
An unnecessary connector appears	Explain which existing action already provides the required data and ask Copilot to remove the extra dependency.
The final outline looks correct	Continue reviewing; the actual fields, expressions, connections, recipients, and test behavior are not configured yet.

Independent practice

Write a prompt for a monthly training-capacity report. Include the schedule, data source, inclusion rule, output, and one explicit boundary. Exchange prompts with another participant and identify one ambiguity before asking Copilot to generate a suggestion.

Lesson summary

Copilot can quickly propose a trigger and action sequence, preserve versions, and respond to refinements. A participant remains responsible for the business rule, connector choice, licensing, recipient safety, expressions, configuration, and testing. The best result came from reviewing each suggestion and explaining the exact correction.

Chapter 9: Move automations with environments and solutions

Estimated time: 60 minutes

Your result: An unmanaged solution package that contains the completed capstone flow and its connection references, plus a safe walkthrough of the import starting point.

What you will learn

By the end of this chapter, you will be able to:

- Explain what an environment is and why organizations separate development from production.
- Describe what a solution can contain.
- Explain how Power Automate, Power Apps, and Copilot Studio can share solution components.
- Create a solution and add an existing cloud flow.
- Publish and export an unmanaged solution.
- Identify the safe import, connection-repair, verification, and testing sequence.

The business problem

A flow can work correctly in one environment and still be difficult to move safely. It may depend on connections, tables, apps, bots, environment variables, or other components. Copying parts one at a time makes it easy to miss a dependency.

Power Platform solutions group related components into a transportable package. A maker can develop and review work in one environment, export the solution, and import it into another environment under controlled conditions.

This chapter uses the completed capstone as the example:

Development environment > Solution > Publish > Export package > Target environment > Import > Repair connections > Verify > Authorized test

The live demonstration stops after opening the import screen. Importing the package back into the same environment is not a valid migration test, and testing the capstone could send email.

Before you begin

This is a trainer-led exercise unless every participant has access to an environment with Microsoft Dataverse and permission to create, export, and import solutions. Power Automate Free access alone does not guarantee that these features or permissions are available.

The worked example uses PA – Weekly training capstone from Chapter 7. Keep the flow turned off. Do not import or test it unless the trainer supplies a separate target environment, approves the destinations, and authorizes the run.

Environments in plain language

An **environment** is a separate workspace for business apps, flows, data, connections, and security. The environment selector at the top of Power Automate shows which workspace you are using.

Environment	Typical purpose	Practical rule
Development	Build and change automations	Use fictional or approved test data.
Test	Validate behavior before release	Reconnect services and perform controlled tests.
Production	Run approved business processes	Restrict changes and monitor real operation.

Separating these stages reduces the chance that unfinished changes will affect real users. The separation is useful only when permissions, connection ownership, data, and release procedures are also controlled.

Power Automate, Power Apps, and Copilot Studio all use the Power Platform. A single solution can therefore package related cloud flows, apps, agents, tables, connection references, environment variables, and other dependencies for the same business process. The exact component types available depend on the environment and licenses.

9.1 Confirm the current environment

1. Check the environment selector before creating or importing anything. Confirm with the trainer that this is the development or training environment.

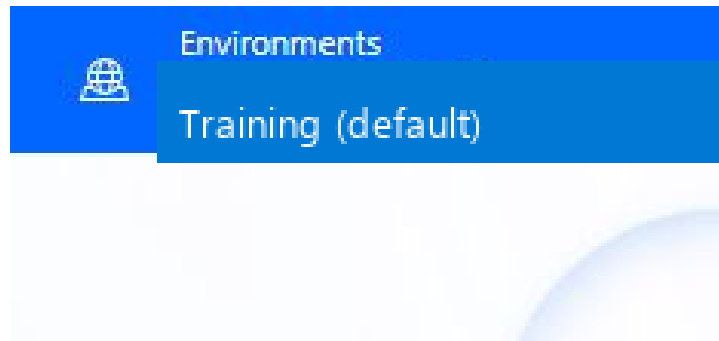


Figure 9.1. The environment selector controls which set of flows, solutions, connections, and data you are viewing.

Checkpoint: If the expected flow or solution is missing, check the environment before rebuilding it.

9.2 Create the solution

1. Open **Solutions**, select **New solution**, and enter:

Field	Value
Display name	Power Automate Training Migration
Name	PowerAutomateTrainingMigration
Publisher	The trainer-approved publisher
Version	1.0.0.0

Then select **Create**.

New solution ✕

Display name *

Power Automate Training Migration

Name *

PowerAutomateTrainingMigration

Publisher *

CDS Default Publisher (Cradf35) ▼ ✎

+ New publisher

Version *

1.0.0.0

More options ▼

Create

Cancel

Figure 9.2. The display name is friendly; the unique name is used internally and normally should not be changed later.

9.3 Review the empty solution

1. Open the new solution and confirm that its object list is empty.

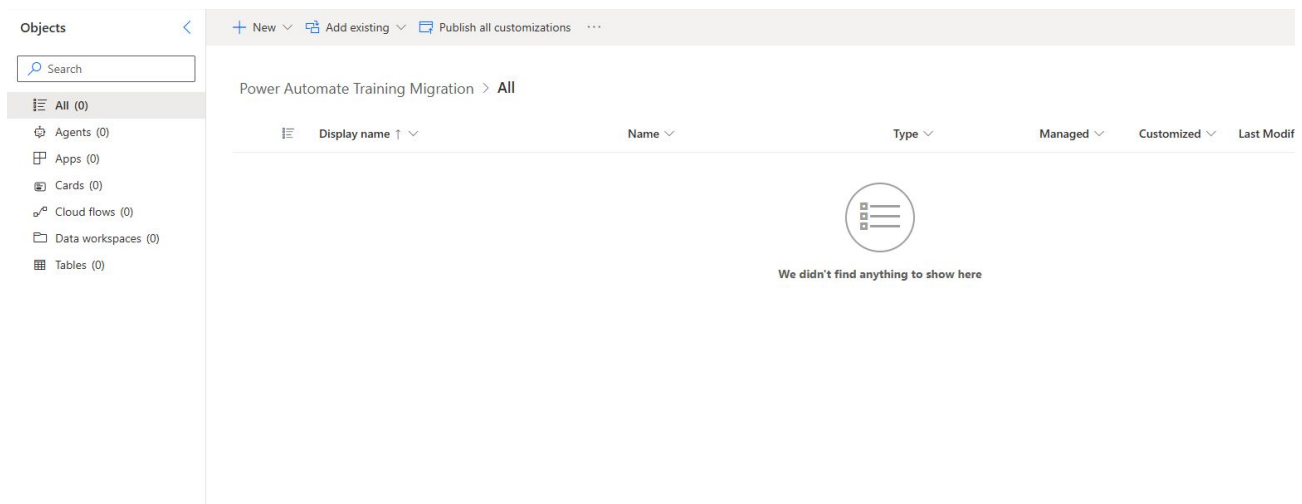


Figure 9.3. A new solution is a container; it does not automatically include existing flows.

9.4 Start adding an existing component

1. Select **Add existing**. Review the component categories and point to **Automation**.

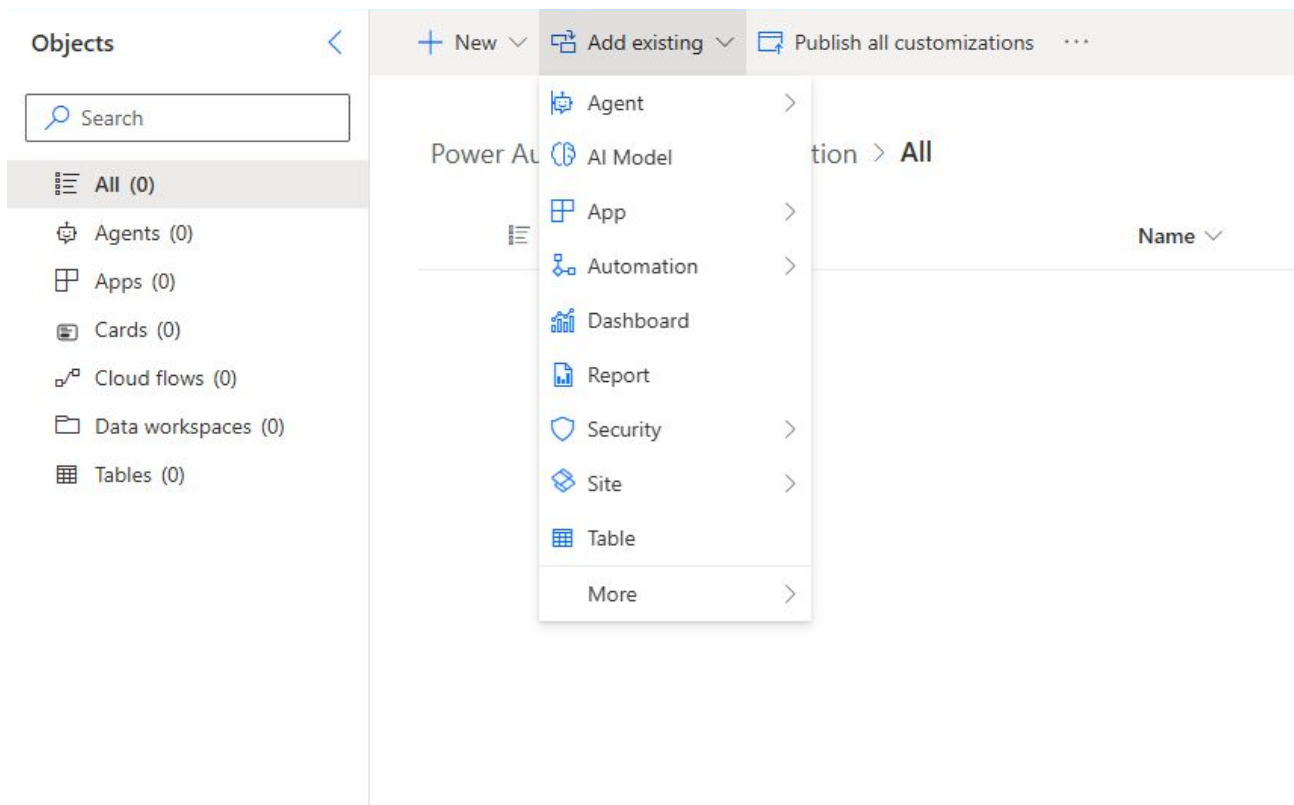


Figure 9.4. Add existing brings a component that already exists in the environment into the solution.

9.5 Choose Cloud flow

1. Under **Automation**, select **Cloud flow**.

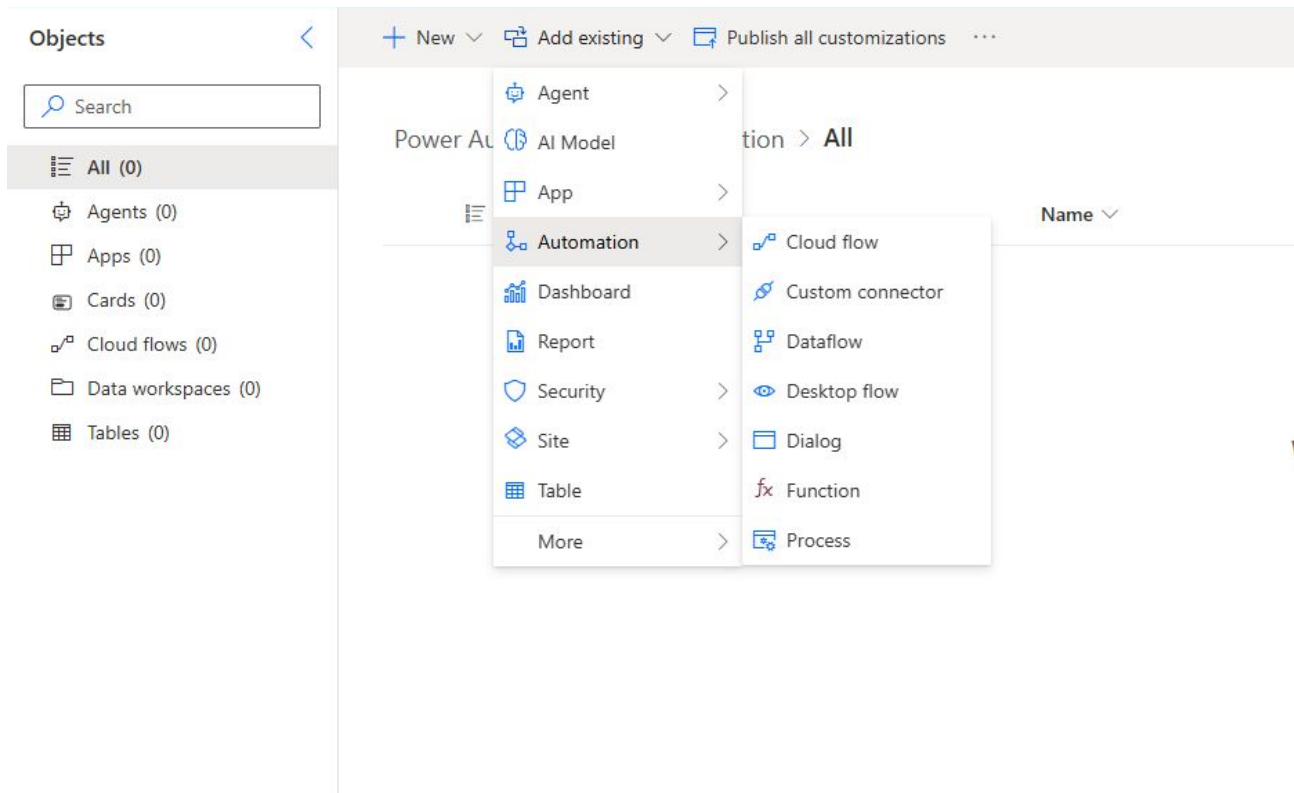
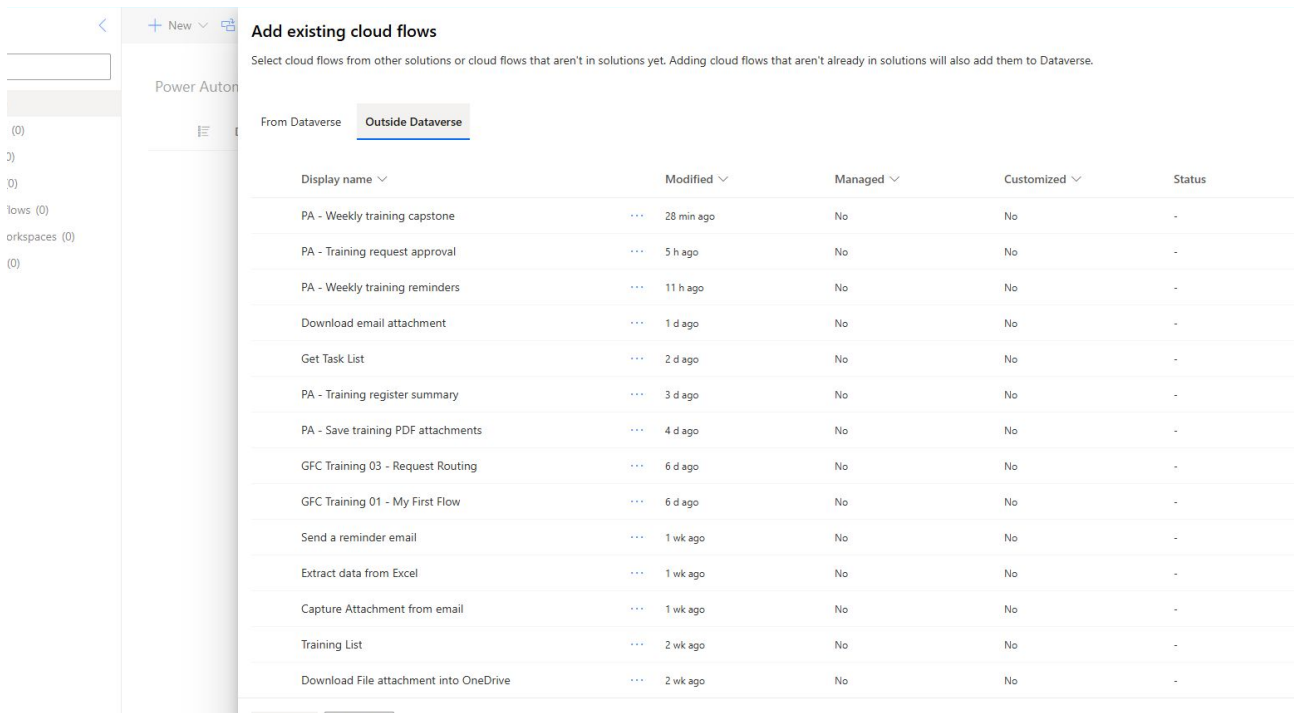


Figure 9.5. Cloud flow opens the list of flows that can be added to the solution.

9.6 Find a flow outside Dataverse

1. Select **Outside Dataverse**. Locate PA – Weekly training capstone in the list.



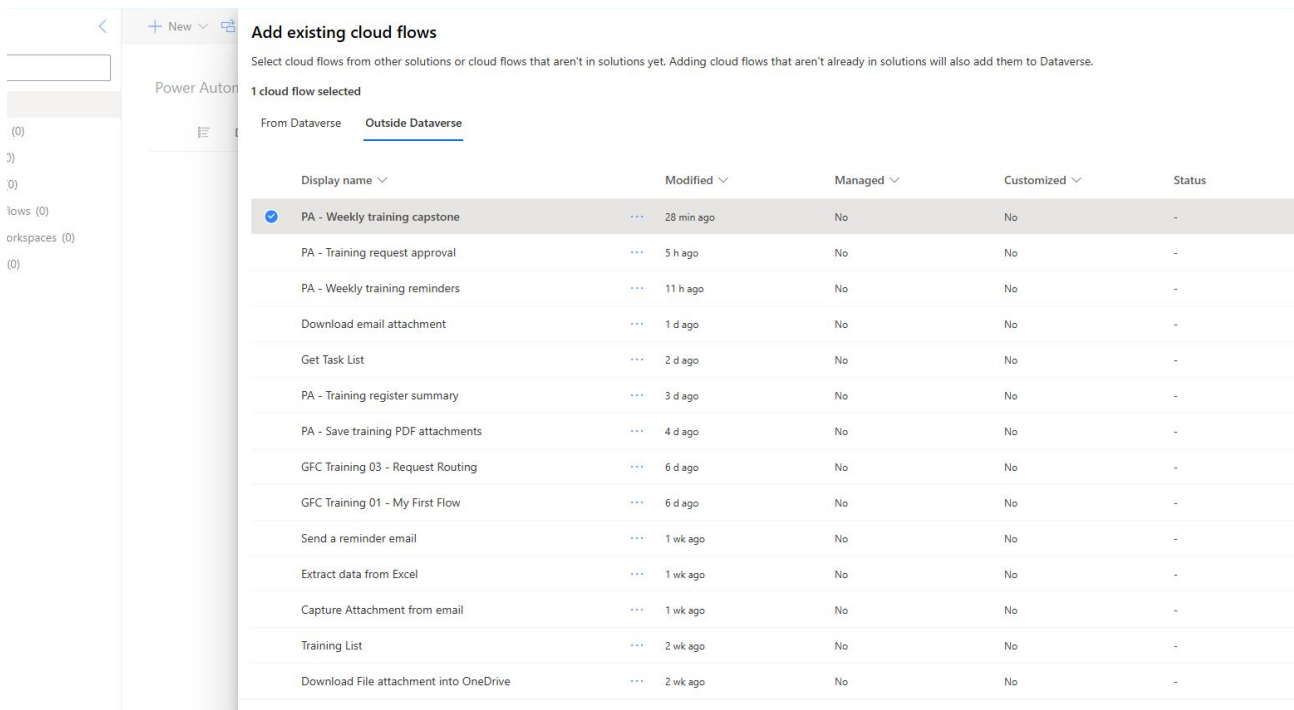
The screenshot shows the 'Add existing cloud flows' dialog in Microsoft Power Automate. The 'Outside Dataverse' tab is selected, and a list of flows is displayed. The flow 'PA - Weekly training capstone' is highlighted.

Display name	Modified	Managed	Customized	Status
PA - Weekly training capstone	28 min ago	No	No	-
PA - Training request approval	5 h ago	No	No	-
PA - Weekly training reminders	11 h ago	No	No	-
Download email attachment	1 d ago	No	No	-
Get Task List	2 d ago	No	No	-
PA - Training register summary	3 d ago	No	No	-
PA - Save training PDF attachments	4 d ago	No	No	-
GFC Training 03 - Request Routing	6 d ago	No	No	-
GFC Training 01 - My First Flow	6 d ago	No	No	-
Send a reminder email	1 wk ago	No	No	-
Extract data from Excel	1 wk ago	No	No	-
Capture Attachment from email	1 wk ago	No	No	-
Training List	2 wk ago	No	No	-
Download File attachment into OneDrive	2 wk ago	No	No	-

Figure 9.6. A non-solution flow appears under **Outside Dataverse** until it is added to a solution.

9.7 Add the completed capstone

1. Select PA – Weekly training capstone, then select **Add**.



The screenshot shows the 'Add existing cloud flows' dialog in Microsoft Power Automate. The 'Outside Dataverse' tab is selected, and the flow 'PA - Weekly training capstone' is selected.

Display name	Modified	Managed	Customized	Status
PA - Weekly training capstone	28 min ago	No	No	-
PA - Training request approval	5 h ago	No	No	-
PA - Weekly training reminders	11 h ago	No	No	-
Download email attachment	1 d ago	No	No	-
Get Task List	2 d ago	No	No	-
PA - Training register summary	3 d ago	No	No	-
PA - Save training PDF attachments	4 d ago	No	No	-
GFC Training 03 - Request Routing	6 d ago	No	No	-
GFC Training 01 - My First Flow	6 d ago	No	No	-
Send a reminder email	1 wk ago	No	No	-
Extract data from Excel	1 wk ago	No	No	-
Capture Attachment from email	1 wk ago	No	No	-
Training List	2 wk ago	No	No	-
Download File attachment into OneDrive	2 wk ago	No	No	-

Figure 9.7. Only the completed capstone is selected for this migration package.

Power Automate may also add connection references for connectors used by the flow. A connection reference is a solution component that points to a connection. The target environment still needs a valid connection and authorized account.

9.8 Verify the solution components

1. Confirm that the solution contains the capstone cloud flow and the required Excel Online (Business) and Office 365 Outlook connection references.

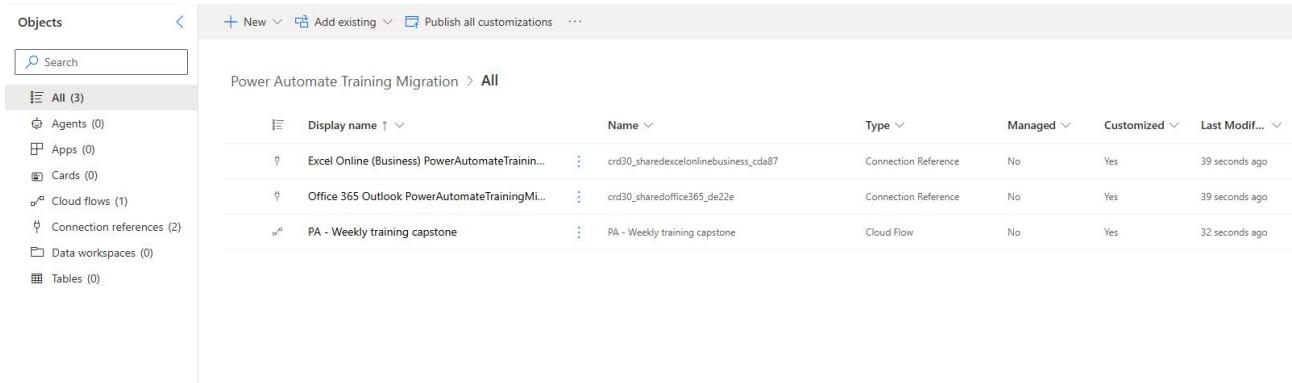


Figure 9.8. The solution contains three objects: the flow and two connection references.

Checkpoint: Do not assume that every dependency is ready just because it appears in the object list. File locations, table names, recipients, connection accounts, environment variables, and permissions may differ in the target environment.

9.9 Review the solution before export

1. Open **Overview** and confirm the solution name, version, package type, publisher, and included items.

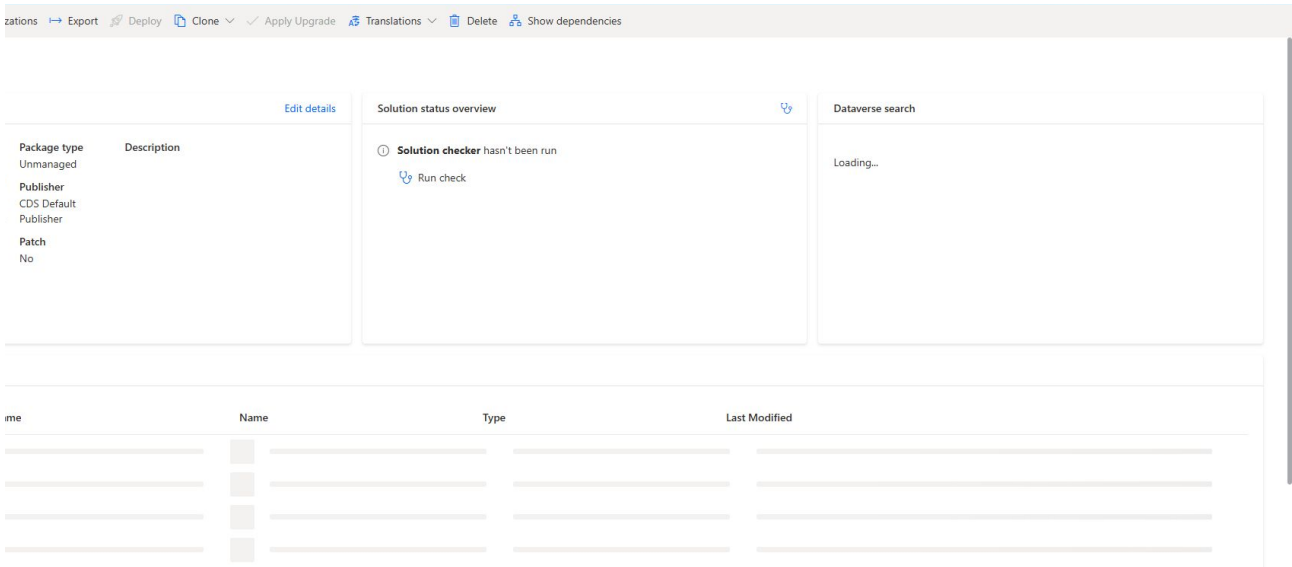


Figure 9.9. Overview is the final review point before publishing and exporting.

9.10 Publish changes

1. Select **Export**. On **Before you export**, select **Publish** so the latest solution changes are included.

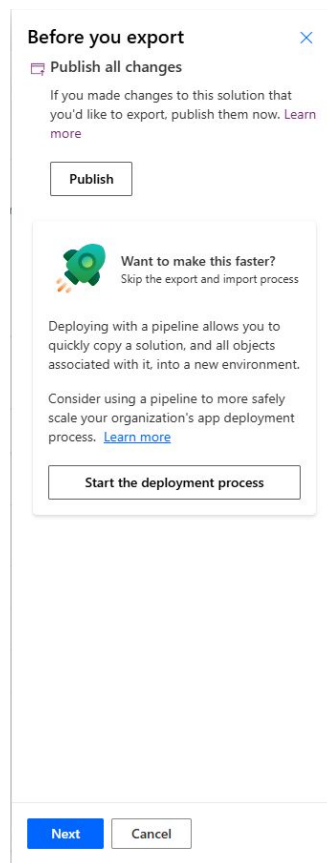


Figure 9.10. Publishing makes the latest customizations available to the exported package.

2. Wait until the panel shows **Published**, then select **Next**.

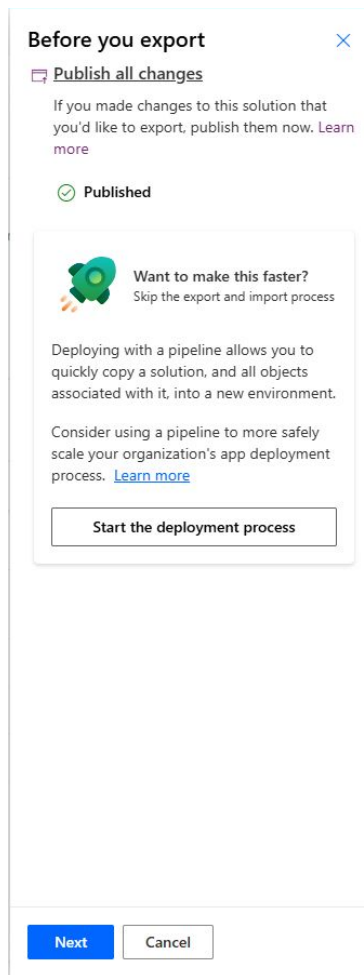
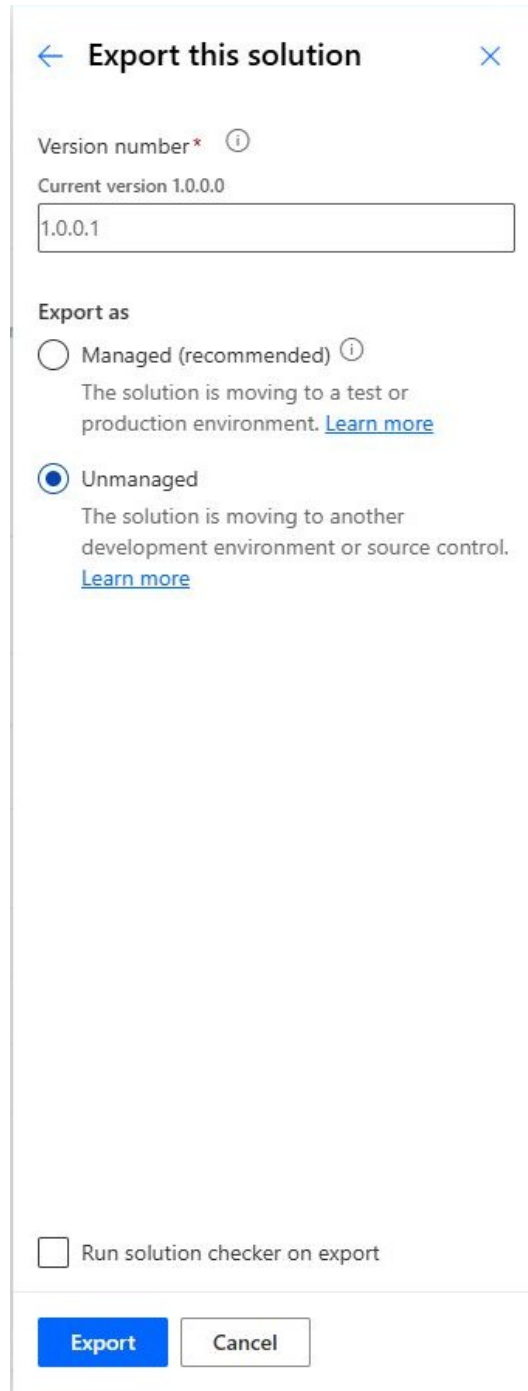


Figure 9.11. The published status confirms that the export can continue.

9.11 Export an unmanaged package

1. Confirm the version, select **Unmanaged**, and select **Export**. Power Automate suggests the next version number (for example, 1.0.0.1), so each exported package keeps its own version. This is why the file in Figure 9.13 ends in 1_0_0_1.



← Export this solution ×

Version number* ⓘ

Current version 1.0.0.0

1.0.0.1

Export as

☐ Managed (recommended) ⓘ

The solution is moving to a test or production environment. [Learn more](#)

☒ Unmanaged

The solution is moving to another development environment or source control. [Learn more](#)

☐ Run solution checker on export

Export Cancel

Figure 9.12. An unmanaged package is appropriate for this learning exercise because its components can be edited after import.

Managed solutions are normally used for controlled distribution where recipients should not directly edit the solution components. The release approach is an organizational design decision; do not switch package types casually in an established deployment process.

9.12 Download and protect the package

1. When the success message appears, select **Download**. Store the ZIP file in the trainer-approved location and keep its filename and version together.

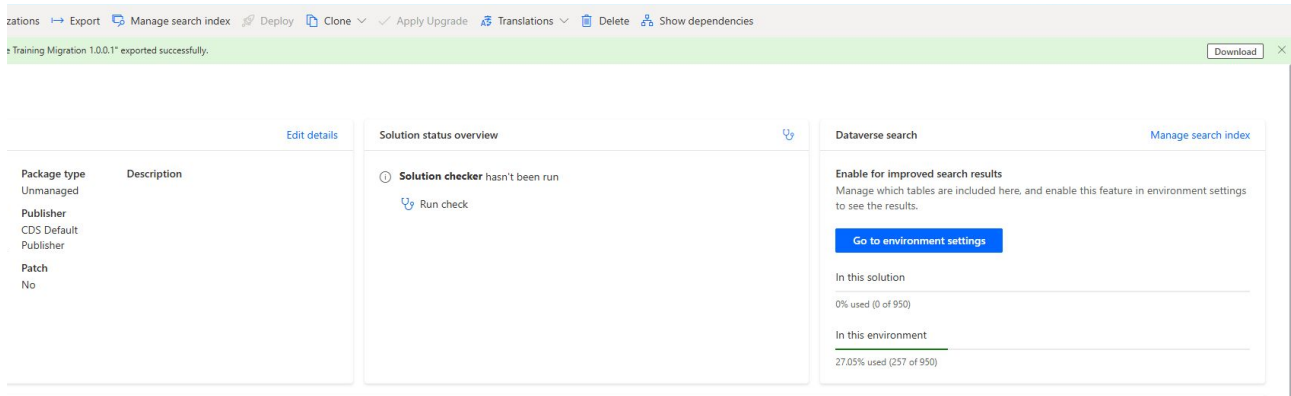


Figure 9.13. The worked example produced **PowerAutomateTrainingMigration_1_0_0_1.zip**.

Do not unzip or rename the contents before import. Treat the package as a release artifact and keep a record of who exported it, from which environment, and why.

9.13 Switch to the target environment

1. Use the environment selector to choose the trainer-approved target environment. Confirm the new environment name before selecting **Solutions**.

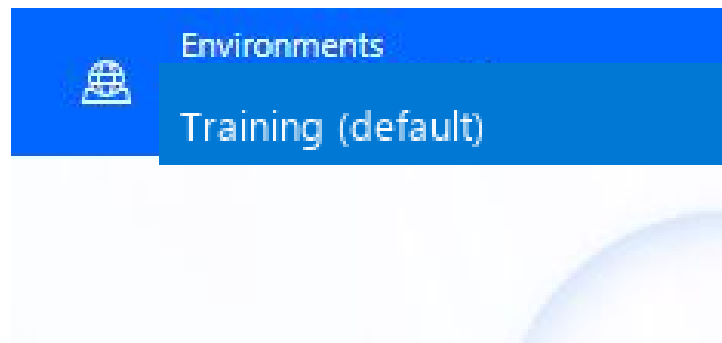


Figure 9.14. Recheck the selected environment before starting an import.

Your trainer may not switch environments during the demonstration if a second training environment is not available. Never use a production environment as a substitute for a training target.

9.14 Start the import

1. In the target environment, open **Solutions**, select **Import solution**, and use **Browse** to select the exported ZIP package. Select **Next** only after confirming the package name and target environment.

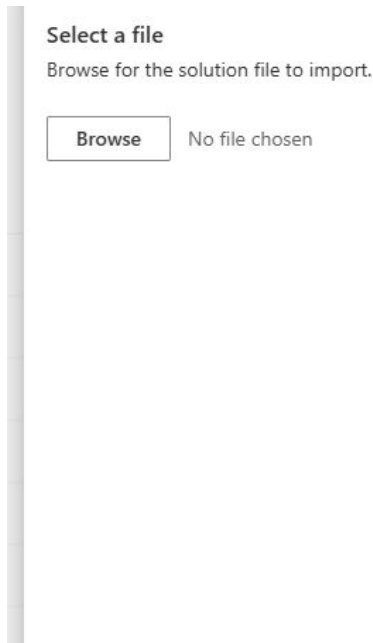


Figure 9.15. The demonstration stops at the import starting point; no file was uploaded and no solution was imported.

9.15 Repair connections and settings

After the package is selected, Power Platform may ask for connection mappings or environment-specific values. Complete these checks before import or first use:

1. Map each connection reference to an approved connection in the target environment.
2. Confirm that the Excel workbook and `tblTraining` table exist in the intended target location.
3. Review recipients, team and channel selections, form references, schedules, time zones, and environment variables.
4. Confirm that the importing account and future flow owner have the required permissions.
5. Keep the imported flow turned off until verification is complete.

9.16 Verify and test safely

After an authorized import, open the solution and confirm that the expected components are present. Run Flow checker, review every connection and configuration field, and use fictional data and approved destinations for the first test.

Testing is a separate release decision. An imported flow can be structurally valid while still pointing to the wrong workbook, mailbox, form, team, channel, or recipient. Obtain trainer authorization immediately before a test that can send messages or modify shared data.

Troubleshooting

Symptom	What to check
Solutions is missing or unavailable	Environment type, Dataverse availability, license, security role, and maker permissions.
The flow is not in the list	Current environment, Outside Dataverse tab, flow ownership, and whether the flow is already in another solution.
Extra components appear	Connection references or other dependencies may be added automatically. Review each one.
Export remains pending	Refresh the Solutions page, open export history, and confirm that publishing completed.
Import says the solution already exists	Confirm that you are in the intended target environment and whether this is an upgrade or an accidental same-environment import.
A connection shows an error after import	Create or select an approved target connection and remap the connection reference.
The flow imports but cannot find the workbook	Verify the target OneDrive or SharePoint location, filename, table name, permissions, and connector account.
The imported flow is off	Leave it off while reviewing settings; turn it on only through the trainer-approved release process.

Independent practice

Create a migration checklist for one of the earlier flows. Name its trigger, connectors, data sources, recipients, environment-specific settings, owner, safe test data, and evidence required before production activation. Do not export, import, or run the flow.

Lesson summary

Environments separate workspaces, permissions, data, and release stages. Solutions package related Power Platform components so a process can move in a controlled way. A safe migration includes dependency review, publishing, versioned export, a confirmed target environment, connection mapping, configuration checks, Flow checker, controlled testing, and a deliberate activation decision.